Análisis Estadístico de Datos - Taller 02

Adriana Porras, Alejandra Niño, Bayron Morales, Juliana Cárdenas

EIA data

U.S. Energy Information Administration (EIA) es una organización que se encarga de recopilar, analizar y difundir información energética de manera independiente e imparcial para promover la formulación de políticas sólidas, mercados eficientes y la comprensión pública de la energía y su interacción con la economía y el medio ambiente.

Los datos que trabajaremos a continuación corresponden a la extracción de diferentes variables relacionadas con energías renovables y combustibles fósiles, las UE son 111 países de todos los continentes del mundo, seleccionados para fines académicos por ser aquellos países que presentan mayor cantidad de información de análisis.

El objetivo de este análisis es encontrar diferentes relaciones entre los países que se están estudiando, de tal forma que podamos caracterizar en términos de importación, exportación, consumo, producción, etc. a los países de estudio.

```
# Instalación de librerias en caso de no tenerlas
# install.packages("plotly")
# install.packages("MASS")
# install.packages("psych")
# install.packages("tidyverse")
# install.packages("factoextra")
# install.packages("GGally")
# install.packages("dplyr")

# Ejecución de librerias
library(tidyverse)
library(plotly)
library(MASS)
library(psych)
library(tidyverse)
library(factoextra)
library(GGally)
library(dplyr)
```

```
# Colores usados en todos los gráficos del taller
rojo        <- "#DA0921"
rojo_oscuro <- "#950606"
```

```
data1 <- read.csv("datos_taller_02.csv", row.names=1, sep=";")

# Revisar variables en la base
names(data1)
```

```
##  [1] "Bunker_fuel_consumption_TBPD"
##  [2] "Bunker_residual_fuel_oil_consumption_TBPD"
##  [3] "Crude_oil_including_lease_condensate_exports_TBPD"
##  [4] "Crude_oil_including_lease_condensate_imports_TBPD"
##  [5] "Crude_oil_including_lease_condensate_production_TBPD"
##  [6] "Crude_oil_including_lease_condensate_reserves_BB"
##  [7] "Distillate_fuel_oil_consumption_TBPD"
##  [8] "Distillate_fuel_oil_production_TBPD"
##  [9] "Jet_fuel_consumption_TBPD"
## [10] "Jet_fuel_production_TBPD"
## [11] "Kerosene_consumption_TBPD"
## [12] "Kerosene_production_TBPD"
## [13] "Liquefied_petroleum_gases_.LPG._consumption_TBPD"
## [14] "Liquefied_petroleum_gases_.LPG._production_TBPD"
## [15] "Motor_gasoline_consumption_TBPD"
## [16] "Motor_gasoline_production_TBPD"
## [17] "Natural_gas_plant_liquids_production_TBPD"
## [18] "Other_liquids_production_TBPD"
## [19] "Other_refined_products_consumption_TBPD"
## [20] "Other_refined_products_production_TBPD"
## [21] "Petroleum_and_other_liquids_CO2_emissions_MMTCD"
## [22] "Petroleum_and_other_liquids_consumption_TBPD"
## [23] "Petroleum_and_other_liquids_production_TBPD"
## [24] "Refined_petroleum_products_consumption_TBPD"
## [25] "Refined_petroleum_products_production_TBPD"
## [26] "Refinery_processing_gain_TBPD"
## [27] "Residual_fuel_oil_consumption_TBPD"
## [28] "Residual_fuel_oil_production_TBPD"
## [29] "Biomass_and_waste_electricity_installed_capacity_MK"
## [30] "Biomass_and_waste_electricity_net_generation_BKWH"
## [31] "Electricity_distribution_losses_BKWH"
## [32] "Electricity_exports_BKWH"
## [33] "Electricity_imports_BKWH"
## [34] "Electricity_installed_capacity_MK"
## [35] "Electricity_net_consumption_BKWH"
## [36] "Electricity_net_generation_BKWH"
## [37] "Electricity_net_imports_BKWH"
## [38] "Fossil_fuels_electricity_installed_capacity_MK"
## [39] "Fossil_fuels_electricity_net_generation_BKWH"
## [40] "Geothermal_electricity_installed_capacity_MK"
## [41] "Geothermal_electricity_net_generation_BKWH"
## [42] "Hydroelectric_pumped_storage_electricity_installed_capacity_MK"
## [43] "Hydroelectric_pumped_storage_electricity_net_generation_BKWH"
## [44] "Hydroelectricity_installed_capacity_MK"
## [45] "Hydroelectricity_net_generation_BKWH"
## [46] "Non.hydro_renewable_electricity_installed_capacity_MK"
## [47] "Non.hydro_renewable_electricity_net_generation_BKWH"
## [48] "Nuclear_electricity_installed_capacity_MK"
## [49] "Nuclear_electricity_net_generation_BKWH"
## [50] "Renewable_electricity_installed_capacity_MK"
## [51] "Renewable_electricity_net_generation_BKWH"
## [52] "Solar_electricity_installed_capacity_MK"
## [53] "Solar_electricity_net_generation_BKWH"
## [54] "Tide_and_wave_electricity_installed_capacity_MK"
```

```
## [55] "Tide_and_wave_electricity_net_generation_BKWH"
## [56] "Wind_electricity_installed_capacity_MK"
## [57] "Wind_electricity_net_generation_BKWH"
```

Normal multivariada

Pregunta 1. Encuentre la estimación del vector de medias y la matriz de varianzas y covarianzas para 5 variables de su interés del conjunto de datos.

Las 5 variables seleccionadas son:

- Electricity_net_consumption_BKWH (35)
- Electricity_net_generation_BKWH (36)
- Electricity_distribution_losses_BKWH (31)
- Fossil_fuels_electricity_net_generation_BKWH (39)
- Renewable_electricity_net_generation_BKWH (51)

```
data2 <- data1[c(35:36,31,39,51)]

# Se ajustan los nombres para facilitar la visualización
names(data2) <- c("Consumo(C)", "Generacion(G)","Perdida(P)",
                  "Fossil_G", "Renewable_G")

names(data2)
```

```
## [1] "Consumo(C)"    "Generacion(G)" "Perdida(P)"    "Fossil_G"
## [5] "Renewable_G"
```

```
# Vector de medias
colMeans(data2)
```

```
##    Consumo(C) Generacion(G)    Perdida(P)      Fossil_G   Renewable_G
##     181.81592     198.29009      17.00450     132.58408      44.77117
```

Es necesario resaltar que las cinco variables seleccionadas manejan la misma escala, ya que se presentan en BKWH (Billion Kilowatt Hours (Miles de millones de kilovatios-hora)). Respecto al vector de medias, la generación neta de electricidad es la que tiene la mayor media (Generación(G): 198.290 BKWH), seguido por el consumo neto de electricidad (Consumo(C): 181.816 BKWH) y la generación neta de electricidad a partir de combustibles fósiles (Fossil_G: 132.584). La generación neta de electricidad renovable tiene una media de 44.771 y la media más baja la tienen las pérdidas en la distribución de electricidad (17.004). Esta última variable representa la diferencia entre la energía eléctrica que se inyecta en la red de distribución y la energia que finalmente se entrega a los consumidores finales.

```
# Matriz de varianzas y covarianzas:
cov(data2)
```

```
##                Consumo(C) Generacion(G) Perdida(P)  Fossil_G Renewable_G
## Consumo(C)      383958.82     408034.46  25162.312 290441.49   82086.952
## Generacion(G)   408034.46     434094.97  27076.711 308733.29   87623.857
## Perdida(P)       25162.31      27076.71   1990.936  19210.37    5578.751
## Fossil_G        290441.49     308733.29  19210.368 223824.71   60829.937
## Renewable_G      82086.95      87623.86   5578.751  60829.94   20864.464
```

La variable que presenta mayor varianza es consumo neto de electricidad (Consumo(C): 383958) y la que presenta menor varianza es pérdidas en la distribución de electricidad (Perdida(P): 1990.936). Sin embargo, se debe recordar que esta se ve afectada por valores atípicos, por lo que puede estar sobrestimada.

Respecto a las covarianza, se observa que todas las variables se relacionan positivamente, es decir, que al aumentar una, las otras también aumentarán. Para poder medir la fuerza de esa relación, se requiere realizar un análisis de correlación, el cual se realizará en el siguiente punto.

ACP

Pregunta 2. Realice un análisis descriptivo univariado y bivariado del conjunto de datos. Considere nuevamente las mismas 5 variables de su interés para este punto.

Análisis descriptivo univariado

Para realizar el análisis descriptivo univariado se van a emplear y analizar sus boxplot.

```
# Crear una función para crear boxplot

boxplot_estadisticos <- function(data2, variable, titulo, subtitulo, etiqueta_x) {

  valores <- data2[[variable]]

  estadisticos <- boxplot.stats(valores)$stats

  etiquetas <- data.frame(
    x = as.numeric(estadisticos),
    y = 1.18,
    etiqueta = c(
      paste0("Mín: ", round(estadisticos[1], 1)),
      paste0("Q1: ", round(estadisticos[2], 1)),
      paste0("Q2: ", round(estadisticos[3], 1)),
      paste0("Q3: ", round(estadisticos[4], 1)),
      paste0("Máx: ", round(estadisticos[5], 1))))

  ggplot(data2, aes(x = .data[[variable]], y = 1)) +

    geom_boxplot(
      fill = rojo,
      color = rojo_oscuro,
      width = 0.35) +

    geom_segment(
      data = etiquetas,
      aes(
```

```
        x = x,
        xend = x,
        y = 0.82,
        yend = 1.18
      ),
      color = rojo_oscuro,
      linewidth = 0.5,
      linetype = "dashed") +

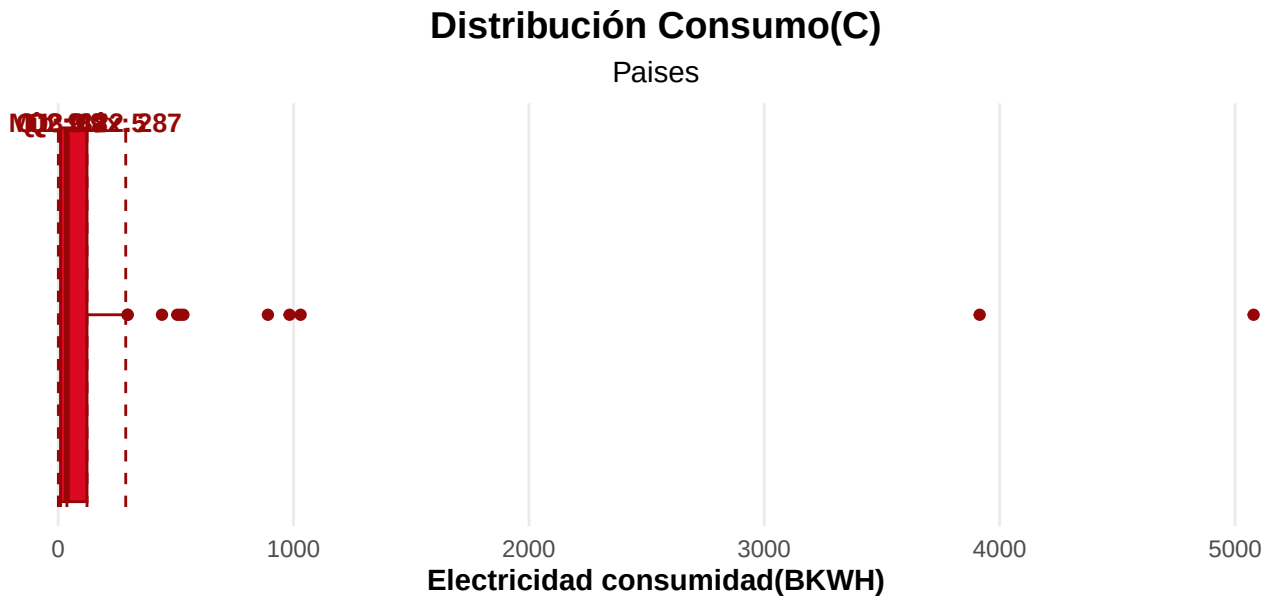
    geom_text(
      data = etiquetas,
      aes(
        x = x,
        y = y,
        label = etiqueta
      ),
      size = 3.5,
      fontface = "bold",
      color = "#950606") +

    labs(
      title = titulo,
      subtitle = subtitulo,
      x = etiqueta_x,
      y = NULL) +

    theme_minimal() +

    theme(
      plot.title = element_text(
        face = "bold",
        size = 14,
        hjust = 0.5),
      plot.subtitle = element_text(
        size = 11,
        hjust = 0.5),
      axis.text.y = element_blank(),
      axis.ticks.y = element_blank(),
      axis.title.y = element_blank(),
      axis.title.x = element_text(face = "bold"),
      panel.grid.minor = element_blank(),
      panel.grid.major.y = element_blank())}
```

```
boxplot_estadisticos(
 data2,
  "Consumo(C)",
  "Distribución Consumo(C)",
  "Paises",
  "Electricidad consumidad(BKWH)")
```




```
caja_Consumo <- boxplot(data2$`Consumo(C)`, plot = FALSE)

caja_Consumo$stats
```

```
##        [,1]
## [1,]   0.80
## [2,]   9.95
## [3,]  37.00
## [4,] 122.50
## [5,] 287.00
```

```
length(caja_Consumo$out)
```

```
## [1] 11
```

```
#install.packages("moments")

# Calcular la asimetría del consumo
asimetria <- moments::skewness(data2$`Consumo(C)`)

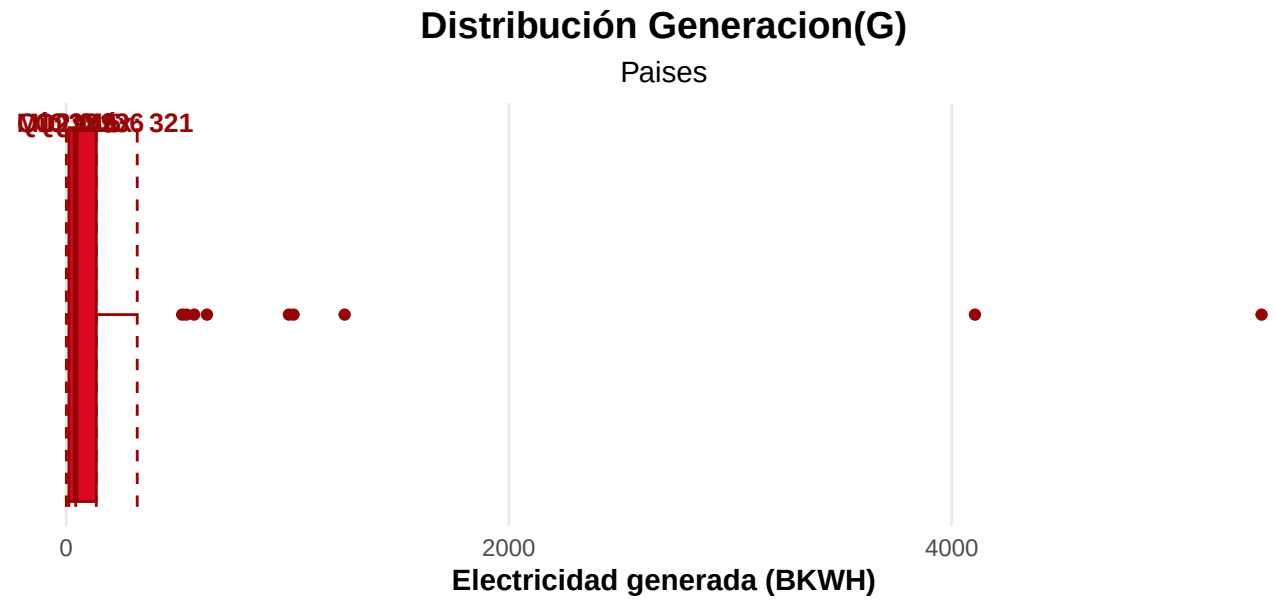
# Mostrar el resultado
print(asimetria)
```

```
## [1] 6.554928
```

En el punto anterior se identificó que la media del consumo neto de electricidad es de 181.81592 BKWH, este valor dista bastante de la mediana (Q2): 37, debido a que la media es sensible a valores atípicos y, en este caso, se observan 11 datos atipicos con valores extremos (superiores a 287, es decir, superiores a 8 veces la mediana) de las 111 observaciones que hay en la data.
Adicional, se calculó el coeficiente de asimetría de la variable y su resultado fue positivo(6.55). Por todo lo anterior, podemos afirmar que los datos siguen una distribución asimétrica positiva o hacia la derecha.

Para finalizar, se puede observar que el 75% de los países tiene un consumo neto de electricidad menor o igual a 122.5 BKWH, con 11 países presentando valores 1.5 veces mayores al rango intercuartilico superior (287). Los mayores consumos de electricidad los presentan China (5079) y Estados Unidos (3915).

```
boxplot_estadisticos(
 data2,
  "Generacion(G)",
  "Distribución Generacion(G)",
  "Paises",
  "Electricidad generada (BKWH)")
```




```
caja_Generacion <- boxplot(data2$`Generacion(G)`, plot = FALSE)

caja_Generacion$stats
```

```
##       [,1]
## [1,]   0.9
## [2,]  12.5
## [3,]  43.0
## [4,] 136.0
## [5,] 321.0
```

```
length(caja_Generacion$out)
```

```
## [1] 9
```

```
asimetria2 <- moments::skewness(data2$`Generacion(G)`)

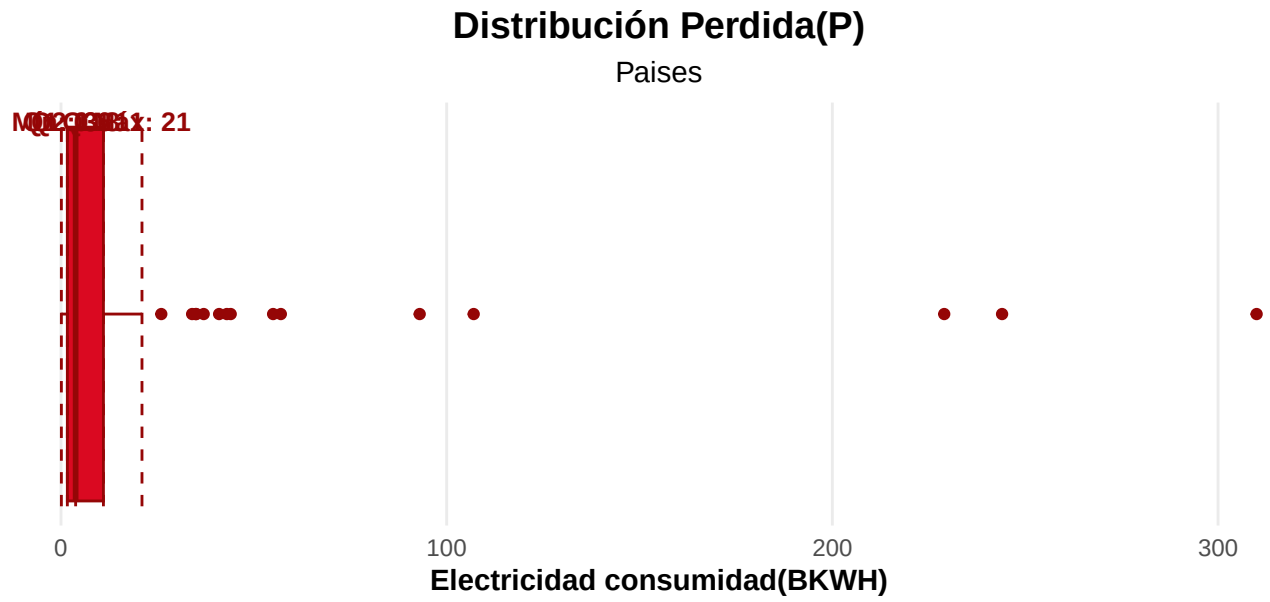
# Mostrar el resultado
print(asimetria2)
```

```
## [1] 6.464784
```

Los datos de la electricidad neta generada son similares a los del consumo, con una media que dista bastante de la mediana (Media: 198.290 y Mediana: 43), con un coeficiente de asimetría de 6.4, se sugiere una distribución con asimetría positiva. Se identifican 9 valores atípicos, los cuales son superiores a 321.

El 75% de los países tiene una generación neta de electricidad menor o igual a 136 BKWH. Los países que más generan electricidad son China (5400) y Estados Unidos (4105), los cuales, son también los mayores consumidores.

```
boxplot_estadisticos(
 data2,
  "Perdida(P)",
  "Distribución Perdida(P)",
  "Paises",
  "Electricidad consumidad(BKWH)")
```




```
caja_Perdida <- boxplot(data2$`Perdida(P)`, plot = FALSE)

caja_Perdida$stats
```

```
##       [,1]
## [1,]  0.10
## [2,]  1.65
## [3,]  3.80
## [4,] 11.00
## [5,] 21.00
```

```
length(caja_Perdida$out)
```

```
## [1] 15
```

```r
asimetria3 <- moments::skewness(data2$`Perdida(P)`)

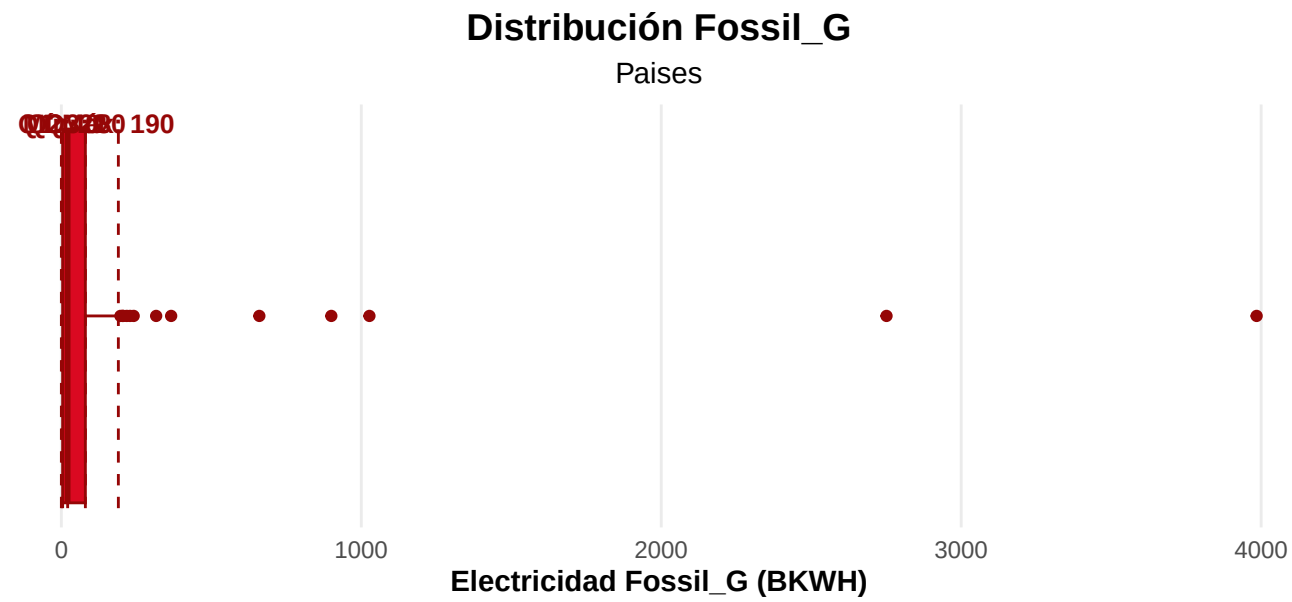
# Mostrar el resultado
print(asimetria3)
```

```
## [1] 4.883258
```

Esta variable no solo es la que presenta la menor media (17.0045) y la menor varianza (1990.936), sino que presenta también la menor mediana (3.80). El límite superior de los datos es 21, es decir, que todos los países -exceptuando los datos atípicos- tienen pérdidas en la distribución de energía iguales o inferiores a 21 BKWH, este valor de 2 cifras es muy inferior a los reportados en las variable de consumo y generación de electricidad, la variable de Pérdida también presenta una distribución con asimetría positiva. Se observan 15 datos atípicos.

En este caso, los países con mayor consumo y generación de electricidad, también son los que presentan mayores pérdidas en la distribución de electricidad: China con 310 BKWH y Estados Unidos con 244 BKWH, siendo estos valores atípicos.

```r
boxplot_estadisticos(
 data2,
  "Fossil_G",
  "Distribución Fossil_G",
  "Paises",
  "Electricidad Fossil_G (BKWH)")
```




```r
caja_Fossil_G <- boxplot(data2$Fossil_G, plot = FALSE)

caja_Fossil_G$stats
```

```
##         [,1]
## [1,]   0.00
```

```
## [2,]   4.25
## [3,]  21.00
## [4,]  80.00
## [5,] 190.00
```

```
length(caja_Fossil_G$out)
```

```
## [1] 13
```

```
asimetria4 <- moments::skewness(data2$`Fossil_G`)

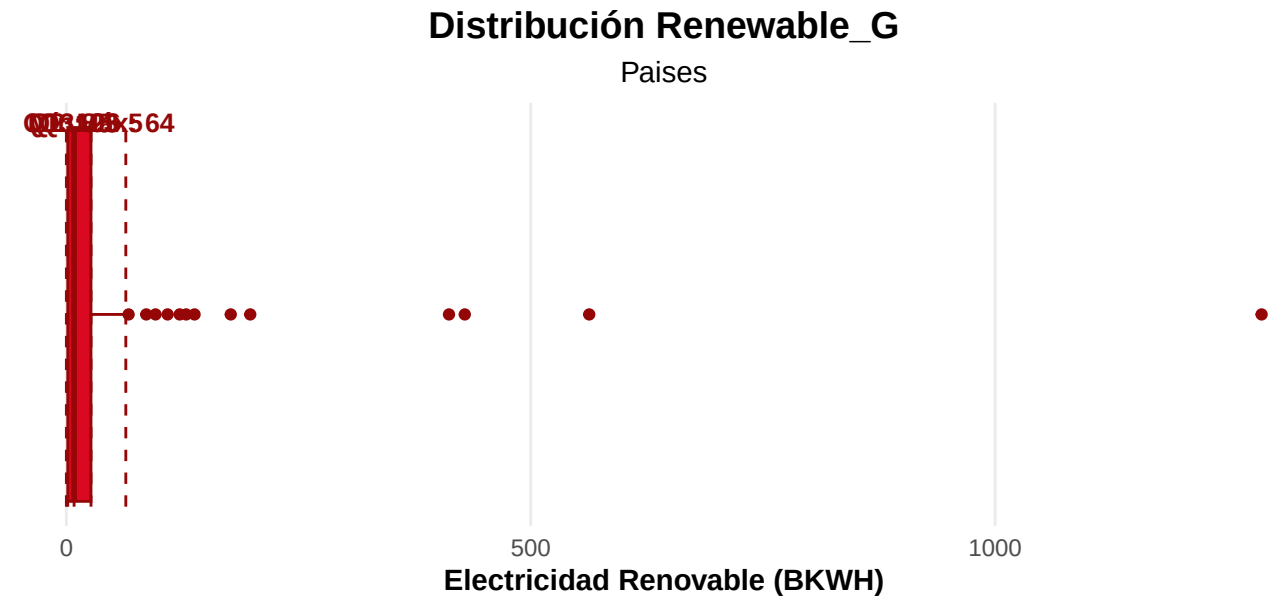
# Mostrar el resultado
print(asimetria4)
```

```
## [1] 6.582706
```

La variable generación de energía con combustible fósil sigue la misma tendencia que las anteriores variables, con una media (132.584) muy superior a la mediana (21), y coeficiente de asimetría mayor a cero, lo que comprueba una distribución asimétrica positiva.

En esta variable, se identifican por primera vez países con 0 generación de electricidad con combustibles fósiles. El 75% o los países presentan una generación de energía con combustible fósil igual o inferior a 80 BKWH. China y Estados Unidos repiten como los países que encabezan la lista, con 3985 BKWH y 2751 BKWH, respectivamente. Si se compara la generación de electricidad con esta variable referente puntualmente a combustibles fósiles, se observa que para china el 74% de la generación es con estos combustibles y en Estados Unidos corresponde al 67%.

```
boxplot_estadisticos(
 data2,
  "Renewable_G",
  "Distribución Renewable_G",
  "Paises",
  "Electricidad Renovable (BKWH)")
```



```r
caja_Renewable_G <- boxplot(data2$Renewable_G, plot = FALSE)

caja_Renewable_G$stats
```

```
##      [,1]
## [1,]  0.0
## [2,]  1.5
## [3,]  8.3
## [4,] 26.5
## [5,] 64.0
```

```r
length(caja_Renewable_G$out)
```

```
## [1] 13
```

```r
asimetria5 <- moments::skewness(data2$`Renewable_G`)

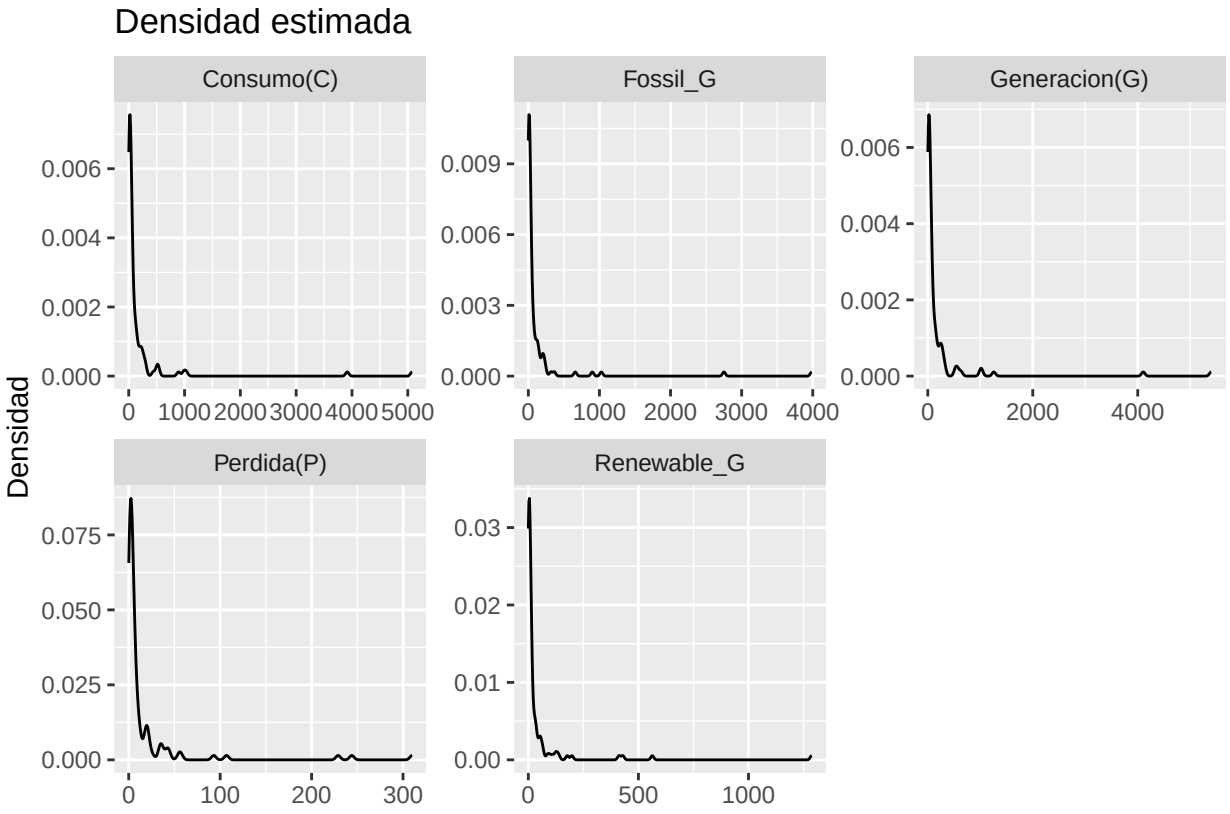
# Mostrar el resultado
print(asimetria5)
```

```
## [1] 6.56162
```

Finalmente, la generación de energía renovable también presenta valores mínimos de 0, una media (44.771) superior a la mediana (8.3), con distribución asimétrica positiva por coefciente mayor a 0, y con la segunda menor varianza (20864.464).

Es necesario mencionar que, si bien en esta variable se mantienen en primer lugar China (1287) y Estados Unidos (563), dentro del top 4 (todos valores atípicos) ingresan Brazil (429) y Canada (412) y salen Japón y Rusia que se mantuvieron en 4to y 5to puesto en todas las otras variables, excepto en pérdidas en la distribución de electricidad, en el que Rusia está de 4to y Japón de 8vo. El 75% de los países presenta una generación de electricidad renovable menor o igual a 26.5, valor 3 veces menor al mismo porcentaje de generación con combustibles fósiles.

```r
gather(data2) %>%
  ggplot(aes(x=value)) +
  geom_density() +
  labs(x= "",y='Densidad',title="Densidad estimada") +
  facet_wrap(~key, scales = "free")
```

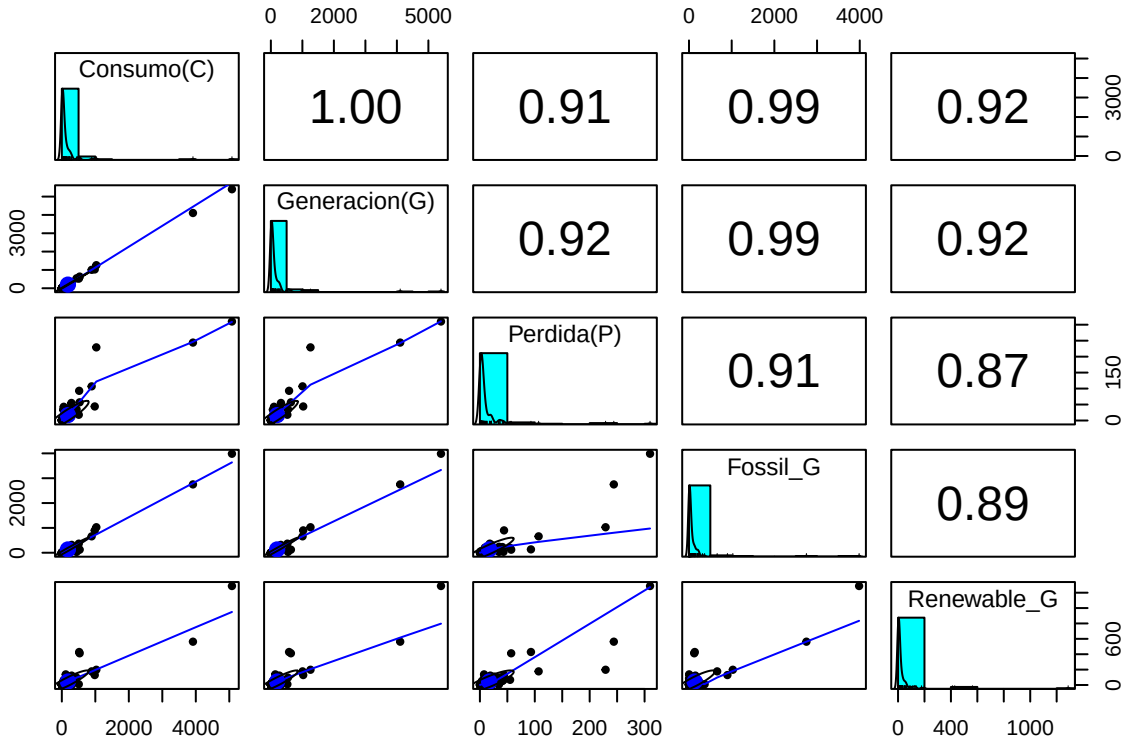


Las gráficas de densidad confirman que las variables siguen una ditribución simétrica positiva, en la que la mayoría de países presenta valores muy bajos y unos pocos países valores altos. También se identifican algunos picos en los datos, empezando todas las variables con valores altos, luego aumentando en mayor o menor medida, y continuando con una caída pronunciada con presencia de algunos picos.

Análisis descriptivo multivariado

Para poder realizar los análisis descriptivos multivariados, se realiza una gráfica que incluye la correlación (de Pearson) y gráficos de dispersión para cada par de variables, y los histogramas de frecuencia de cada variable.

```
pairs.panels(data2,
             scale = FALSE,      # If TRUE, scales the correlation text font
             density = TRUE,     # If TRUE, adds density plots and histograms
             method = "pearson", # Correlation method (also "spearman" or "kendall")
             cor = TRUE,         # If TRUE, reports correlations
             jiggle = FALSE)     # If TRUE, data points are jittered
```

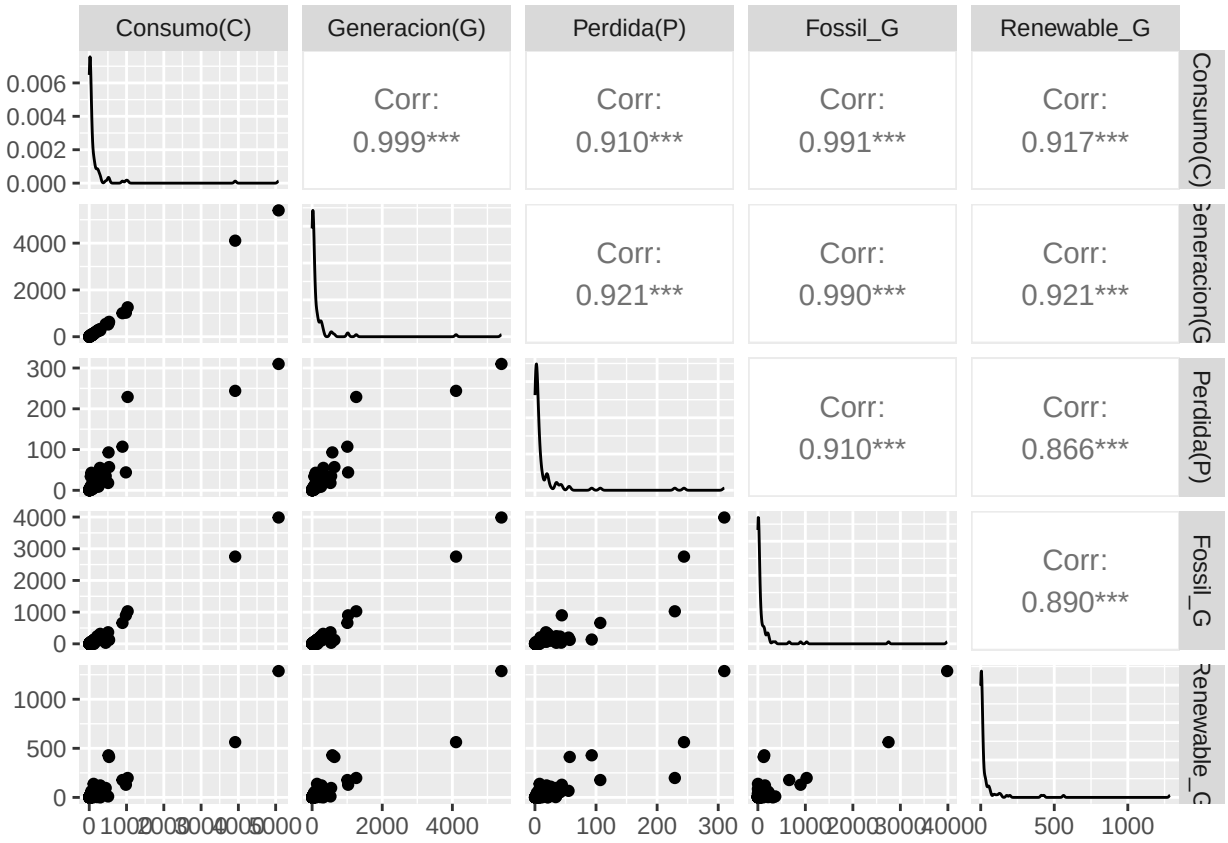


La gráfica permite identificar correlaciones positivas altas para todas las variables. Consumo con Generación, y Consumo y Generación con generación a partir de Combustibles Fósiles tienen correlaciones altísimas (0.99 y 1) y las gráficas de dispersión muestran líneas longitudinales crecientes, es decir que, al aumentar una, casi siempre va a aumentar la otra. Dependiendo del análisis, se debería evaluar quedarse con una sola variable, ya que parecen estar midiendo casi lo mismo y podrían sobrestimar el modelo.

La menor correlación se observa entre las Pérdidas de electricidad con la Generación de electricidad Renovable.

La variables de Pérdidas de distribución de electricidad y generación de electricidad con fuentes Fosiles presentan una correlación alta (0.91), sin embargo, en la gráfica de distribución no parece que presentarán una relación tan alta, esto se debe probablemente a que algunos puntos no sigan una correlación lineal aparente.

```
#complemento usando ggpairs, nos damos cuenta nos brinda la misma información ya analizada
GGally::ggpairs(data2)
```



Pregunta 3. El conjunto de datos incluye 57 variables, analizarlas de forma individual podría representar un gasto computacional y de tiempo bastante agotador; por lo cuál una técnica de reducción de dimensionalidad sería ideal. Ejecute un ACP sobre estos datos, compare la contribución de las variables sobre el primer plano factorial de un ACP normado (escalado) y uno sin normalizar (sin escalar).

Teniendo en cuenta que el flujo completo para realizar un ACP arranca con la exploración de las variables y conocimiento de la data, luego observando las correlaciones entre las variables y el verificar no tener variables cuantitativas(57 variables numericas, correcto), procedemos a detectar outliers con la distancia de Mahalanobis antes de ejecutar ACP, ya que como vimos en el análisis de los puntos anteriores, hay paises con observaciones muy superiores. Se realiza este análisis debido a que un outlier aumenta artificialmente la varianza en su dirección, pudiendo sesgar los ejes principales de la metodología ACP.
Exploración de las 57 variables:

```
dim(data1)
```

```
## [1] 111  57
```

```
# Tipos de variables
table(sapply(data1, class))
```

```
##
## numeric
##      57
```

```r
# Verificamos que no hay valores faltantes por columna
sort(colSums(is.na(data1)), decreasing = TRUE)
```

```
##                                   Bunker_fuel_consumption_TBPD
##                                                              0
##                        Bunker_residual_fuel_oil_consumption_TBPD
##                                                              0
##                 Crude_oil_including_lease_condensate_exports_TBPD
##                                                              0
##                 Crude_oil_including_lease_condensate_imports_TBPD
##                                                              0
##              Crude_oil_including_lease_condensate_production_TBPD
##                                                              0
##                  Crude_oil_including_lease_condensate_reserves_BB
##                                                              0
##                               Distillate_fuel_oil_consumption_TBPD
##                                                              0
##                                Distillate_fuel_oil_production_TBPD
##                                                              0
##                                         Jet_fuel_consumption_TBPD
##                                                              0
##                                          Jet_fuel_production_TBPD
##                                                              0
##                                         Kerosene_consumption_TBPD
##                                                              0
##                                          Kerosene_production_TBPD
##                                                              0
##                  Liquefied_petroleum_gases_.LPG._consumption_TBPD
##                                                              0
##                   Liquefied_petroleum_gases_.LPG._production_TBPD
##                                                              0
##                                       Motor_gasoline_consumption_TBPD
##                                                              0
##                                        Motor_gasoline_production_TBPD
##                                                              0
##                         Natural_gas_plant_liquids_production_TBPD
##                                                              0
##                                       Other_liquids_production_TBPD
##                                                              0
##                           Other_refined_products_consumption_TBPD
##                                                              0
##                            Other_refined_products_production_TBPD
##                                                              0
##                   Petroleum_and_other_liquids_CO2_emissions_MMTCD
##                                                              0
##                      Petroleum_and_other_liquids_consumption_TBPD
##                                                              0
##                       Petroleum_and_other_liquids_production_TBPD
##                                                              0
##                       Refined_petroleum_products_consumption_TBPD
##                                                              0
##                        Refined_petroleum_products_production_TBPD
##                                                              0
```

```
##                                Refinery_processing_gain_TBPD
##                                                            0
##                             Residual_fuel_oil_consumption_TBPD
##                                                            0
##                              Residual_fuel_oil_production_TBPD
##                                                            0
##            Biomass_and_waste_electricity_installed_capacity_MK
##                                                            0
##              Biomass_and_waste_electricity_net_generation_BKWH
##                                                            0
##                           Electricity_distribution_losses_BKWH
##                                                            0
##                                      Electricity_exports_BKWH
##                                                            0
##                                      Electricity_imports_BKWH
##                                                            0
##                               Electricity_installed_capacity_MK
##                                                            0
##                                Electricity_net_consumption_BKWH
##                                                            0
##                                 Electricity_net_generation_BKWH
##                                                            0
##                                    Electricity_net_imports_BKWH
##                                                            0
##                  Fossil_fuels_electricity_installed_capacity_MK
##                                                            0
##                    Fossil_fuels_electricity_net_generation_BKWH
##                                                            0
##                    Geothermal_electricity_installed_capacity_MK
##                                                            0
##                      Geothermal_electricity_net_generation_BKWH
##                                                            0
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK
##                                                            0
##   Hydroelectric_pumped_storage_electricity_net_generation_BKWH
##                                                            0
##                           Hydroelectricity_installed_capacity_MK
##                                                            0
##                             Hydroelectricity_net_generation_BKWH
##                                                            0
##          Non.hydro_renewable_electricity_installed_capacity_MK
##                                                            0
##            Non.hydro_renewable_electricity_net_generation_BKWH
##                                                            0
##                        Nuclear_electricity_installed_capacity_MK
##                                                            0
##                          Nuclear_electricity_net_generation_BKWH
##                                                            0
##                      Renewable_electricity_installed_capacity_MK
##                                                            0
##                        Renewable_electricity_net_generation_BKWH
##                                                            0
##                          Solar_electricity_installed_capacity_MK
##                                                            0
```

```
##                          Solar_electricity_net_generation_BKWH
##                                                              0
##                 Tide_and_wave_electricity_installed_capacity_MK
##                                                              0
##                   Tide_and_wave_electricity_net_generation_BKWH
##                                                              0
##                         Wind_electricity_installed_capacity_MK
##                                                              0
##                           Wind_electricity_net_generation_BKWH
##                                                              0
```

```
# Variables con varianza cero
nombres <- names(data1)[
  sapply(data1, function(x) {
    is.numeric(x) &&
      (all(is.na(x)) || var(x, na.rm = TRUE) == 0)
  })
]

nombres
```

```
## character(0)
```

```
# Columnas duplicadas exactamente
duplicadas <- duplicated(as.list(data1))
names(data1)[duplicadas]
```

```
## [1] "Refined_petroleum_products_consumption_TBPD"
```

```
# Identificar los índices de las columnas duplicadas
duplicadas_idx <- which(duplicated(as.list(data1)))

# Buscar la columna original idéntica
for (j in duplicadas_idx) {
  anteriores <- which(vapply(
    data1[seq_len(j - 1)],
    function(x) identical(x, data1[[j]]),
    logical(1)
  ))

  cat("Columna duplicada:", names(data1)[j], "\n")
  cat("Columna original:", names(data1)[anteriores], "\n\n")
}
```

```
## Columna duplicada: Refined_petroleum_products_consumption_TBPD
## Columna original: Petroleum_and_other_liquids_consumption_TBPD
```

Despues de entender el conjunto de datos, y realizar el procedimiento para evaluar los outliers mediante Mahalanobis, se estaba presentando un error de singularidad, no se estaba pudiendo generar la matriz de covarianzas y el problema radicaba en que 2 columnas estaban duplicadas, eran identicas y redundantes, lo que imposibilita la generación de la matriz. Luego, procedimos a eliminar la columna duplicada. Pero, al continuar con el procedimiento mencionado de Mahalanobis este arrojaba sugerencia de 42 paises outliera,

como representa mas del 30% de la muestra, y la guí del taller pide eliminar paises solo en el punto 7. Se decidió continuar con la data sin la columna duplicada y aplicar ACP con y sin escalar, como se muestra en los siguientes códigos.

```
# Crear una copia sin la variable duplicada
data_sin_dup <- subset(
  data1,
  select = -Refined_petroleum_products_consumption_TBPD
)

# Verificar las dimensiones
dim(data_sin_dup)
```

```
## [1] 111  56
```

```
# Comprobar valores faltantes
cat("\nColumnas con valores NA:\n")
```

```
##
## Columnas con valores NA:
```

```
print(sum(colSums(is.na(data_sin_dup)) > 0))
```

```
## [1] 0
```

```
cat("\nTotal de valores NA:\n")
```

```
##
## Total de valores NA:
```

```
print(sum(is.na(data_sin_dup)))
```

```
## [1] 0
```

```
# Asegurar que todas las columnas sean numéricas
cat("\nTipos de variables:\n")
```

```
##
## Tipos de variables:
```

```
print(table(sapply(data_sin_dup, class)))
```

```
##
## numeric
##      56
```

```
# Calcular el rango de los datos centrados, soLO CENTRADOS, sin escalar
X <- scale(data_sin_dup, center = TRUE, scale = FALSE)

cat("\nRango de X:\n")
```

```
##
## Rango de X:
```

```
print(qr(X, tol = 1e-10)$rank)
```

```
## [1] 56
```

```
# Calcular el rango de la matriz de covarianzas
cat("\nRango de la matriz de covarianzas:\n")
```

```
##
## Rango de la matriz de covarianzas:
```

```
print(qr(cov(data_sin_dup), tol = 1e-10)$rank)
```

```
## [1] 56
```

Ahora que verificamos independencia procedemos a desarrollar ACP sobre los datos sin duplicados centrando con la funcion de prcomp.

```
# Descomposición QR
qr_X <- qr(X, tol = 1e-10)

cat("Rango de los datos centrados:", qr_X$rank, "\n")
```

```
## Rango de los datos centrados: 56
```

```
cat("Número de columnas:", ncol(X), "\n")
```

```
## Número de columnas: 56
```

```
# Identificar columnas dependientes solo si existen
if (qr_X$rank < ncol(X)) {
  dependientes <- qr_X$pivot[
    (qr_X$rank + 1):ncol(X)
  ]

  print(names(data_sin_dup)[dependientes])
} else {
  cat("No se detectaron columnas linealmente dependientes.\n")
}
```

```
## No se detectaron columnas linealmente dependientes.
```

```
# 1. ACP sin escalar
acp_sin_escalar <- prcomp(
  data_sin_dup,
  center = TRUE,
  scale. = FALSE
```

```
)

# 2.ACP normado
acp_escalado <- prcomp(
  data_sin_dup,
  center = TRUE,
  scale. = TRUE
)

# Varianza explicada por las dos primeras componentes
summary(acp_sin_escalar)$importance[, 1:2]
```

```
##                             PC1        PC2
## Standard deviation     4383.38377 1977.51968
## Proportion of Variance    0.80428    0.16369
## Cumulative Proportion     0.80428    0.96798
```

```
summary(acp_escalado)$importance[, 1:2]
```

```
##                            PC1      PC2
## Standard deviation     5.860516 2.095872
## Proportion of Variance 0.613320 0.078440
## Cumulative Proportion  0.613320 0.691760
```

En el ACP sin escalar, la primera componente principal explica el 80,43 % de la varianza total, mientras que la segunda explica el 16,37 %. En conjunto, ambas componentes representan el 96,80 % de la variabilidad de los datos. Este resultado indica que el primer plano factorial resume una proporción muy elevada de la varianza original. Sin embargo, al conservarse las escalas originales, y como se observa a partir de las magnitudes de la desviación estandar de las componentes PC1 Y PC2, las variables con mayor varianza absoluta ejercen una influencia predominante en la construcción de los ejes.

Por otra parte, en el ACP normado, la primera componente explica el 61,33 % de la varianza y la segunda el 7,84 %, acumulando un 69,18 % en el plano factorial. La reducción de la proporción de varianza explicada por las dos primeras componentes indica que, al estandarizar las variables, la variabilidad se distribuye de manera diferente entre los ejes principales.

En el ACP normado, cada variable se centra y se divide por su desviación estándar. Así, las variables quedan expresadas en una escala comparable y ninguna domina únicamente por tener valores numéricos más grandes.

La comparación evidencia que el ACP sin escalar concentra una mayor proporción de la varianza en las dos primeras componentes, mientras que el ACP normado permite analizar las relaciones entre las variables sin que sus diferencias de escala determinen directamente su influencia. Por tanto, la selección de cuál usar debe responder al objetivo de estudio y considerar las contribuciones de las variables a los ejes del plano factorial.

Ahora se busca como contribuyen las variables sobre el primer plano factorial.

```
# Contribuciones porcentuales por variable y componente

contrib_sin <- acp_sin_escalar$rotation[, 1:2]^2 * 100
contrib_norm <- acp_escalado$rotation[, 1:2]^2 * 100

# Contribución conjunta al primer plano factorial
# Promedio de las contribuciones a PC1 y PC2
```

```r
plano_sin <- rowMeans(contrib_sin)
plano_norm <- rowMeans(contrib_norm)

# Tabla comparativa
comparacion <- data.frame(
  Variable = rownames(acp_sin_escalar$rotation),
  Sin_escalar_PC1 = contrib_sin[, 1],
  Sin_escalar_PC2 = contrib_sin[, 2],
  Sin_escalar_plano = plano_sin,
  Normado_PC1 = contrib_norm[, 1],
  Normado_PC2 = contrib_norm[, 2],
  Normado_plano = plano_norm
)

# Diez variables que más contribuyen al plano en cada ACP
head(comparacion[
  order(comparacion$Sin_escalar_plano, decreasing = TRUE),
], 10)
```

```
##                                                
```

```
## Petroleum_and_other_liquids_production_TBPD                  22.443472
## Crude_oil_including_lease_condensate_production_TBPD         18.324129
## Refined_petroleum_products_production_TBPD                   15.737790
## Petroleum_and_other_liquids_consumption_TBPD                 15.344394
## Crude_oil_including_lease_condensate_exports_TBPD             9.398207
## Crude_oil_including_lease_condensate_imports_TBPD             5.597275
## Motor_gasoline_production_TBPD                                2.700484
## Motor_gasoline_consumption_TBPD                               2.287159
## Electricity_net_generation_BKWH                               1.506508
## Electricity_net_consumption_BKWH                              1.353373
##                                                          Normado_PC1 Normado_PC2
## Petroleum_and_other_liquids_production_TBPD            1.38840749   7.9690232
## Crude_oil_including_lease_condensate_production_TBPD   1.04972050   7.6612820
## Refined_petroleum_products_production_TBPD             2.78105137   0.5849247
## Petroleum_and_other_liquids_consumption_TBPD           2.82221800   0.4044817
## Crude_oil_including_lease_condensate_exports_TBPD      0.04579476   8.0624336
## Crude_oil_including_lease_condensate_imports_TBPD      2.49815183   0.7696938
## Motor_gasoline_production_TBPD                         2.34542297   1.8605465
## Motor_gasoline_consumption_TBPD                        2.36315332   1.9132555
## Electricity_net_generation_BKWH                        2.67873675   0.9158719
## Electricity_net_consumption_BKWH                       2.68272281   0.8909640
##                                                          Normado_plano
## Petroleum_and_other_liquids_production_TBPD                   4.678715
## Crude_oil_including_lease_condensate_production_TBPD          4.355501
## Refined_petroleum_products_production_TBPD                    1.682988
## Petroleum_and_other_liquids_consumption_TBPD                  1.613350
## Crude_oil_including_lease_condensate_exports_TBPD             4.054114
## Crude_oil_including_lease_condensate_imports_TBPD             1.633923
## Motor_gasoline_production_TBPD                                2.102985
## Motor_gasoline_consumption_TBPD                               2.138204
## Electricity_net_generation_BKWH                               1.797304
## Electricity_net_consumption_BKWH                              1.786843
```

```
head(comparacion[
  order(comparacion$Normado_plano, decreasing = TRUE),
], 10)
```

```
##                                                
```

```
## Hydroelectricity_installed_capacity_MK                  2.037943e-03
## Hydroelectricity_net_generation_BKWH                    2.626548e-02
## Solar_electricity_installed_capacity_MK                 3.424570e-05
## Crude_oil_including_lease_condensate_reserves_BB        1.411703e-03
## Renewable_electricity_net_generation_BKWH               5.640664e-02
##                                                      Sin_escalar_PC2
## Natural_gas_plant_liquids_production_TBPD               1.217529e-01
## Petroleum_and_other_liquids_production_TBPD             2.606181e+01
## Crude_oil_including_lease_condensate_production_TBPD    2.629921e+01
## Crude_oil_including_lease_condensate_exports_TBPD       1.809345e+01
## Renewable_electricity_installed_capacity_MK             5.906511e-03
## Hydroelectricity_installed_capacity_MK                  1.446987e-03
## Hydroelectricity_net_generation_BKWH                    1.659250e-02
## Solar_electricity_installed_capacity_MK                 9.770576e-05
## Crude_oil_including_lease_condensate_reserves_BB        2.405556e-02
## Renewable_electricity_net_generation_BKWH               4.663922e-02
##                                                     Sin_escalar_plano
## Natural_gas_plant_liquids_production_TBPD               2.667170e-01
## Petroleum_and_other_liquids_production_TBPD             2.244347e+01
## Crude_oil_including_lease_condensate_production_TBPD    1.832413e+01
## Crude_oil_including_lease_condensate_exports_TBPD       9.398207e+00
## Renewable_electricity_installed_capacity_MK             5.748031e-03
## Hydroelectricity_installed_capacity_MK                  1.742465e-03
## Hydroelectricity_net_generation_BKWH                    2.142899e-02
## Solar_electricity_installed_capacity_MK                 6.597573e-05
## Crude_oil_including_lease_condensate_reserves_BB        1.273363e-02
## Renewable_electricity_net_generation_BKWH               5.152293e-02
##                                                     Normado_PC1 Normado_PC2
## Natural_gas_plant_liquids_production_TBPD            1.40775717    9.556008
## Petroleum_and_other_liquids_production_TBPD          1.38840749    7.969023
## Crude_oil_including_lease_condensate_production_TBPD 1.04972050    7.661282
## Crude_oil_including_lease_condensate_exports_TBPD    0.04579476    8.062434
## Renewable_electricity_installed_capacity_MK          2.17249553    3.410624
## Hydroelectricity_installed_capacity_MK               1.76606576    3.700433
## Hydroelectricity_net_generation_BKWH                 1.61352001    3.802997
## Solar_electricity_installed_capacity_MK              1.74052094    3.648149
## Crude_oil_including_lease_condensate_reserves_BB     0.06849591    4.981325
## Renewable_electricity_net_generation_BKWH            2.08169421    2.927302
##                                                     Normado_plano
## Natural_gas_plant_liquids_production_TBPD               5.481883
## Petroleum_and_other_liquids_production_TBPD             4.678715
## Crude_oil_including_lease_condensate_production_TBPD    4.355501
## Crude_oil_including_lease_condensate_exports_TBPD       4.054114
## Renewable_electricity_installed_capacity_MK             2.791560
## Hydroelectricity_installed_capacity_MK                  2.733250
## Hydroelectricity_net_generation_BKWH                    2.708258
## Solar_electricity_installed_capacity_MK                 2.694335
## Crude_oil_including_lease_condensate_reserves_BB        2.524910
## Renewable_electricity_net_generation_BKWH               2.504498
```

Al observar los resultados del primer plano factorial sin escalar, observamos variables importantes de producción, consumo, comercio de petróleo y generación de electricidad que tienen una influencia significativa al conservar su escala. Mientras que en el ACP escalado toman mas relevancia indicadores de capacidad instalada, producción petrolera y generación de energía.

Pregunta 4. Una vez calculado el PCA normado, ¿qué número de componentes principales deberíamos seleccionar? ¿Bajo que criterio seleccionó este número de componentes?

Usando el gráfico de sedimentación o screeplot y el criterio "del codo" en el PCA normado podemos observar que el mejor k es de 2, estas 2 componentes logran resumir las 56 variables sin pérdida valiosa de información, vemos adicionalemente que un k de 3 tiene un menor aporte porque la curva comienza a estabilizarse ahí.

```
round(acp_escalado$rotation, 3)
```

```
##                                                              PC1    PC2
## Bunker_fuel_consumption_TBPD                               0.102  0.156
## Bunker_residual_fuel_oil_consumption_TBPD                  0.051  0.083
## Crude_oil_including_lease_condensate_exports_TBPD          0.021  0.284
## Crude_oil_including_lease_condensate_imports_TBPD          0.158 -0.088
## Crude_oil_including_lease_condensate_production_TBPD       0.102  0.277
## Crude_oil_including_lease_condensate_reserves_BB           0.026  0.223
## Distillate_fuel_oil_consumption_TBPD                       0.168 -0.020
## Distillate_fuel_oil_production_TBPD                        0.168  0.023
## Jet_fuel_consumption_TBPD                                  0.158  0.117
## Jet_fuel_production_TBPD                                   0.159  0.102
## Kerosene_consumption_TBPD                                  0.044 -0.093
## Kerosene_production_TBPD                                   0.045 -0.024
## Liquefied_petroleum_gases_.LPG._consumption_TBPD           0.162  0.042
## Liquefied_petroleum_gases_.LPG._production_TBPD            0.155 -0.019
## Motor_gasoline_consumption_TBPD                            0.154  0.138
## Motor_gasoline_production_TBPD                             0.153  0.136
## Natural_gas_plant_liquids_production_TBPD                  0.119  0.309
## Other_liquids_production_TBPD                              0.149  0.083
## Other_refined_products_consumption_TBPD                    0.163  0.000
## Other_refined_products_production_TBPD                     0.163 -0.036
## Petroleum_and_other_liquids_CO2_emissions_MMTCD            0.168  0.061
## Petroleum_and_other_liquids_consumption_TBPD               0.168  0.064
## Petroleum_and_other_liquids_production_TBPD                0.118  0.282
## Refined_petroleum_products_production_TBPD                 0.167  0.076
## Refinery_processing_gain_TBPD                              0.149  0.126
## Residual_fuel_oil_consumption_TBPD                         0.091  0.047
## Residual_fuel_oil_production_TBPD                          0.084  0.173
## Biomass_and_waste_electricity_installed_capacity_MK        0.139  0.019
## Biomass_and_waste_electricity_net_generation_BKWH          0.154 -0.057
## Electricity_distribution_losses_BKWH                       0.153 -0.068
## Electricity_exports_BKWH                                   0.041 -0.081
## Electricity_imports_BKWH                                   0.093  0.061
## Electricity_installed_capacity_MK                          0.164 -0.094
## Electricity_net_consumption_BKWH                           0.164 -0.094
## Electricity_net_generation_BKWH                            0.164 -0.096
## Electricity_net_imports_BKWH                               0.041  0.123
## Fossil_fuels_electricity_installed_capacity_MK             0.164 -0.072
## Fossil_fuels_electricity_net_generation_BKWH               0.159 -0.104
## Geothermal_electricity_installed_capacity_MK               0.086  0.154
## Geothermal_electricity_net_generation_BKWH                 0.087  0.160
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK  0.135 -0.130
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.154  0.047
## Hydroelectricity_installed_capacity_MK                     0.133 -0.192
## Hydroelectricity_net_generation_BKWH                       0.127 -0.195
```

```
## Non.hydro_renewable_electricity_installed_capacity_MK           0.158 -0.150
## Non.hydro_renewable_electricity_net_generation_BKWH             0.164 -0.065
## Nuclear_electricity_installed_capacity_MK                       0.129  0.108
## Nuclear_electricity_net_generation_BKWH                         0.133  0.123
## Renewable_electricity_installed_capacity_MK                     0.147 -0.185
## Renewable_electricity_net_generation_BKWH                       0.144 -0.171
## Solar_electricity_installed_capacity_MK                         0.132 -0.191
## Solar_electricity_net_generation_BKWH                           0.140 -0.136
## Tide_and_wave_electricity_installed_capacity_MK                 0.019 -0.082
## Tide_and_wave_electricity_net_generation_BKWH                   0.018 -0.086
## Wind_electricity_installed_capacity_MK                          0.153 -0.157
## Wind_electricity_net_generation_BKWH                            0.161 -0.065
##                                                                   PC3    PC4
## Bunker_fuel_consumption_TBPD                                   -0.057  0.097
## Bunker_residual_fuel_oil_consumption_TBPD                      -0.094  0.168
## Crude_oil_including_lease_condensate_exports_TBPD              -0.338  0.065
## Crude_oil_including_lease_condensate_imports_TBPD               0.047  0.073
## Crude_oil_including_lease_condensate_production_TBPD           -0.241 -0.006
## Crude_oil_including_lease_condensate_reserves_BB               -0.300  0.059
## Distillate_fuel_oil_consumption_TBPD                           -0.013 -0.002
## Distillate_fuel_oil_production_TBPD                            -0.016  0.009
## Jet_fuel_consumption_TBPD                                       0.116  0.011
## Jet_fuel_production_TBPD                                        0.081  0.029
## Kerosene_consumption_TBPD                                      -0.094  0.205
## Kerosene_production_TBPD                                       -0.153  0.224
## Liquefied_petroleum_gases_.LPG._consumption_TBPD               -0.063  0.003
## Liquefied_petroleum_gases_.LPG._production_TBPD                -0.114 -0.050
## Motor_gasoline_consumption_TBPD                                 0.126 -0.061
## Motor_gasoline_production_TBPD                                  0.138 -0.039
## Natural_gas_plant_liquids_production_TBPD                       0.033 -0.010
## Other_liquids_production_TBPD                                   0.129 -0.108
## Other_refined_products_consumption_TBPD                        -0.095  0.033
## Other_refined_products_production_TBPD                         -0.080 -0.004
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 0.014 -0.003
## Petroleum_and_other_liquids_consumption_TBPD                    0.016 -0.005
## Petroleum_and_other_liquids_production_TBPD                    -0.172 -0.016
## Refined_petroleum_products_production_TBPD                      0.016 -0.004
## Refinery_processing_gain_TBPD                                   0.176 -0.040
## Residual_fuel_oil_consumption_TBPD                             -0.239  0.159
## Residual_fuel_oil_production_TBPD                              -0.262  0.091
## Biomass_and_waste_electricity_installed_capacity_MK             0.120 -0.041
## Biomass_and_waste_electricity_net_generation_BKWH               0.078 -0.044
## Electricity_distribution_losses_BKWH                           -0.100 -0.065
## Electricity_exports_BKWH                                       -0.017  0.266
## Electricity_imports_BKWH                                        0.205  0.040
## Electricity_installed_capacity_MK                              -0.047 -0.059
## Electricity_net_consumption_BKWH                               -0.050 -0.060
## Electricity_net_generation_BKWH                                -0.057 -0.057
## Electricity_net_imports_BKWH                                    0.186 -0.205
## Fossil_fuels_electricity_installed_capacity_MK                 -0.043 -0.074
## Fossil_fuels_electricity_net_generation_BKWH                   -0.073 -0.075
## Geothermal_electricity_installed_capacity_MK                    0.232 -0.018
## Geothermal_electricity_net_generation_BKWH                      0.231 -0.023
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK  0.028  0.148
```

```
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.082 -0.093
## Hydroelectricity_installed_capacity_MK                         -0.166 -0.123
## Hydroelectricity_net_generation_BKWH                           -0.175 -0.121
## Non.hydro_renewable_electricity_installed_capacity_MK           0.013 -0.025
## Non.hydro_renewable_electricity_net_generation_BKWH             0.077 -0.041
## Nuclear_electricity_installed_capacity_MK                       0.144  0.178
## Nuclear_electricity_net_generation_BKWH                         0.159  0.150
## Renewable_electricity_installed_capacity_MK                    -0.106 -0.091
## Renewable_electricity_net_generation_BKWH                      -0.117 -0.106
## Solar_electricity_installed_capacity_MK                         0.021  0.141
## Solar_electricity_net_generation_BKWH                           0.058  0.134
## Tide_and_wave_electricity_installed_capacity_MK                 0.103  0.469
## Tide_and_wave_electricity_net_generation_BKWH                   0.095  0.499
## Wind_electricity_installed_capacity_MK                         -0.018 -0.081
## Wind_electricity_net_generation_BKWH                            0.057 -0.078
##                                                                   PC5    PC6
## Bunker_fuel_consumption_TBPD                                   -0.121  0.475
## Bunker_residual_fuel_oil_consumption_TBPD                      -0.175  0.585
## Crude_oil_including_lease_condensate_exports_TBPD               0.073 -0.135
## Crude_oil_including_lease_condensate_imports_TBPD              -0.109  0.003
## Crude_oil_including_lease_condensate_production_TBPD            0.091 -0.093
## Crude_oil_including_lease_condensate_reserves_BB                0.075 -0.153
## Distillate_fuel_oil_consumption_TBPD                            0.017 -0.039
## Distillate_fuel_oil_production_TBPD                            -0.012 -0.015
## Jet_fuel_consumption_TBPD                                      -0.012  0.029
## Jet_fuel_production_TBPD                                       -0.036  0.064
## Kerosene_consumption_TBPD                                      -0.414 -0.283
## Kerosene_production_TBPD                                       -0.386 -0.275
## Liquefied_petroleum_gases_.LPG._consumption_TBPD               -0.062 -0.075
## Liquefied_petroleum_gases_.LPG._production_TBPD                 0.023  0.021
## Motor_gasoline_consumption_TBPD                                 0.009 -0.033
## Motor_gasoline_production_TBPD                                  0.002 -0.018
## Natural_gas_plant_liquids_production_TBPD                       0.057 -0.077
## Other_liquids_production_TBPD                                   0.056  0.047
## Other_refined_products_consumption_TBPD                        -0.019 -0.014
## Other_refined_products_production_TBPD                         -0.001  0.050
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                -0.029 -0.015
## Petroleum_and_other_liquids_consumption_TBPD                   -0.021 -0.017
## Petroleum_and_other_liquids_production_TBPD                     0.086 -0.084
## Refined_petroleum_products_production_TBPD                     -0.011 -0.006
## Refinery_processing_gain_TBPD                                  -0.008 -0.014
## Residual_fuel_oil_consumption_TBPD                             -0.223  0.326
## Residual_fuel_oil_production_TBPD                               0.008 -0.054
## Biomass_and_waste_electricity_installed_capacity_MK             0.020  0.017
## Biomass_and_waste_electricity_net_generation_BKWH              -0.010  0.010
## Electricity_distribution_losses_BKWH                            0.004 -0.039
## Electricity_exports_BKWH                                        0.402 -0.054
## Electricity_imports_BKWH                                        0.047  0.036
## Electricity_installed_capacity_MK                               0.022  0.005
## Electricity_net_consumption_BKWH                                0.021  0.007
## Electricity_net_generation_BKWH                                 0.026  0.002
## Electricity_net_imports_BKWH                                   -0.324  0.080
## Fossil_fuels_electricity_installed_capacity_MK                 -0.015  0.003
## Fossil_fuels_electricity_net_generation_BKWH                   -0.032 -0.003
```

```
## Geothermal_electricity_installed_capacity_MK                    -0.114 -0.127
## Geothermal_electricity_net_generation_BKWH                      -0.098 -0.119
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.173 -0.139
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.061  0.002
## Hydroelectricity_installed_capacity_MK                          0.115  0.041
## Hydroelectricity_net_generation_BKWH                            0.134  0.035
## Non.hydro_renewable_electricity_installed_capacity_MK           0.007 -0.001
## Non.hydro_renewable_electricity_net_generation_BKWH             0.002 -0.006
## Nuclear_electricity_installed_capacity_MK                       0.215 -0.010
## Nuclear_electricity_net_generation_BKWH                         0.193 -0.010
## Renewable_electricity_installed_capacity_MK                     0.080  0.027
## Renewable_electricity_net_generation_BKWH                       0.106  0.026
## Solar_electricity_installed_capacity_MK                        -0.114 -0.078
## Solar_electricity_net_generation_BKWH                          -0.134 -0.089
## Tide_and_wave_electricity_installed_capacity_MK                 0.085  0.037
## Tide_and_wave_electricity_net_generation_BKWH                   0.168  0.031
## Wind_electricity_installed_capacity_MK                          0.052  0.029
## Wind_electricity_net_generation_BKWH                            0.046  0.015
##                                                                   PC7    PC8
## Bunker_fuel_consumption_TBPD                                   -0.107  0.037
## Bunker_residual_fuel_oil_consumption_TBPD                      -0.112  0.065
## Crude_oil_including_lease_condensate_exports_TBPD               0.142 -0.012
## Crude_oil_including_lease_condensate_imports_TBPD              -0.075  0.011
## Crude_oil_including_lease_condensate_production_TBPD           0.101 -0.037
## Crude_oil_including_lease_condensate_reserves_BB               0.108  0.060
## Distillate_fuel_oil_consumption_TBPD                           0.010 -0.001
## Distillate_fuel_oil_production_TBPD                            0.009 -0.008
## Jet_fuel_consumption_TBPD                                     -0.083  0.048
## Jet_fuel_production_TBPD                                      -0.070 -0.036
## Kerosene_consumption_TBPD                                     -0.127  0.157
## Kerosene_production_TBPD                                      -0.101  0.219
## Liquefied_petroleum_gases_.LPG._consumption_TBPD              -0.006 -0.072
## Liquefied_petroleum_gases_.LPG._production_TBPD                0.034 -0.068
## Motor_gasoline_consumption_TBPD                               -0.076  0.023
## Motor_gasoline_production_TBPD                                -0.090  0.034
## Natural_gas_plant_liquids_production_TBPD                     -0.058  0.064
## Other_liquids_production_TBPD                                  0.057  0.079
## Other_refined_products_consumption_TBPD                        0.021 -0.069
## Other_refined_products_production_TBPD                         0.066 -0.065
## Petroleum_and_other_liquids_CO2_emissions_MMTCD               -0.046  0.001
## Petroleum_and_other_liquids_consumption_TBPD                  -0.034 -0.007
## Petroleum_and_other_liquids_production_TBPD                    0.071 -0.013
## Refined_petroleum_products_production_TBPD                    -0.018 -0.008
## Refinery_processing_gain_TBPD                                 -0.100  0.050
## Residual_fuel_oil_consumption_TBPD                            -0.040 -0.009
## Residual_fuel_oil_production_TBPD                              0.138 -0.072
## Biomass_and_waste_electricity_installed_capacity_MK            0.234  0.246
## Biomass_and_waste_electricity_net_generation_BKWH              0.191  0.192
## Electricity_distribution_losses_BKWH                           0.020  0.000
## Electricity_exports_BKWH                                      -0.185  0.356
## Electricity_imports_BKWH                                       0.383  0.423
## Electricity_installed_capacity_MK                             -0.032 -0.075
## Electricity_net_consumption_BKWH                              -0.042 -0.086
## Electricity_net_generation_BKWH                               -0.047 -0.082
```

```
## Electricity_net_imports_BKWH                                    0.492  0.035
## Fossil_fuels_electricity_installed_capacity_MK                 -0.048 -0.102
## Fossil_fuels_electricity_net_generation_BKWH                   -0.054 -0.121
## Geothermal_electricity_installed_capacity_MK                   -0.067 -0.356
## Geothermal_electricity_net_generation_BKWH                     -0.048 -0.367
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.108  0.091
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.082 -0.008
## Hydroelectricity_installed_capacity_MK                          0.067 -0.060
## Hydroelectricity_net_generation_BKWH                            0.074 -0.027
## Non.hydro_renewable_electricity_installed_capacity_MK           0.038  0.009
## Non.hydro_renewable_electricity_net_generation_BKWH             0.027  0.054
## Nuclear_electricity_installed_capacity_MK                      -0.170  0.047
## Nuclear_electricity_net_generation_BKWH                        -0.166  0.049
## Renewable_electricity_installed_capacity_MK                     0.059 -0.037
## Renewable_electricity_net_generation_BKWH                       0.066 -0.006
## Solar_electricity_installed_capacity_MK                         0.107 -0.003
## Solar_electricity_net_generation_BKWH                           0.062  0.031
## Tide_and_wave_electricity_installed_capacity_MK                 0.364 -0.281
## Tide_and_wave_electricity_net_generation_BKWH                   0.216 -0.246
## Wind_electricity_installed_capacity_MK                         -0.024 -0.020
## Wind_electricity_net_generation_BKWH                           -0.053  0.033
##                                                                   PC9   PC10
## Bunker_fuel_consumption_TBPD                                   -0.042  0.074
## Bunker_residual_fuel_oil_consumption_TBPD                      -0.166 -0.005
## Crude_oil_including_lease_condensate_exports_TBPD              -0.104 -0.139
## Crude_oil_including_lease_condensate_imports_TBPD               0.092 -0.040
## Crude_oil_including_lease_condensate_production_TBPD            0.000 -0.007
## Crude_oil_including_lease_condensate_reserves_BB               -0.239 -0.370
## Distillate_fuel_oil_consumption_TBPD                           -0.007 -0.001
## Distillate_fuel_oil_production_TBPD                             0.093  0.062
## Jet_fuel_consumption_TBPD                                       0.040 -0.030
## Jet_fuel_production_TBPD                                        0.139 -0.036
## Kerosene_consumption_TBPD                                       0.081  0.200
## Kerosene_production_TBPD                                       -0.030 -0.043
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                0.087  0.065
## Liquefied_petroleum_gases_.LPG._production_TBPD                 0.181  0.265
## Motor_gasoline_consumption_TBPD                                 0.030 -0.073
## Motor_gasoline_production_TBPD                                  0.065 -0.086
## Natural_gas_plant_liquids_production_TBPD                       0.003 -0.167
## Other_liquids_production_TBPD                                   0.016 -0.064
## Other_refined_products_consumption_TBPD                         0.066 -0.039
## Other_refined_products_production_TBPD                          0.125  0.118
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 0.011 -0.027
## Petroleum_and_other_liquids_consumption_TBPD                    0.026 -0.033
## Petroleum_and_other_liquids_production_TBPD                     0.005 -0.041
## Refined_petroleum_products_production_TBPD                      0.106  0.050
## Refinery_processing_gain_TBPD                                   0.073 -0.118
## Residual_fuel_oil_consumption_TBPD                             -0.172 -0.038
## Residual_fuel_oil_production_TBPD                               0.159  0.488
## Biomass_and_waste_electricity_installed_capacity_MK            -0.036  0.195
## Biomass_and_waste_electricity_net_generation_BKWH              -0.031  0.082
## Electricity_distribution_losses_BKWH                            0.089  0.165
## Electricity_exports_BKWH                                       -0.301  0.203
## Electricity_imports_BKWH                                       -0.191  0.138
```

```
## Electricity_installed_capacity_MK                                  0.018 -0.037
## Electricity_net_consumption_BKWH                                   0.036 -0.041
## Electricity_net_generation_BKWH                                    0.037 -0.026
## Electricity_net_imports_BKWH                                       0.112 -0.066
## Fossil_fuels_electricity_installed_capacity_MK                     0.055 -0.062
## Fossil_fuels_electricity_net_generation_BKWH                       0.068 -0.053
## Geothermal_electricity_installed_capacity_MK                      -0.438  0.213
## Geothermal_electricity_net_generation_BKWH                        -0.432  0.220
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.038 -0.013
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.066  0.091
## Hydroelectricity_installed_capacity_MK                            -0.107  0.070
## Hydroelectricity_net_generation_BKWH                              -0.145  0.073
## Non.hydro_renewable_electricity_installed_capacity_MK             -0.085 -0.112
## Non.hydro_renewable_electricity_net_generation_BKWH               -0.069 -0.084
## Nuclear_electricity_installed_capacity_MK                          0.134  0.065
## Nuclear_electricity_net_generation_BKWH                            0.129  0.033
## Renewable_electricity_installed_capacity_MK                       -0.104  0.005
## Renewable_electricity_net_generation_BKWH                         -0.133  0.034
## Solar_electricity_installed_capacity_MK                           -0.153 -0.148
## Solar_electricity_net_generation_BKWH                             -0.135 -0.149
## Tide_and_wave_electricity_installed_capacity_MK                    0.089 -0.110
## Tide_and_wave_electricity_net_generation_BKWH                      0.119 -0.056
## Wind_electricity_installed_capacity_MK                            -0.049 -0.154
## Wind_electricity_net_generation_BKWH                              -0.030 -0.164
##                                                                     PC11   PC12
## Bunker_fuel_consumption_TBPD                                      -0.048 -0.078
## Bunker_residual_fuel_oil_consumption_TBPD                          0.027 -0.046
## Crude_oil_including_lease_condensate_exports_TBPD                 -0.065 -0.085
## Crude_oil_including_lease_condensate_imports_TBPD                  0.127 -0.067
## Crude_oil_including_lease_condensate_production_TBPD              -0.113 -0.037
## Crude_oil_including_lease_condensate_reserves_BB                   0.227  0.126
## Distillate_fuel_oil_consumption_TBPD                               0.102 -0.093
## Distillate_fuel_oil_production_TBPD                                0.047 -0.122
## Jet_fuel_consumption_TBPD                                         -0.029  0.017
## Jet_fuel_production_TBPD                                           0.039  0.106
## Kerosene_consumption_TBPD                                          0.212  0.264
## Kerosene_production_TBPD                                           0.123 -0.384
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                   0.052 -0.008
## Liquefied_petroleum_gases_.LPG._production_TBPD                   -0.149 -0.111
## Motor_gasoline_consumption_TBPD                                    0.023  0.137
## Motor_gasoline_production_TBPD                                    -0.004  0.082
## Natural_gas_plant_liquids_production_TBPD                          0.007  0.020
## Other_liquids_production_TBPD                                      0.173  0.197
## Other_refined_products_consumption_TBPD                            0.045  0.151
## Other_refined_products_production_TBPD                             0.075  0.028
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                    0.051  0.045
## Petroleum_and_other_liquids_consumption_TBPD                       0.053  0.076
## Petroleum_and_other_liquids_production_TBPD                       -0.077 -0.011
## Refined_petroleum_products_production_TBPD                        -0.001 -0.002
## Refinery_processing_gain_TBPD                                      0.042  0.061
## Residual_fuel_oil_consumption_TBPD                                 0.059  0.130
## Residual_fuel_oil_production_TBPD                                 -0.309 -0.042
## Biomass_and_waste_electricity_installed_capacity_MK                0.330 -0.041
## Biomass_and_waste_electricity_net_generation_BKWH                  0.212 -0.052
```

```
## Electricity_distribution_losses_BKWH                              0.220 -0.246
## Electricity_exports_BKWH                                         -0.074  0.001
## Electricity_imports_BKWH                                         -0.214  0.053
## Electricity_installed_capacity_MK                                -0.066  0.020
## Electricity_net_consumption_BKWH                                 -0.041  0.057
## Electricity_net_generation_BKWH                                  -0.022  0.037
## Electricity_net_imports_BKWH                                     -0.116  0.035
## Fossil_fuels_electricity_installed_capacity_MK                   -0.099  0.009
## Fossil_fuels_electricity_net_generation_BKWH                     -0.059  0.020
## Geothermal_electricity_installed_capacity_MK                      0.050 -0.027
## Geothermal_electricity_net_generation_BKWH                        0.039 -0.059
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.254  0.393
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.238 -0.089
## Hydroelectricity_installed_capacity_MK                            0.076  0.156
## Hydroelectricity_net_generation_BKWH                              0.114  0.217
## Non.hydro_renewable_electricity_installed_capacity_MK            -0.062 -0.207
## Non.hydro_renewable_electricity_net_generation_BKWH               0.004 -0.193
## Nuclear_electricity_installed_capacity_MK                         0.008 -0.045
## Nuclear_electricity_net_generation_BKWH                           0.005 -0.028
## Renewable_electricity_installed_capacity_MK                       0.028  0.026
## Renewable_electricity_net_generation_BKWH                         0.091  0.119
## Solar_electricity_installed_capacity_MK                          -0.307  0.000
## Solar_electricity_net_generation_BKWH                            -0.286 -0.071
## Tide_and_wave_electricity_installed_capacity_MK                   0.093  0.016
## Tide_and_wave_electricity_net_generation_BKWH                     0.174 -0.006
## Wind_electricity_installed_capacity_MK                           -0.045 -0.298
## Wind_electricity_net_generation_BKWH                             -0.034 -0.274
##                                                                    PC13   PC14
## Bunker_fuel_consumption_TBPD                                      0.014 -0.139
## Bunker_residual_fuel_oil_consumption_TBPD                         0.018 -0.056
## Crude_oil_including_lease_condensate_exports_TBPD                -0.266  0.083
## Crude_oil_including_lease_condensate_imports_TBPD                 0.170  0.123
## Crude_oil_including_lease_condensate_production_TBPD             -0.155 -0.006
## Crude_oil_including_lease_condensate_reserves_BB                  0.583 -0.240
## Distillate_fuel_oil_consumption_TBPD                              0.035  0.044
## Distillate_fuel_oil_production_TBPD                               0.111  0.044
## Jet_fuel_consumption_TBPD                                        -0.039 -0.125
## Jet_fuel_production_TBPD                                          0.083  0.013
## Kerosene_consumption_TBPD                                         0.077 -0.219
## Kerosene_production_TBPD                                         -0.181  0.319
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                  0.011  0.163
## Liquefied_petroleum_gases_.LPG._production_TBPD                   0.057 -0.101
## Motor_gasoline_consumption_TBPD                                  -0.031 -0.085
## Motor_gasoline_production_TBPD                                    0.043 -0.083
## Natural_gas_plant_liquids_production_TBPD                        -0.140  0.041
## Other_liquids_production_TBPD                                    -0.253  0.040
## Other_refined_products_consumption_TBPD                          -0.053  0.248
## Other_refined_products_production_TBPD                            0.026  0.092
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                   0.001  0.015
## Petroleum_and_other_liquids_consumption_TBPD                     -0.019  0.040
## Petroleum_and_other_liquids_production_TBPD                      -0.162  0.001
## Refined_petroleum_products_production_TBPD                        0.067 -0.018
## Refinery_processing_gain_TBPD                                     0.029 -0.063
## Residual_fuel_oil_consumption_TBPD                               -0.072  0.167
```

```
## Residual_fuel_oil_production_TBPD                                  0.102 -0.175
## Biomass_and_waste_electricity_installed_capacity_MK               -0.255 -0.192
## Biomass_and_waste_electricity_net_generation_BKWH                 -0.147 -0.182
## Electricity_distribution_losses_BKWH                               0.256 -0.056
## Electricity_exports_BKWH                                           0.103  0.141
## Electricity_imports_BKWH                                           0.270  0.312
## Electricity_installed_capacity_MK                                  0.045  0.024
## Electricity_net_consumption_BKWH                                   0.038  0.068
## Electricity_net_generation_BKWH                                    0.050  0.058
## Electricity_net_imports_BKWH                                       0.133  0.130
## Fossil_fuels_electricity_installed_capacity_MK                     0.088  0.044
## Fossil_fuels_electricity_net_generation_BKWH                       0.097  0.098
## Geothermal_electricity_installed_capacity_MK                       0.031  0.055
## Geothermal_electricity_net_generation_BKWH                         0.030  0.043
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.042  0.081
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.004 -0.213
## Hydroelectricity_installed_capacity_MK                            -0.069  0.027
## Hydroelectricity_net_generation_BKWH                              -0.100  0.008
## Non.hydro_renewable_electricity_installed_capacity_MK             -0.017 -0.083
## Non.hydro_renewable_electricity_net_generation_BKWH               -0.032 -0.078
## Nuclear_electricity_installed_capacity_MK                         -0.002 -0.073
## Nuclear_electricity_net_generation_BKWH                           -0.001 -0.066
## Renewable_electricity_installed_capacity_MK                       -0.053 -0.013
## Renewable_electricity_net_generation_BKWH                         -0.088 -0.015
## Solar_electricity_installed_capacity_MK                           -0.103 -0.362
## Solar_electricity_net_generation_BKWH                             -0.105 -0.307
## Tide_and_wave_electricity_installed_capacity_MK                   -0.004 -0.005
## Tide_and_wave_electricity_net_generation_BKWH                     -0.032  0.065
## Wind_electricity_installed_capacity_MK                             0.058  0.044
## Wind_electricity_net_generation_BKWH                               0.035  0.015
##                                                                     PC15   PC16
## Bunker_fuel_consumption_TBPD                                       0.068 -0.050
## Bunker_residual_fuel_oil_consumption_TBPD                          0.067  0.005
## Crude_oil_including_lease_condensate_exports_TBPD                  0.272  0.313
## Crude_oil_including_lease_condensate_imports_TBPD                  0.199 -0.100
## Crude_oil_including_lease_condensate_production_TBPD               0.080  0.089
## Crude_oil_including_lease_condensate_reserves_BB                  -0.293 -0.031
## Distillate_fuel_oil_consumption_TBPD                              -0.042  0.175
## Distillate_fuel_oil_production_TBPD                                0.073 -0.125
## Jet_fuel_consumption_TBPD                                          0.015  0.016
## Jet_fuel_production_TBPD                                           0.105 -0.155
## Kerosene_consumption_TBPD                                          0.263  0.253
## Kerosene_production_TBPD                                          -0.331 -0.295
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                   0.139  0.032
## Liquefied_petroleum_gases_.LPG._production_TBPD                    0.029 -0.056
## Motor_gasoline_consumption_TBPD                                   -0.019 -0.097
## Motor_gasoline_production_TBPD                                     0.018 -0.140
## Natural_gas_plant_liquids_production_TBPD                          0.248 -0.276
## Other_liquids_production_TBPD                                     -0.260 -0.011
## Other_refined_products_consumption_TBPD                            0.039 -0.124
## Other_refined_products_production_TBPD                            -0.025 -0.154
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                    0.004  0.024
## Petroleum_and_other_liquids_consumption_TBPD                      -0.002 -0.006
## Petroleum_and_other_liquids_production_TBPD                        0.089  0.022
```

```
## Refined_petroleum_products_production_TBPD                        0.002 -0.152
## Refinery_processing_gain_TBPD                                     0.052 -0.142
## Residual_fuel_oil_consumption_TBPD                               -0.169  0.164
## Residual_fuel_oil_production_TBPD                                -0.310 -0.155
## Biomass_and_waste_electricity_installed_capacity_MK              -0.079  0.065
## Biomass_and_waste_electricity_net_generation_BKWH                 0.017  0.153
## Electricity_distribution_losses_BKWH                              0.137  0.144
## Electricity_exports_BKWH                                          0.189 -0.099
## Electricity_imports_BKWH                                          0.125  0.021
## Electricity_installed_capacity_MK                                 0.003  0.069
## Electricity_net_consumption_BKWH                                  0.034  0.060
## Electricity_net_generation_BKWH                                   0.042  0.064
## Electricity_net_imports_BKWH                                     -0.071  0.110
## Fossil_fuels_electricity_installed_capacity_MK                    0.047  0.082
## Fossil_fuels_electricity_net_generation_BKWH                      0.104  0.099
## Geothermal_electricity_installed_capacity_MK                     -0.025  0.038
## Geothermal_electricity_net_generation_BKWH                       -0.010  0.019
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.067  0.094
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH      0.283 -0.253
## Hydroelectricity_installed_capacity_MK                           -0.077 -0.082
## Hydroelectricity_net_generation_BKWH                             -0.033 -0.177
## Non.hydro_renewable_electricity_installed_capacity_MK             0.013  0.070
## Non.hydro_renewable_electricity_net_generation_BKWH              -0.022  0.043
## Nuclear_electricity_installed_capacity_MK                        -0.217  0.238
## Nuclear_electricity_net_generation_BKWH                          -0.191  0.157
## Renewable_electricity_installed_capacity_MK                      -0.047 -0.025
## Renewable_electricity_net_generation_BKWH                        -0.032 -0.127
## Solar_electricity_installed_capacity_MK                           0.078 -0.021
## Solar_electricity_net_generation_BKWH                            -0.008 -0.181
## Tide_and_wave_electricity_installed_capacity_MK                   0.117 -0.201
## Tide_and_wave_electricity_net_generation_BKWH                    -0.036  0.117
## Wind_electricity_installed_capacity_MK                            0.002  0.102
## Wind_electricity_net_generation_BKWH                             -0.042  0.034
##                                                                    PC17   PC18
## Bunker_fuel_consumption_TBPD                                      0.198 -0.147
## Bunker_residual_fuel_oil_consumption_TBPD                         0.180 -0.024
## Crude_oil_including_lease_condensate_exports_TBPD                 0.246 -0.054
## Crude_oil_including_lease_condensate_imports_TBPD                 0.063 -0.118
## Crude_oil_including_lease_condensate_production_TBPD              0.057  0.067
## Crude_oil_including_lease_condensate_reserves_BB                  0.029 -0.201
## Distillate_fuel_oil_consumption_TBPD                             -0.191  0.097
## Distillate_fuel_oil_production_TBPD                              -0.052 -0.006
## Jet_fuel_consumption_TBPD                                         0.094  0.018
## Jet_fuel_production_TBPD                                          0.116 -0.117
## Kerosene_consumption_TBPD                                         0.163  0.242
## Kerosene_production_TBPD                                          0.171  0.047
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 -0.331 -0.223
## Liquefied_petroleum_gases_.LPG._production_TBPD                   0.112 -0.085
## Motor_gasoline_consumption_TBPD                                  -0.045  0.131
## Motor_gasoline_production_TBPD                                   -0.011  0.110
## Natural_gas_plant_liquids_production_TBPD                        -0.099  0.020
## Other_liquids_production_TBPD                                    -0.050 -0.027
## Other_refined_products_consumption_TBPD                          -0.093 -0.092
## Other_refined_products_production_TBPD                            0.071 -0.143
```

```
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                -0.151  0.105
## Petroleum_and_other_liquids_consumption_TBPD                   -0.130  0.057
## Petroleum_and_other_liquids_production_TBPD                     0.030  0.060
## Refined_petroleum_products_production_TBPD                      0.000  0.007
## Refinery_processing_gain_TBPD                                   0.041  0.095
## Residual_fuel_oil_consumption_TBPD                             -0.481  0.248
## Residual_fuel_oil_production_TBPD                              -0.143  0.008
## Biomass_and_waste_electricity_installed_capacity_MK            -0.094 -0.381
## Biomass_and_waste_electricity_net_generation_BKWH              -0.058 -0.195
## Electricity_distribution_losses_BKWH                            0.021  0.095
## Electricity_exports_BKWH                                       -0.113 -0.036
## Electricity_imports_BKWH                                        0.017  0.154
## Electricity_installed_capacity_MK                              -0.004  0.020
## Electricity_net_consumption_BKWH                               -0.008 -0.015
## Electricity_net_generation_BKWH                                -0.008 -0.011
## Electricity_net_imports_BKWH                                    0.121  0.176
## Fossil_fuels_electricity_installed_capacity_MK                 -0.051 -0.021
## Fossil_fuels_electricity_net_generation_BKWH                   -0.081 -0.096
## Geothermal_electricity_installed_capacity_MK                    0.070 -0.042
## Geothermal_electricity_net_generation_BKWH                      0.042 -0.027
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK  0.109 -0.289
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.240  0.285
## Hydroelectricity_installed_capacity_MK                          0.151  0.169
## Hydroelectricity_net_generation_BKWH                            0.217  0.177
## Non.hydro_renewable_electricity_installed_capacity_MK          -0.043 -0.048
## Non.hydro_renewable_electricity_net_generation_BKWH             0.012  0.017
## Nuclear_electricity_installed_capacity_MK                       0.087  0.218
## Nuclear_electricity_net_generation_BKWH                         0.090  0.196
## Renewable_electricity_installed_capacity_MK                     0.082  0.096
## Renewable_electricity_net_generation_BKWH                       0.175  0.144
## Solar_electricity_installed_capacity_MK                        -0.170 -0.027
## Solar_electricity_net_generation_BKWH                          -0.087  0.093
## Tide_and_wave_electricity_installed_capacity_MK                 0.084  0.027
## Tide_and_wave_electricity_net_generation_BKWH                  -0.022  0.051
## Wind_electricity_installed_capacity_MK                          0.013  0.012
## Wind_electricity_net_generation_BKWH                            0.063  0.101
##                                                                  PC19   PC20
## Bunker_fuel_consumption_TBPD                                    0.086 -0.127
## Bunker_residual_fuel_oil_consumption_TBPD                       0.058  0.001
## Crude_oil_including_lease_condensate_exports_TBPD               0.077  0.119
## Crude_oil_including_lease_condensate_imports_TBPD               0.059  0.200
## Crude_oil_including_lease_condensate_production_TBPD           -0.098 -0.101
## Crude_oil_including_lease_condensate_reserves_BB                0.040  0.037
## Distillate_fuel_oil_consumption_TBPD                            0.108 -0.162
## Distillate_fuel_oil_production_TBPD                             0.014 -0.058
## Jet_fuel_consumption_TBPD                                      -0.072  0.159
## Jet_fuel_production_TBPD                                       -0.086  0.116
## Kerosene_consumption_TBPD                                      -0.281 -0.038
## Kerosene_production_TBPD                                        0.059 -0.032
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                0.377  0.131
## Liquefied_petroleum_gases_.LPG._production_TBPD                 0.042 -0.113
## Motor_gasoline_consumption_TBPD                                -0.143 -0.049
## Motor_gasoline_production_TBPD                                 -0.104 -0.019
## Natural_gas_plant_liquids_production_TBPD                       0.116 -0.070
```

```
## Other_liquids_production_TBPD                                  0.012 -0.513
## Other_refined_products_consumption_TBPD                       -0.092  0.321
## Other_refined_products_production_TBPD                         0.013 -0.018
## Petroleum_and_other_liquids_CO2_emissions_MMTCD               -0.035 -0.022
## Petroleum_and_other_liquids_consumption_TBPD                  -0.034  0.033
## Petroleum_and_other_liquids_production_TBPD                   -0.063 -0.128
## Refined_petroleum_products_production_TBPD                    -0.054 -0.018
## Refinery_processing_gain_TBPD                                 -0.046 -0.049
## Residual_fuel_oil_consumption_TBPD                            -0.123  0.046
## Residual_fuel_oil_production_TBPD                             -0.181  0.096
## Biomass_and_waste_electricity_installed_capacity_MK           -0.100  0.153
## Biomass_and_waste_electricity_net_generation_BKWH             -0.045  0.093
## Electricity_distribution_losses_BKWH                           0.275 -0.408
## Electricity_exports_BKWH                                      -0.118 -0.095
## Electricity_imports_BKWH                                       0.006 -0.001
## Electricity_installed_capacity_MK                             -0.018 -0.007
## Electricity_net_consumption_BKWH                              -0.032  0.034
## Electricity_net_generation_BKWH                               -0.013  0.003
## Electricity_net_imports_BKWH                                   0.117  0.081
## Fossil_fuels_electricity_installed_capacity_MK                -0.075 -0.045
## Fossil_fuels_electricity_net_generation_BKWH                  -0.073 -0.061
## Geothermal_electricity_installed_capacity_MK                   0.000  0.014
## Geothermal_electricity_net_generation_BKWH                    -0.021 -0.005
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK  0.290 -0.084
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.264  0.173
## Hydroelectricity_installed_capacity_MK                         0.098  0.029
## Hydroelectricity_net_generation_BKWH                           0.097  0.090
## Non.hydro_renewable_electricity_installed_capacity_MK         -0.105  0.045
## Non.hydro_renewable_electricity_net_generation_BKWH           -0.115  0.066
## Nuclear_electricity_installed_capacity_MK                      0.285  0.243
## Nuclear_electricity_net_generation_BKWH                        0.215  0.203
## Renewable_electricity_installed_capacity_MK                    0.029  0.037
## Renewable_electricity_net_generation_BKWH                      0.045  0.088
## Solar_electricity_installed_capacity_MK                        0.100  0.021
## Solar_electricity_net_generation_BKWH                          0.208 -0.073
## Tide_and_wave_electricity_installed_capacity_MK               -0.145 -0.094
## Tide_and_wave_electricity_net_generation_BKWH                  0.028 -0.071
## Wind_electricity_installed_capacity_MK                        -0.172  0.022
## Wind_electricity_net_generation_BKWH                          -0.212  0.082
##                                                                 PC21   PC22
## Bunker_fuel_consumption_TBPD                                   0.019 -0.024
## Bunker_residual_fuel_oil_consumption_TBPD                      0.036 -0.064
## Crude_oil_including_lease_condensate_exports_TBPD              0.109 -0.041
## Crude_oil_including_lease_condensate_imports_TBPD              0.143  0.000
## Crude_oil_including_lease_condensate_production_TBPD           0.035  0.053
## Crude_oil_including_lease_condensate_reserves_BB               0.046 -0.014
## Distillate_fuel_oil_consumption_TBPD                          -0.106 -0.133
## Distillate_fuel_oil_production_TBPD                            0.040  0.113
## Jet_fuel_consumption_TBPD                                     -0.249 -0.341
## Jet_fuel_production_TBPD                                      -0.005 -0.248
## Kerosene_consumption_TBPD                                      0.033 -0.047
## Kerosene_production_TBPD                                       0.071 -0.032
## Liquefied_petroleum_gases_.LPG._consumption_TBPD              -0.178 -0.108
## Liquefied_petroleum_gases_.LPG._production_TBPD                0.099  0.038
```

```
## Motor_gasoline_consumption_TBPD                                  -0.047  0.015
## Motor_gasoline_production_TBPD                                   -0.032  0.070
## Natural_gas_plant_liquids_production_TBPD                        -0.295  0.065
## Other_liquids_production_TBPD                                     0.402 -0.212
## Other_refined_products_consumption_TBPD                           0.198 -0.003
## Other_refined_products_production_TBPD                            0.167  0.002
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                  -0.085 -0.052
## Petroleum_and_other_liquids_consumption_TBPD                     -0.040 -0.049
## Petroleum_and_other_liquids_production_TBPD                       0.009  0.043
## Refined_petroleum_products_production_TBPD                        0.026  0.035
## Refinery_processing_gain_TBPD                                    -0.014  0.075
## Residual_fuel_oil_consumption_TBPD                               -0.094  0.228
## Residual_fuel_oil_production_TBPD                                -0.098 -0.127
## Biomass_and_waste_electricity_installed_capacity_MK              -0.084  0.151
## Biomass_and_waste_electricity_net_generation_BKWH                 0.063  0.089
## Electricity_distribution_losses_BKWH                             -0.226  0.135
## Electricity_exports_BKWH                                          0.085 -0.048
## Electricity_imports_BKWH                                          0.072 -0.053
## Electricity_installed_capacity_MK                                 0.106  0.018
## Electricity_net_consumption_BKWH                                  0.147  0.034
## Electricity_net_generation_BKWH                                   0.123  0.041
## Electricity_net_imports_BKWH                                     -0.013 -0.022
## Fossil_fuels_electricity_installed_capacity_MK                    0.166  0.035
## Fossil_fuels_electricity_net_generation_BKWH                      0.182  0.017
## Geothermal_electricity_installed_capacity_MK                      0.059 -0.050
## Geothermal_electricity_net_generation_BKWH                        0.017  0.076
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.231  0.114
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.291 -0.135
## Hydroelectricity_installed_capacity_MK                           -0.083 -0.103
## Hydroelectricity_net_generation_BKWH                             -0.129  0.046
## Non.hydro_renewable_electricity_installed_capacity_MK            -0.002 -0.027
## Non.hydro_renewable_electricity_net_generation_BKWH              -0.076 -0.007
## Nuclear_electricity_installed_capacity_MK                         0.164  0.167
## Nuclear_electricity_net_generation_BKWH                           0.139  0.158
## Renewable_electricity_installed_capacity_MK                      -0.056 -0.079
## Renewable_electricity_net_generation_BKWH                        -0.123  0.034
## Solar_electricity_installed_capacity_MK                           0.156 -0.098
## Solar_electricity_net_generation_BKWH                             0.079  0.024
## Tide_and_wave_electricity_installed_capacity_MK                  -0.040  0.438
## Tide_and_wave_electricity_net_generation_BKWH                    -0.101 -0.502
## Wind_electricity_installed_capacity_MK                           -0.047 -0.024
## Wind_electricity_net_generation_BKWH                             -0.171 -0.057
##                                                                    PC23   PC24
## Bunker_fuel_consumption_TBPD                                      0.016  0.043
## Bunker_residual_fuel_oil_consumption_TBPD                         0.022  0.050
## Crude_oil_including_lease_condensate_exports_TBPD                -0.266  0.147
## Crude_oil_including_lease_condensate_imports_TBPD                -0.053  0.272
## Crude_oil_including_lease_condensate_production_TBPD             -0.051 -0.083
## Crude_oil_including_lease_condensate_reserves_BB                  0.069 -0.031
## Distillate_fuel_oil_consumption_TBPD                             -0.068  0.201
## Distillate_fuel_oil_production_TBPD                              -0.034  0.092
## Jet_fuel_consumption_TBPD                                        -0.066 -0.021
## Jet_fuel_production_TBPD                                         -0.106 -0.097
## Kerosene_consumption_TBPD                                         0.244  0.080
```

```
## Kerosene_production_TBPD                                          -0.081 -0.120
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                   0.023  0.188
## Liquefied_petroleum_gases_.LPG._production_TBPD                    0.130 -0.203
## Motor_gasoline_consumption_TBPD                                   -0.321  0.049
## Motor_gasoline_production_TBPD                                    -0.189 -0.044
## Natural_gas_plant_liquids_production_TBPD                          0.589 -0.162
## Other_liquids_production_TBPD                                      0.111  0.152
## Other_refined_products_consumption_TBPD                            0.202  0.184
## Other_refined_products_production_TBPD                             0.082 -0.058
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                   -0.123  0.116
## Petroleum_and_other_liquids_consumption_TBPD                      -0.103  0.116
## Petroleum_and_other_liquids_production_TBPD                        0.047 -0.085
## Refined_petroleum_products_production_TBPD                        -0.078 -0.016
## Refinery_processing_gain_TBPD                                     -0.185 -0.063
## Residual_fuel_oil_consumption_TBPD                                -0.008 -0.145
## Residual_fuel_oil_production_TBPD                                 -0.020  0.081
## Biomass_and_waste_electricity_installed_capacity_MK               -0.138  0.006
## Biomass_and_waste_electricity_net_generation_BKWH                  0.115 -0.249
## Electricity_distribution_losses_BKWH                              -0.078  0.059
## Electricity_exports_BKWH                                          -0.044  0.003
## Electricity_imports_BKWH                                           0.001 -0.034
## Electricity_installed_capacity_MK                                 -0.020 -0.125
## Electricity_net_consumption_BKWH                                  -0.004 -0.100
## Electricity_net_generation_BKWH                                   -0.010 -0.090
## Electricity_net_imports_BKWH                                       0.063  0.001
## Fossil_fuels_electricity_installed_capacity_MK                    -0.043 -0.162
## Fossil_fuels_electricity_net_generation_BKWH                      -0.028 -0.138
## Geothermal_electricity_installed_capacity_MK                       0.097 -0.134
## Geothermal_electricity_net_generation_BKWH                        -0.032  0.104
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.218 -0.352
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.221 -0.388
## Hydroelectricity_installed_capacity_MK                            -0.002 -0.021
## Hydroelectricity_net_generation_BKWH                               0.014  0.097
## Non.hydro_renewable_electricity_installed_capacity_MK              0.062 -0.024
## Non.hydro_renewable_electricity_net_generation_BKWH                0.034 -0.074
## Nuclear_electricity_installed_capacity_MK                          0.134  0.001
## Nuclear_electricity_net_generation_BKWH                            0.046 -0.022
## Renewable_electricity_installed_capacity_MK                        0.022 -0.022
## Renewable_electricity_net_generation_BKWH                          0.020  0.056
## Solar_electricity_installed_capacity_MK                            0.085  0.124
## Solar_electricity_net_generation_BKWH                             -0.013  0.187
## Tide_and_wave_electricity_installed_capacity_MK                   -0.076  0.052
## Tide_and_wave_electricity_net_generation_BKWH                      0.008 -0.222
## Wind_electricity_installed_capacity_MK                             0.079 -0.081
## Wind_electricity_net_generation_BKWH                               0.012 -0.049
##                                                                     PC25   PC26
## Bunker_fuel_consumption_TBPD                                      -0.141  0.089
## Bunker_residual_fuel_oil_consumption_TBPD                          0.107  0.201
## Crude_oil_including_lease_condensate_exports_TBPD                  0.091 -0.144
## Crude_oil_including_lease_condensate_imports_TBPD                  0.184 -0.295
## Crude_oil_including_lease_condensate_production_TBPD               0.021  0.076
## Crude_oil_including_lease_condensate_reserves_BB                  -0.018  0.052
## Distillate_fuel_oil_consumption_TBPD                              -0.217  0.015
## Distillate_fuel_oil_production_TBPD                                0.047 -0.028
```

```
## Jet_fuel_consumption_TBPD                                      -0.334 -0.264
## Jet_fuel_production_TBPD                                       -0.063 -0.186
## Kerosene_consumption_TBPD                                       0.067  0.200
## Kerosene_production_TBPD                                       -0.212  0.040
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                0.174  0.396
## Liquefied_petroleum_gases_.LPG._production_TBPD                -0.004 -0.007
## Motor_gasoline_consumption_TBPD                                 0.027  0.157
## Motor_gasoline_production_TBPD                                  0.093  0.009
## Natural_gas_plant_liquids_production_TBPD                      -0.109 -0.001
## Other_liquids_production_TBPD                                   0.215 -0.064
## Other_refined_products_consumption_TBPD                         0.052 -0.120
## Other_refined_products_production_TBPD                          0.102  0.023
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                -0.082  0.047
## Petroleum_and_other_liquids_consumption_TBPD                   -0.036  0.045
## Petroleum_and_other_liquids_production_TBPD                     0.020  0.057
## Refined_petroleum_products_production_TBPD                      0.073 -0.017
## Refinery_processing_gain_TBPD                                   0.123 -0.009
## Residual_fuel_oil_consumption_TBPD                              0.042 -0.271
## Residual_fuel_oil_production_TBPD                               0.146 -0.052
## Biomass_and_waste_electricity_installed_capacity_MK            -0.241  0.169
## Biomass_and_waste_electricity_net_generation_BKWH               0.147 -0.214
## Electricity_distribution_losses_BKWH                           -0.013 -0.329
## Electricity_exports_BKWH                                       -0.025 -0.008
## Electricity_imports_BKWH                                       -0.048  0.054
## Electricity_installed_capacity_MK                              -0.148  0.109
## Electricity_net_consumption_BKWH                               -0.157  0.054
## Electricity_net_generation_BKWH                                -0.148  0.028
## Electricity_net_imports_BKWH                                   -0.032  0.022
## Fossil_fuels_electricity_installed_capacity_MK                 -0.213  0.115
## Fossil_fuels_electricity_net_generation_BKWH                   -0.219  0.050
## Geothermal_electricity_installed_capacity_MK                    0.056 -0.230
## Geothermal_electricity_net_generation_BKWH                     -0.040  0.192
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK  0.244 -0.026
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.090  0.064
## Hydroelectricity_installed_capacity_MK                         -0.070  0.108
## Hydroelectricity_net_generation_BKWH                           -0.015 -0.057
## Non.hydro_renewable_electricity_installed_capacity_MK           0.085  0.048
## Non.hydro_renewable_electricity_net_generation_BKWH             0.205  0.006
## Nuclear_electricity_installed_capacity_MK                      -0.046  0.038
## Nuclear_electricity_net_generation_BKWH                         0.008  0.017
## Renewable_electricity_installed_capacity_MK                    -0.014  0.088
## Renewable_electricity_net_generation_BKWH                       0.044 -0.044
## Solar_electricity_installed_capacity_MK                        -0.189 -0.186
## Solar_electricity_net_generation_BKWH                           0.079  0.048
## Tide_and_wave_electricity_installed_capacity_MK                -0.073  0.045
## Tide_and_wave_electricity_net_generation_BKWH                   0.075  0.059
## Wind_electricity_installed_capacity_MK                          0.235  0.118
## Wind_electricity_net_generation_BKWH                            0.254  0.084
##                                                                  PC27   PC28
## Bunker_fuel_consumption_TBPD                                    0.009  0.153
## Bunker_residual_fuel_oil_consumption_TBPD                      -0.095  0.095
## Crude_oil_including_lease_condensate_exports_TBPD               0.134 -0.116
## Crude_oil_including_lease_condensate_imports_TBPD               0.090 -0.116
## Crude_oil_including_lease_condensate_production_TBPD           -0.098  0.088
```

```
## Crude_oil_including_lease_condensate_reserves_BB               -0.015 -0.003
## Distillate_fuel_oil_consumption_TBPD                           -0.373  0.043
## Distillate_fuel_oil_production_TBPD                            -0.082  0.067
## Jet_fuel_consumption_TBPD                                      -0.286 -0.135
## Jet_fuel_production_TBPD                                        0.113 -0.375
## Kerosene_consumption_TBPD                                       0.007 -0.065
## Kerosene_production_TBPD                                        0.050  0.058
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                0.159 -0.089
## Liquefied_petroleum_gases_.LPG._production_TBPD                -0.008  0.027
## Motor_gasoline_consumption_TBPD                                 0.081  0.058
## Motor_gasoline_production_TBPD                                  0.191  0.230
## Natural_gas_plant_liquids_production_TBPD                       0.063 -0.092
## Other_liquids_production_TBPD                                  -0.104 -0.189
## Other_refined_products_consumption_TBPD                        -0.261  0.496
## Other_refined_products_production_TBPD                         -0.081 -0.330
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                -0.121  0.053
## Petroleum_and_other_liquids_consumption_TBPD                   -0.112  0.110
## Petroleum_and_other_liquids_production_TBPD                    -0.066  0.052
## Refined_petroleum_products_production_TBPD                      0.052  0.028
## Refinery_processing_gain_TBPD                                   0.204  0.161
## Residual_fuel_oil_consumption_TBPD                              0.165 -0.215
## Residual_fuel_oil_production_TBPD                               0.010 -0.027
## Biomass_and_waste_electricity_installed_capacity_MK             0.129 -0.079
## Biomass_and_waste_electricity_net_generation_BKWH               0.009  0.239
## Electricity_distribution_losses_BKWH                            0.062  0.101
## Electricity_exports_BKWH                                       -0.001 -0.022
## Electricity_imports_BKWH                                        0.040 -0.019
## Electricity_installed_capacity_MK                               0.052 -0.023
## Electricity_net_consumption_BKWH                                0.060 -0.003
## Electricity_net_generation_BKWH                                 0.060  0.004
## Electricity_net_imports_BKWH                                    0.008  0.003
## Fossil_fuels_electricity_installed_capacity_MK                  0.062 -0.026
## Fossil_fuels_electricity_net_generation_BKWH                    0.083  0.018
## Geothermal_electricity_installed_capacity_MK                    0.134  0.110
## Geothermal_electricity_net_generation_BKWH                     -0.119 -0.108
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.314  0.011
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.283 -0.069
## Hydroelectricity_installed_capacity_MK                          0.139  0.059
## Hydroelectricity_net_generation_BKWH                            0.018 -0.030
## Non.hydro_renewable_electricity_installed_capacity_MK          -0.036 -0.053
## Non.hydro_renewable_electricity_net_generation_BKWH            -0.071 -0.009
## Nuclear_electricity_installed_capacity_MK                      -0.040 -0.101
## Nuclear_electricity_net_generation_BKWH                         0.017 -0.018
## Renewable_electricity_installed_capacity_MK                     0.071  0.014
## Renewable_electricity_net_generation_BKWH                      -0.005 -0.026
## Solar_electricity_installed_capacity_MK                         0.000  0.037
## Solar_electricity_net_generation_BKWH                           0.199 -0.042
## Tide_and_wave_electricity_installed_capacity_MK                -0.217 -0.137
## Tide_and_wave_electricity_net_generation_BKWH                   0.220  0.198
## Wind_electricity_installed_capacity_MK                         -0.082 -0.089
## Wind_electricity_net_generation_BKWH                           -0.153 -0.108
##                                                                  PC29   PC30
## Bunker_fuel_consumption_TBPD                                   -0.175 -0.166
## Bunker_residual_fuel_oil_consumption_TBPD                       0.175  0.133
```

```
## Crude_oil_including_lease_condensate_exports_TBPD              -0.001 -0.024
## Crude_oil_including_lease_condensate_imports_TBPD              -0.215  0.244
## Crude_oil_including_lease_condensate_production_TBPD           -0.019  0.083
## Crude_oil_including_lease_condensate_reserves_BB               -0.024 -0.032
## Distillate_fuel_oil_consumption_TBPD                           -0.149  0.398
## Distillate_fuel_oil_production_TBPD                            -0.081  0.340
## Jet_fuel_consumption_TBPD                                       0.022 -0.266
## Jet_fuel_production_TBPD                                        0.115 -0.020
## Kerosene_consumption_TBPD                                      -0.061 -0.039
## Kerosene_production_TBPD                                        0.105 -0.063
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                0.110 -0.333
## Liquefied_petroleum_gases_.LPG._production_TBPD                -0.186 -0.089
## Motor_gasoline_consumption_TBPD                                 0.124 -0.051
## Motor_gasoline_production_TBPD                                  0.014  0.036
## Natural_gas_plant_liquids_production_TBPD                      -0.023  0.135
## Other_liquids_production_TBPD                                  -0.066 -0.208
## Other_refined_products_consumption_TBPD                        -0.050 -0.232
## Other_refined_products_production_TBPD                          0.213  0.217
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 0.011  0.025
## Petroleum_and_other_liquids_consumption_TBPD                    0.010 -0.020
## Petroleum_and_other_liquids_production_TBPD                    -0.022  0.075
## Refined_petroleum_products_production_TBPD                      0.025  0.141
## Refinery_processing_gain_TBPD                                   0.020  0.030
## Residual_fuel_oil_consumption_TBPD                             -0.057 -0.018
## Residual_fuel_oil_production_TBPD                               0.032 -0.007
## Biomass_and_waste_electricity_installed_capacity_MK            -0.322  0.063
## Biomass_and_waste_electricity_net_generation_BKWH               0.442  0.046
## Electricity_distribution_losses_BKWH                            0.139 -0.243
## Electricity_exports_BKWH                                        0.029 -0.079
## Electricity_imports_BKWH                                       -0.010 -0.032
## Electricity_installed_capacity_MK                              -0.018 -0.019
## Electricity_net_consumption_BKWH                                0.026 -0.014
## Electricity_net_generation_BKWH                                 0.034 -0.029
## Electricity_net_imports_BKWH                                   -0.003  0.003
## Fossil_fuels_electricity_installed_capacity_MK                  0.009 -0.043
## Fossil_fuels_electricity_net_generation_BKWH                    0.032 -0.030
## Geothermal_electricity_installed_capacity_MK                   -0.350 -0.077
## Geothermal_electricity_net_generation_BKWH                      0.361  0.092
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.107  0.073
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.113 -0.072
## Hydroelectricity_installed_capacity_MK                         -0.095  0.042
## Hydroelectricity_net_generation_BKWH                            0.064  0.008
## Non.hydro_renewable_electricity_installed_capacity_MK          -0.046  0.011
## Non.hydro_renewable_electricity_net_generation_BKWH             0.044 -0.080
## Nuclear_electricity_installed_capacity_MK                       0.010  0.014
## Nuclear_electricity_net_generation_BKWH                        -0.007 -0.042
## Renewable_electricity_installed_capacity_MK                    -0.072  0.030
## Renewable_electricity_net_generation_BKWH                       0.061 -0.015
## Solar_electricity_installed_capacity_MK                         0.206  0.059
## Solar_electricity_net_generation_BKWH                          -0.155 -0.018
## Tide_and_wave_electricity_installed_capacity_MK                -0.034 -0.240
## Tide_and_wave_electricity_net_generation_BKWH                  -0.002  0.159
## Wind_electricity_installed_capacity_MK                         -0.074 -0.016
## Wind_electricity_net_generation_BKWH                           -0.121 -0.155
```

```
##                                                                PC31   PC32
## Bunker_fuel_consumption_TBPD                                 -0.075  0.004
## Bunker_residual_fuel_oil_consumption_TBPD                     0.003 -0.050
## Crude_oil_including_lease_condensate_exports_TBPD            -0.082 -0.023
## Crude_oil_including_lease_condensate_imports_TBPD            -0.250  0.157
## Crude_oil_including_lease_condensate_production_TBPD          0.134  0.017
## Crude_oil_including_lease_condensate_reserves_BB             -0.004 -0.010
## Distillate_fuel_oil_consumption_TBPD                         -0.126 -0.240
## Distillate_fuel_oil_production_TBPD                          -0.091 -0.065
## Jet_fuel_consumption_TBPD                                    -0.045 -0.045
## Jet_fuel_production_TBPD                                      0.065  0.038
## Kerosene_consumption_TBPD                                     0.003 -0.016
## Kerosene_production_TBPD                                     -0.044 -0.046
## Liquefied_petroleum_gases_.LPG._consumption_TBPD             -0.078 -0.252
## Liquefied_petroleum_gases_.LPG._production_TBPD               0.053 -0.045
## Motor_gasoline_consumption_TBPD                               0.183 -0.119
## Motor_gasoline_production_TBPD                               -0.102  0.021
## Natural_gas_plant_liquids_production_TBPD                    -0.124  0.113
## Other_liquids_production_TBPD                                -0.191  0.087
## Other_refined_products_consumption_TBPD                       0.237  0.248
## Other_refined_products_production_TBPD                        0.551 -0.193
## Petroleum_and_other_liquids_CO2_emissions_MMTCD               0.055 -0.086
## Petroleum_and_other_liquids_consumption_TBPD                  0.086 -0.075
## Petroleum_and_other_liquids_production_TBPD                   0.078  0.036
## Refined_petroleum_products_production_TBPD                    0.018 -0.039
## Refinery_processing_gain_TBPD                                 0.028  0.001
## Residual_fuel_oil_consumption_TBPD                            0.057  0.068
## Residual_fuel_oil_production_TBPD                            -0.256  0.069
## Biomass_and_waste_electricity_installed_capacity_MK           0.145  0.236
## Biomass_and_waste_electricity_net_generation_BKWH            -0.275 -0.264
## Electricity_distribution_losses_BKWH                          0.221  0.244
## Electricity_exports_BKWH                                      0.022 -0.034
## Electricity_imports_BKWH                                      0.011  0.004
## Electricity_installed_capacity_MK                            -0.084  0.012
## Electricity_net_consumption_BKWH                             -0.124  0.006
## Electricity_net_generation_BKWH                              -0.101  0.022
## Electricity_net_imports_BKWH                                  0.024  0.018
## Fossil_fuels_electricity_installed_capacity_MK               -0.133 -0.002
## Fossil_fuels_electricity_net_generation_BKWH                 -0.120  0.041
## Geothermal_electricity_installed_capacity_MK                  0.117 -0.426
## Geothermal_electricity_net_generation_BKWH                   -0.100  0.453
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.027  0.135
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH  -0.054  0.095
## Hydroelectricity_installed_capacity_MK                       -0.029  0.047
## Hydroelectricity_net_generation_BKWH                         -0.075 -0.065
## Non.hydro_renewable_electricity_installed_capacity_MK         0.094  0.048
## Non.hydro_renewable_electricity_net_generation_BKWH          -0.071 -0.013
## Nuclear_electricity_installed_capacity_MK                     0.034  0.007
## Nuclear_electricity_net_generation_BKWH                       0.017  0.022
## Renewable_electricity_installed_capacity_MK                   0.028  0.043
## Renewable_electricity_net_generation_BKWH                    -0.082 -0.053
## Solar_electricity_installed_capacity_MK                       0.163  0.007
## Solar_electricity_net_generation_BKWH                         0.064  0.037
## Tide_and_wave_electricity_installed_capacity_MK              -0.148 -0.169
```

```
## Tide_and_wave_electricity_net_generation_BKWH                    0.045  0.169
## Wind_electricity_installed_capacity_MK                           0.040  0.034
## Wind_electricity_net_generation_BKWH                             0.001  0.064
##                                                                   PC33   PC34
## Bunker_fuel_consumption_TBPD                                     0.175  0.468
## Bunker_residual_fuel_oil_consumption_TBPD                       -0.306 -0.343
## Crude_oil_including_lease_condensate_exports_TBPD                0.064  0.092
## Crude_oil_including_lease_condensate_imports_TBPD               -0.050  0.099
## Crude_oil_including_lease_condensate_production_TBPD            -0.052 -0.130
## Crude_oil_including_lease_condensate_reserves_BB                 0.082 -0.012
## Distillate_fuel_oil_consumption_TBPD                             0.029 -0.048
## Distillate_fuel_oil_production_TBPD                              0.178 -0.097
## Jet_fuel_consumption_TBPD                                        0.147 -0.003
## Jet_fuel_production_TBPD                                        -0.046 -0.309
## Kerosene_consumption_TBPD                                        0.017  0.015
## Kerosene_production_TBPD                                        -0.018 -0.026
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 0.042 -0.146
## Liquefied_petroleum_gases_.LPG._production_TBPD                  0.565 -0.324
## Motor_gasoline_consumption_TBPD                                  0.007  0.265
## Motor_gasoline_production_TBPD                                  -0.036 -0.107
## Natural_gas_plant_liquids_production_TBPD                       -0.128  0.142
## Other_liquids_production_TBPD                                   -0.051 -0.055
## Other_refined_products_consumption_TBPD                          0.001 -0.040
## Other_refined_products_production_TBPD                          -0.007  0.309
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                  0.018  0.079
## Petroleum_and_other_liquids_consumption_TBPD                     0.035  0.071
## Petroleum_and_other_liquids_production_TBPD                     -0.062 -0.095
## Refined_petroleum_products_production_TBPD                       0.034 -0.045
## Refinery_processing_gain_TBPD                                    0.069 -0.138
## Residual_fuel_oil_consumption_TBPD                               0.211 -0.035
## Residual_fuel_oil_production_TBPD                               -0.294  0.135
## Biomass_and_waste_electricity_installed_capacity_MK             -0.123 -0.076
## Biomass_and_waste_electricity_net_generation_BKWH                0.104  0.089
## Electricity_distribution_losses_BKWH                            -0.223 -0.006
## Electricity_exports_BKWH                                         0.035 -0.004
## Electricity_imports_BKWH                                         0.005  0.006
## Electricity_installed_capacity_MK                               -0.100 -0.043
## Electricity_net_consumption_BKWH                                -0.055  0.108
## Electricity_net_generation_BKWH                                 -0.067  0.103
## Electricity_net_imports_BKWH                                     0.006  0.028
## Fossil_fuels_electricity_installed_capacity_MK                  -0.111 -0.037
## Fossil_fuels_electricity_net_generation_BKWH                    -0.111  0.125
## Geothermal_electricity_installed_capacity_MK                    -0.225 -0.016
## Geothermal_electricity_net_generation_BKWH                       0.246  0.003
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.072  0.050
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.064  0.073
## Hydroelectricity_installed_capacity_MK                          -0.051 -0.088
## Hydroelectricity_net_generation_BKWH                             0.063  0.031
## Non.hydro_renewable_electricity_installed_capacity_MK           -0.039  0.014
## Non.hydro_renewable_electricity_net_generation_BKWH              0.013  0.030
## Nuclear_electricity_installed_capacity_MK                       -0.080 -0.033
## Nuclear_electricity_net_generation_BKWH                          0.002  0.039
## Renewable_electricity_installed_capacity_MK                     -0.054 -0.045
## Renewable_electricity_net_generation_BKWH                        0.054  0.030
```

```
## Solar_electricity_installed_capacity_MK                          -0.126 -0.146
## Solar_electricity_net_generation_BKWH                             0.145  0.094
## Tide_and_wave_electricity_installed_capacity_MK                  -0.088 -0.059
## Tide_and_wave_electricity_net_generation_BKWH                     0.098  0.101
## Wind_electricity_installed_capacity_MK                            0.013  0.081
## Wind_electricity_net_generation_BKWH                             -0.066 -0.013
##                                                                    PC35   PC36
## Bunker_fuel_consumption_TBPD                                     -0.077 -0.385
## Bunker_residual_fuel_oil_consumption_TBPD                         0.034  0.336
## Crude_oil_including_lease_condensate_exports_TBPD                 0.008  0.242
## Crude_oil_including_lease_condensate_imports_TBPD                -0.005  0.042
## Crude_oil_including_lease_condensate_production_TBPD              0.026 -0.307
## Crude_oil_including_lease_condensate_reserves_BB                 -0.005  0.033
## Distillate_fuel_oil_consumption_TBPD                              0.151 -0.081
## Distillate_fuel_oil_production_TBPD                              -0.146 -0.031
## Jet_fuel_consumption_TBPD                                         0.084  0.144
## Jet_fuel_production_TBPD                                          0.073 -0.382
## Kerosene_consumption_TBPD                                        -0.007 -0.042
## Kerosene_production_TBPD                                         -0.069 -0.011
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 -0.075 -0.126
## Liquefied_petroleum_gases_.LPG._production_TBPD                  -0.097  0.327
## Motor_gasoline_consumption_TBPD                                  -0.143  0.283
## Motor_gasoline_production_TBPD                                   -0.039 -0.052
## Natural_gas_plant_liquids_production_TBPD                        -0.015  0.205
## Other_liquids_production_TBPD                                    -0.084  0.043
## Other_refined_products_consumption_TBPD                           0.121  0.001
## Other_refined_products_production_TBPD                            0.174  0.067
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                  -0.071  0.055
## Petroleum_and_other_liquids_consumption_TBPD                      0.005  0.091
## Petroleum_and_other_liquids_production_TBPD                       0.012 -0.215
## Refined_petroleum_products_production_TBPD                       -0.027 -0.024
## Refinery_processing_gain_TBPD                                    -0.010 -0.041
## Residual_fuel_oil_consumption_TBPD                                0.002 -0.024
## Residual_fuel_oil_production_TBPD                                -0.014  0.066
## Biomass_and_waste_electricity_installed_capacity_MK              -0.053  0.084
## Biomass_and_waste_electricity_net_generation_BKWH                 0.198 -0.087
## Electricity_distribution_losses_BKWH                              0.036  0.035
## Electricity_exports_BKWH                                          0.042  0.005
## Electricity_imports_BKWH                                         -0.027 -0.016
## Electricity_installed_capacity_MK                                 0.119  0.051
## Electricity_net_consumption_BKWH                                 -0.065  0.035
## Electricity_net_generation_BKWH                                  -0.058  0.039
## Electricity_net_imports_BKWH                                     -0.040  0.008
## Fossil_fuels_electricity_installed_capacity_MK                    0.152  0.113
## Fossil_fuels_electricity_net_generation_BKWH                     -0.026  0.061
## Geothermal_electricity_installed_capacity_MK                     -0.042  0.021
## Geothermal_electricity_net_generation_BKWH                        0.044 -0.023
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.042  0.004
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.045 -0.002
## Hydroelectricity_installed_capacity_MK                            0.313 -0.093
## Hydroelectricity_net_generation_BKWH                             -0.279 -0.003
## Non.hydro_renewable_electricity_installed_capacity_MK            -0.268 -0.073
## Non.hydro_renewable_electricity_net_generation_BKWH               0.177  0.020
## Nuclear_electricity_installed_capacity_MK                        -0.085 -0.022
```

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## Nuclear_electricity_net_generation_BKWH                          0.024 -0.056
## Renewable_electricity_installed_capacity_MK                      0.099 -0.081
## Renewable_electricity_net_generation_BKWH                       -0.176  0.000
## Solar_electricity_installed_capacity_MK                         -0.348 -0.066
## Solar_electricity_net_generation_BKWH                            0.456  0.040
## Tide_and_wave_electricity_installed_capacity_MK                  0.033 -0.016
## Tide_and_wave_electricity_net_generation_BKWH                   -0.078  0.068
## Wind_electricity_installed_capacity_MK                          -0.241 -0.103
## Wind_electricity_net_generation_BKWH                             0.112  0.058
##                                                                   PC37   PC38
## Bunker_fuel_consumption_TBPD                                     0.110  0.058
## Bunker_residual_fuel_oil_consumption_TBPD                       -0.111 -0.034
## Crude_oil_including_lease_condensate_exports_TBPD                0.237  0.060
## Crude_oil_including_lease_condensate_imports_TBPD               -0.218  0.041
## Crude_oil_including_lease_condensate_production_TBPD            -0.242 -0.072
## Crude_oil_including_lease_condensate_reserves_BB                -0.005 -0.007
## Distillate_fuel_oil_consumption_TBPD                            -0.020  0.390
## Distillate_fuel_oil_production_TBPD                              0.251 -0.628
## Jet_fuel_consumption_TBPD                                        0.142 -0.147
## Jet_fuel_production_TBPD                                        -0.244 -0.027
## Kerosene_consumption_TBPD                                        0.056  0.006
## Kerosene_production_TBPD                                        -0.020  0.022
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 0.056  0.000
## Liquefied_petroleum_gases_.LPG._production_TBPD                 -0.127  0.133
## Motor_gasoline_consumption_TBPD                                 -0.255 -0.213
## Motor_gasoline_production_TBPD                                   0.235  0.248
## Natural_gas_plant_liquids_production_TBPD                        0.021  0.026
## Other_liquids_production_TBPD                                    0.067 -0.041
## Other_refined_products_consumption_TBPD                          0.009  0.046
## Other_refined_products_production_TBPD                           0.211  0.161
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 -0.155 -0.118
## Petroleum_and_other_liquids_consumption_TBPD                    -0.087  0.009
## Petroleum_and_other_liquids_production_TBPD                     -0.175 -0.042
## Refined_petroleum_products_production_TBPD                       0.181 -0.029
## Refinery_processing_gain_TBPD                                    0.259  0.312
## Residual_fuel_oil_consumption_TBPD                               0.052  0.006
## Residual_fuel_oil_production_TBPD                                0.029  0.068
## Biomass_and_waste_electricity_installed_capacity_MK             -0.010  0.034
## Biomass_and_waste_electricity_net_generation_BKWH               -0.102 -0.063
## Electricity_distribution_losses_BKWH                            -0.033 -0.020
## Electricity_exports_BKWH                                         0.027 -0.002
## Electricity_imports_BKWH                                        -0.015  0.025
## Electricity_installed_capacity_MK                                0.036  0.034
## Electricity_net_consumption_BKWH                                -0.054 -0.020
## Electricity_net_generation_BKWH                                 -0.055 -0.020
## Electricity_net_imports_BKWH                                     0.002 -0.021
## Fossil_fuels_electricity_installed_capacity_MK                  -0.020  0.110
## Fossil_fuels_electricity_net_generation_BKWH                    -0.013 -0.065
## Geothermal_electricity_installed_capacity_MK                    -0.027 -0.041
## Geothermal_electricity_net_generation_BKWH                       0.030  0.046
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK   0.017 -0.027
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH     0.023  0.026
## Hydroelectricity_installed_capacity_MK                           0.239 -0.214
## Hydroelectricity_net_generation_BKWH                            -0.171  0.144
```

```
## Non.hydro_renewable_electricity_installed_capacity_MK          -0.006  0.051
## Non.hydro_renewable_electricity_net_generation_BKWH             0.018 -0.008
## Nuclear_electricity_installed_capacity_MK                       0.087 -0.047
## Nuclear_electricity_net_generation_BKWH                        -0.115 -0.005
## Renewable_electricity_installed_capacity_MK                     0.167 -0.144
## Renewable_electricity_net_generation_BKWH                      -0.125  0.123
## Solar_electricity_installed_capacity_MK                         0.208  0.031
## Solar_electricity_net_generation_BKWH                          -0.260 -0.058
## Tide_and_wave_electricity_installed_capacity_MK                 0.055  0.008
## Tide_and_wave_electricity_net_generation_BKWH                  -0.055  0.008
## Wind_electricity_installed_capacity_MK                         -0.068  0.007
## Wind_electricity_net_generation_BKWH                            0.133  0.025
##                                                                  PC39   PC40
## Bunker_fuel_consumption_TBPD                                    0.116  0.044
## Bunker_residual_fuel_oil_consumption_TBPD                      -0.071 -0.063
## Crude_oil_including_lease_condensate_exports_TBPD               0.241 -0.101
## Crude_oil_including_lease_condensate_imports_TBPD              -0.258  0.418
## Crude_oil_including_lease_condensate_production_TBPD           -0.311  0.102
## Crude_oil_including_lease_condensate_reserves_BB               -0.015  0.009
## Distillate_fuel_oil_consumption_TBPD                            0.161 -0.083
## Distillate_fuel_oil_production_TBPD                             0.062 -0.238
## Jet_fuel_consumption_TBPD                                      -0.399  0.065
## Jet_fuel_production_TBPD                                        0.413 -0.185
## Kerosene_consumption_TBPD                                       0.012 -0.036
## Kerosene_production_TBPD                                        0.008  0.040
## Liquefied_petroleum_gases_.LPG._consumption_TBPD               -0.147  0.043
## Liquefied_petroleum_gases_.LPG._production_TBPD                 0.076  0.127
## Motor_gasoline_consumption_TBPD                                 0.200  0.225
## Motor_gasoline_production_TBPD                                 -0.196 -0.095
## Natural_gas_plant_liquids_production_TBPD                       0.153  0.022
## Other_liquids_production_TBPD                                  -0.010  0.006
## Other_refined_products_consumption_TBPD                         0.154 -0.105
## Other_refined_products_production_TBPD                         -0.168  0.009
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 0.105  0.055
## Petroleum_and_other_liquids_consumption_TBPD                    0.113  0.057
## Petroleum_and_other_liquids_production_TBPD                    -0.228  0.088
## Refined_petroleum_products_production_TBPD                     -0.054 -0.114
## Refinery_processing_gain_TBPD                                  -0.009  0.050
## Residual_fuel_oil_consumption_TBPD                             -0.034  0.029
## Residual_fuel_oil_production_TBPD                               0.089 -0.013
## Biomass_and_waste_electricity_installed_capacity_MK            -0.038 -0.091
## Biomass_and_waste_electricity_net_generation_BKWH               0.060  0.094
## Electricity_distribution_losses_BKWH                            0.034 -0.033
## Electricity_exports_BKWH                                       -0.026  0.012
## Electricity_imports_BKWH                                        0.010 -0.007
## Electricity_installed_capacity_MK                               0.106  0.181
## Electricity_net_consumption_BKWH                               -0.127 -0.215
## Electricity_net_generation_BKWH                                -0.114 -0.205
## Electricity_net_imports_BKWH                                   -0.021  0.001
## Fossil_fuels_electricity_installed_capacity_MK                  0.114  0.166
## Fossil_fuels_electricity_net_generation_BKWH                   -0.148 -0.240
## Geothermal_electricity_installed_capacity_MK                    0.002 -0.038
## Geothermal_electricity_net_generation_BKWH                     -0.003  0.038
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK  0.047  0.003
```

```
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.023 -0.017
## Hydroelectricity_installed_capacity_MK                           0.088  0.288
## Hydroelectricity_net_generation_BKWH                            -0.076 -0.170
## Non.hydro_renewable_electricity_installed_capacity_MK            0.085  0.082
## Non.hydro_renewable_electricity_net_generation_BKWH              0.000 -0.104
## Nuclear_electricity_installed_capacity_MK                        0.032 -0.022
## Nuclear_electricity_net_generation_BKWH                          0.019  0.028
## Renewable_electricity_installed_capacity_MK                      0.066  0.223
## Renewable_electricity_net_generation_BKWH                       -0.046 -0.171
## Solar_electricity_installed_capacity_MK                          0.019  0.158
## Solar_electricity_net_generation_BKWH                           -0.066 -0.275
## Tide_and_wave_electricity_installed_capacity_MK                  0.035  0.081
## Tide_and_wave_electricity_net_generation_BKWH                   -0.060 -0.062
## Wind_electricity_installed_capacity_MK                           0.086  0.079
## Wind_electricity_net_generation_BKWH                            -0.019 -0.138
##                                                                    PC41   PC42
## Bunker_fuel_consumption_TBPD                                     0.123  0.038
## Bunker_residual_fuel_oil_consumption_TBPD                       -0.088 -0.033
## Crude_oil_including_lease_condensate_exports_TBPD               -0.070 -0.016
## Crude_oil_including_lease_condensate_imports_TBPD                0.112 -0.055
## Crude_oil_including_lease_condensate_production_TBPD             0.081  0.030
## Crude_oil_including_lease_condensate_reserves_BB                 0.011 -0.015
## Distillate_fuel_oil_consumption_TBPD                             0.027 -0.109
## Distillate_fuel_oil_production_TBPD                              0.042  0.253
## Jet_fuel_consumption_TBPD                                       -0.225  0.097
## Jet_fuel_production_TBPD                                         0.013 -0.072
## Kerosene_consumption_TBPD                                       -0.011  0.021
## Kerosene_production_TBPD                                         0.009 -0.025
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 0.039  0.031
## Liquefied_petroleum_gases_.LPG._production_TBPD                  0.074 -0.184
## Motor_gasoline_consumption_TBPD                                  0.160 -0.089
## Motor_gasoline_production_TBPD                                  -0.153 -0.022
## Natural_gas_plant_liquids_production_TBPD                       -0.018 -0.041
## Other_liquids_production_TBPD                                   -0.022  0.012
## Other_refined_products_consumption_TBPD                         -0.059  0.049
## Other_refined_products_production_TBPD                          -0.017  0.038
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 -0.150 -0.076
## Petroleum_and_other_liquids_consumption_TBPD                     0.048 -0.044
## Petroleum_and_other_liquids_production_TBPD                      0.056  0.018
## Refined_petroleum_products_production_TBPD                      -0.054  0.049
## Refinery_processing_gain_TBPD                                   -0.080 -0.034
## Residual_fuel_oil_consumption_TBPD                               0.028  0.003
## Residual_fuel_oil_production_TBPD                               -0.065 -0.021
## Biomass_and_waste_electricity_installed_capacity_MK              0.020  0.047
## Biomass_and_waste_electricity_net_generation_BKWH               -0.062 -0.031
## Electricity_distribution_losses_BKWH                            -0.023  0.035
## Electricity_exports_BKWH                                         0.006  0.035
## Electricity_imports_BKWH                                        -0.007 -0.043
## Electricity_installed_capacity_MK                               -0.002  0.347
## Electricity_net_consumption_BKWH                                 0.021 -0.202
## Electricity_net_generation_BKWH                                  0.017 -0.189
## Electricity_net_imports_BKWH                                     0.009  0.037
## Fossil_fuels_electricity_installed_capacity_MK                   0.053  0.581
## Fossil_fuels_electricity_net_generation_BKWH                    -0.006 -0.315
```

```
## Geothermal_electricity_installed_capacity_MK                     0.002  0.023
## Geothermal_electricity_net_generation_BKWH                      -0.007 -0.023
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.009 -0.003
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.015 -0.006
## Hydroelectricity_installed_capacity_MK                          0.021 -0.246
## Hydroelectricity_net_generation_BKWH                            0.022  0.196
## Non.hydro_renewable_electricity_installed_capacity_MK          -0.258 -0.039
## Non.hydro_renewable_electricity_net_generation_BKWH             0.285 -0.021
## Nuclear_electricity_installed_capacity_MK                      -0.026 -0.063
## Nuclear_electricity_net_generation_BKWH                         0.020  0.036
## Renewable_electricity_installed_capacity_MK                    -0.151 -0.189
## Renewable_electricity_net_generation_BKWH                       0.094  0.163
## Solar_electricity_installed_capacity_MK                         0.284 -0.042
## Solar_electricity_net_generation_BKWH                          -0.209  0.063
## Tide_and_wave_electricity_installed_capacity_MK                -0.058 -0.095
## Tide_and_wave_electricity_net_generation_BKWH                   0.066  0.103
## Wind_electricity_installed_capacity_MK                         -0.474  0.040
## Wind_electricity_net_generation_BKWH                            0.500 -0.040
##                                                                  PC43   PC44
## Bunker_fuel_consumption_TBPD                                    0.048  0.021
## Bunker_residual_fuel_oil_consumption_TBPD                      -0.043 -0.024
## Crude_oil_including_lease_condensate_exports_TBPD              -0.026  0.055
## Crude_oil_including_lease_condensate_imports_TBPD               0.056  0.007
## Crude_oil_including_lease_condensate_production_TBPD            0.034 -0.037
## Crude_oil_including_lease_condensate_reserves_BB                0.006 -0.001
## Distillate_fuel_oil_consumption_TBPD                           -0.130 -0.082
## Distillate_fuel_oil_production_TBPD                            -0.062 -0.115
## Jet_fuel_consumption_TBPD                                      -0.059 -0.119
## Jet_fuel_production_TBPD                                        0.000  0.028
## Kerosene_consumption_TBPD                                      -0.036 -0.013
## Kerosene_production_TBPD                                        0.002  0.006
## Liquefied_petroleum_gases_.LPG._consumption_TBPD               -0.035 -0.026
## Liquefied_petroleum_gases_.LPG._production_TBPD                 0.021  0.082
## Motor_gasoline_consumption_TBPD                                -0.113 -0.156
## Motor_gasoline_production_TBPD                                 -0.081  0.465
## Natural_gas_plant_liquids_production_TBPD                      -0.032  0.019
## Other_liquids_production_TBPD                                  -0.014  0.006
## Other_refined_products_consumption_TBPD                        -0.008 -0.045
## Other_refined_products_production_TBPD                         -0.001  0.048
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 0.476  0.487
## Petroleum_and_other_liquids_consumption_TBPD                   -0.093 -0.098
## Petroleum_and_other_liquids_production_TBPD                     0.030 -0.057
## Refined_petroleum_products_production_TBPD                     -0.050  0.171
## Refinery_processing_gain_TBPD                                   0.159 -0.514
## Residual_fuel_oil_consumption_TBPD                             -0.031 -0.026
## Residual_fuel_oil_production_TBPD                               0.000 -0.088
## Biomass_and_waste_electricity_installed_capacity_MK            -0.009  0.001
## Biomass_and_waste_electricity_net_generation_BKWH               0.039 -0.007
## Electricity_distribution_losses_BKWH                           -0.021 -0.007
## Electricity_exports_BKWH                                       -0.002  0.158
## Electricity_imports_BKWH                                        0.004 -0.154
## Electricity_installed_capacity_MK                               0.058 -0.023
## Electricity_net_consumption_BKWH                               -0.049 -0.037
## Electricity_net_generation_BKWH                                -0.054 -0.040
```

```
## Electricity_net_imports_BKWH                                    -0.015  0.181
## Fossil_fuels_electricity_installed_capacity_MK                   0.048  0.045
## Fossil_fuels_electricity_net_generation_BKWH                     0.026 -0.075
## Geothermal_electricity_installed_capacity_MK                     0.001 -0.003
## Geothermal_electricity_net_generation_BKWH                      -0.004  0.002
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK   0.001  0.013
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    -0.008  0.001
## Hydroelectricity_installed_capacity_MK                          -0.062  0.032
## Hydroelectricity_net_generation_BKWH                             0.029 -0.029
## Non.hydro_renewable_electricity_installed_capacity_MK           -0.053  0.003
## Non.hydro_renewable_electricity_net_generation_BKWH              0.056  0.043
## Nuclear_electricity_installed_capacity_MK                        0.506 -0.125
## Nuclear_electricity_net_generation_BKWH                         -0.611  0.159
## Renewable_electricity_installed_capacity_MK                     -0.014  0.024
## Renewable_electricity_net_generation_BKWH                        0.040 -0.019
## Solar_electricity_installed_capacity_MK                         -0.060  0.046
## Solar_electricity_net_generation_BKWH                            0.057 -0.060
## Tide_and_wave_electricity_installed_capacity_MK                 -0.016  0.002
## Tide_and_wave_electricity_net_generation_BKWH                    0.033 -0.004
## Wind_electricity_installed_capacity_MK                          -0.097 -0.081
## Wind_electricity_net_generation_BKWH                             0.066  0.066
##                                                                   PC45   PC46
## Bunker_fuel_consumption_TBPD                                     0.003  0.026
## Bunker_residual_fuel_oil_consumption_TBPD                       -0.002 -0.033
## Crude_oil_including_lease_condensate_exports_TBPD               -0.016 -0.001
## Crude_oil_including_lease_condensate_imports_TBPD                0.016  0.018
## Crude_oil_including_lease_condensate_production_TBPD             0.019  0.011
## Crude_oil_including_lease_condensate_reserves_BB                -0.004 -0.006
## Distillate_fuel_oil_consumption_TBPD                             0.092  0.064
## Distillate_fuel_oil_production_TBPD                             -0.023 -0.048
## Jet_fuel_consumption_TBPD                                       -0.009  0.030
## Jet_fuel_production_TBPD                                         0.050  0.036
## Kerosene_consumption_TBPD                                        0.000  0.001
## Kerosene_production_TBPD                                         0.005  0.011
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 0.005  0.010
## Liquefied_petroleum_gases_.LPG._production_TBPD                  0.002  0.032
## Motor_gasoline_consumption_TBPD                                  0.165  0.192
## Motor_gasoline_production_TBPD                                   0.075  0.366
## Natural_gas_plant_liquids_production_TBPD                        0.007 -0.001
## Other_liquids_production_TBPD                                   -0.013  0.002
## Other_refined_products_consumption_TBPD                          0.022  0.037
## Other_refined_products_production_TBPD                          -0.054 -0.012
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 -0.359 -0.378
## Petroleum_and_other_liquids_consumption_TBPD                     0.087  0.114
## Petroleum_and_other_liquids_production_TBPD                      0.007 -0.012
## Refined_petroleum_products_production_TBPD                       0.016  0.139
## Refinery_processing_gain_TBPD                                   -0.078 -0.469
## Residual_fuel_oil_consumption_TBPD                               0.021  0.029
## Residual_fuel_oil_production_TBPD                               -0.002 -0.080
## Biomass_and_waste_electricity_installed_capacity_MK              0.014 -0.032
## Biomass_and_waste_electricity_net_generation_BKWH                0.059  0.001
## Electricity_distribution_losses_BKWH                             0.008  0.036
## Electricity_exports_BKWH                                         0.455 -0.264
## Electricity_imports_BKWH                                        -0.420  0.253
```

```
## Electricity_installed_capacity_MK                                  0.044  0.046
## Electricity_net_consumption_BKWH                                   0.012 -0.087
## Electricity_net_generation_BKWH                                   -0.004 -0.054
## Electricity_net_imports_BKWH                                       0.499 -0.296
## Fossil_fuels_electricity_installed_capacity_MK                    -0.066  0.007
## Fossil_fuels_electricity_net_generation_BKWH                       0.022 -0.029
## Geothermal_electricity_installed_capacity_MK                       0.002 -0.007
## Geothermal_electricity_net_generation_BKWH                         0.010  0.004
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK     0.015 -0.006
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH       0.010 -0.002
## Hydroelectricity_installed_capacity_MK                             0.051  0.024
## Hydroelectricity_net_generation_BKWH                               0.022  0.010
## Non.hydro_renewable_electricity_installed_capacity_MK             -0.033  0.040
## Non.hydro_renewable_electricity_net_generation_BKWH               -0.181  0.063
## Nuclear_electricity_installed_capacity_MK                          0.203  0.268
## Nuclear_electricity_net_generation_BKWH                           -0.239 -0.298
## Renewable_electricity_installed_capacity_MK                       -0.082 -0.032
## Renewable_electricity_net_generation_BKWH                         -0.040 -0.027
## Solar_electricity_installed_capacity_MK                           -0.025 -0.029
## Solar_electricity_net_generation_BKWH                              0.051  0.023
## Tide_and_wave_electricity_installed_capacity_MK                    0.007 -0.001
## Tide_and_wave_electricity_net_generation_BKWH                     -0.001  0.000
## Wind_electricity_installed_capacity_MK                             0.121  0.029
## Wind_electricity_net_generation_BKWH                               0.054 -0.074
##                                                                     PC47   PC48
## Bunker_fuel_consumption_TBPD                                       0.003  0.007
## Bunker_residual_fuel_oil_consumption_TBPD                         -0.002 -0.005
## Crude_oil_including_lease_condensate_exports_TBPD                  0.010 -0.004
## Crude_oil_including_lease_condensate_imports_TBPD                  0.000  0.008
## Crude_oil_including_lease_condensate_production_TBPD              -0.014 -0.003
## Crude_oil_including_lease_condensate_reserves_BB                   0.001  0.000
## Distillate_fuel_oil_consumption_TBPD                              -0.004  0.023
## Distillate_fuel_oil_production_TBPD                               -0.031 -0.005
## Jet_fuel_consumption_TBPD                                         -0.016 -0.016
## Jet_fuel_production_TBPD                                           0.004  0.020
## Kerosene_consumption_TBPD                                         -0.001  0.001
## Kerosene_production_TBPD                                           0.002  0.002
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                  -0.007 -0.003
## Liquefied_petroleum_gases_.LPG._production_TBPD                    0.019  0.010
## Motor_gasoline_consumption_TBPD                                   -0.020  0.023
## Motor_gasoline_production_TBPD                                     0.027  0.014
## Natural_gas_plant_liquids_production_TBPD                          0.005  0.004
## Other_liquids_production_TBPD                                     -0.002 -0.005
## Other_refined_products_consumption_TBPD                            0.003  0.003
## Other_refined_products_production_TBPD                             0.013 -0.013
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                    0.083 -0.071
## Petroleum_and_other_liquids_consumption_TBPD                      -0.008  0.032
## Petroleum_and_other_liquids_production_TBPD                       -0.010  0.008
## Refined_petroleum_products_production_TBPD                        -0.005 -0.030
## Refinery_processing_gain_TBPD                                     -0.015  0.003
## Residual_fuel_oil_consumption_TBPD                                -0.007  0.003
## Residual_fuel_oil_production_TBPD                                 -0.002  0.004
## Biomass_and_waste_electricity_installed_capacity_MK                0.095  0.023
## Biomass_and_waste_electricity_net_generation_BKWH                  0.065 -0.156
```

```
## Electricity_distribution_losses_BKWH                                -0.003 -0.003
## Electricity_exports_BKWH                                            -0.082  0.090
## Electricity_imports_BKWH                                             0.077 -0.082
## Electricity_installed_capacity_MK                                   -0.070 -0.084
## Electricity_net_consumption_BKWH                                    -0.026 -0.020
## Electricity_net_generation_BKWH                                      0.017  0.036
## Electricity_net_imports_BKWH                                        -0.090  0.099
## Fossil_fuels_electricity_installed_capacity_MK                       0.018  0.051
## Fossil_fuels_electricity_net_generation_BKWH                         0.015 -0.007
## Geothermal_electricity_installed_capacity_MK                         0.018 -0.004
## Geothermal_electricity_net_generation_BKWH                           0.013 -0.025
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.001  0.002
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.001 -0.001
## Hydroelectricity_installed_capacity_MK                           0.105 -0.308
## Hydroelectricity_net_generation_BKWH                            -0.026  0.012
## Non.hydro_renewable_electricity_installed_capacity_MK           -0.722 -0.340
## Non.hydro_renewable_electricity_net_generation_BKWH             -0.235  0.574
## Nuclear_electricity_installed_capacity_MK                       -0.004  0.026
## Nuclear_electricity_net_generation_BKWH                          0.014 -0.027
## Renewable_electricity_installed_capacity_MK                     -0.133  0.495
## Renewable_electricity_net_generation_BKWH                        0.031 -0.021
## Solar_electricity_installed_capacity_MK                          0.206  0.036
## Solar_electricity_net_generation_BKWH                            0.020 -0.071
## Tide_and_wave_electricity_installed_capacity_MK                  0.001  0.001
## Tide_and_wave_electricity_net_generation_BKWH                    0.000 -0.003
## Wind_electricity_installed_capacity_MK                           0.528  0.117
## Wind_electricity_net_generation_BKWH                             0.144 -0.349
##                                                                   PC49   PC50
## Bunker_fuel_consumption_TBPD                                     0.010  0.001
## Bunker_residual_fuel_oil_consumption_TBPD                       -0.007 -0.001
## Crude_oil_including_lease_condensate_exports_TBPD                0.001  0.007
## Crude_oil_including_lease_condensate_imports_TBPD               -0.002 -0.004
## Crude_oil_including_lease_condensate_production_TBPD             0.012 -0.012
## Crude_oil_including_lease_condensate_reserves_BB                -0.001  0.000
## Distillate_fuel_oil_consumption_TBPD                             0.015  0.003
## Distillate_fuel_oil_production_TBPD                              0.008 -0.012
## Jet_fuel_consumption_TBPD                                       -0.014 -0.003
## Jet_fuel_production_TBPD                                         0.011 -0.005
## Kerosene_consumption_TBPD                                        0.002 -0.001
## Kerosene_production_TBPD                                         0.001 -0.002
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 0.003 -0.001
## Liquefied_petroleum_gases_.LPG._production_TBPD                  0.005  0.001
## Motor_gasoline_consumption_TBPD                                  0.018 -0.010
## Motor_gasoline_production_TBPD                                   0.019 -0.013
## Natural_gas_plant_liquids_production_TBPD                        0.002  0.001
## Other_liquids_production_TBPD                                    0.002 -0.001
## Other_refined_products_consumption_TBPD                          0.004  0.009
## Other_refined_products_production_TBPD                          -0.008  0.001
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                 -0.037  0.059
## Petroleum_and_other_liquids_consumption_TBPD                    -0.007 -0.062
## Petroleum_and_other_liquids_production_TBPD                     -0.016  0.002
## Refined_petroleum_products_production_TBPD                       0.003  0.031
## Refinery_processing_gain_TBPD                                   -0.033  0.010
## Residual_fuel_oil_consumption_TBPD                               0.003 -0.001
```

```
## Residual_fuel_oil_production_TBPD                                -0.006 -0.001
## Biomass_and_waste_electricity_installed_capacity_MK              -0.019  0.011
## Biomass_and_waste_electricity_net_generation_BKWH                 0.144 -0.020
## Electricity_distribution_losses_BKWH                             -0.003  0.001
## Electricity_exports_BKWH                                         -0.044 -0.012
## Electricity_imports_BKWH                                          0.040  0.010
## Electricity_installed_capacity_MK                                 0.006  0.686
## Electricity_net_consumption_BKWH                                 -0.056  0.030
## Electricity_net_generation_BKWH                                  -0.035  0.001
## Electricity_net_imports_BKWH                                     -0.047 -0.013
## Fossil_fuels_electricity_installed_capacity_MK                    0.024 -0.474
## Fossil_fuels_electricity_net_generation_BKWH                      0.047 -0.022
## Geothermal_electricity_installed_capacity_MK                      0.001  0.000
## Geothermal_electricity_net_generation_BKWH                        0.026 -0.003
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK    0.000 -0.017
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH     -0.001 -0.001
## Hydroelectricity_installed_capacity_MK                           -0.389 -0.023
## Hydroelectricity_net_generation_BKWH                              0.258  0.289
## Non.hydro_renewable_electricity_installed_capacity_MK            -0.054 -0.093
## Non.hydro_renewable_electricity_net_generation_BKWH              -0.435  0.174
## Nuclear_electricity_installed_capacity_MK                        -0.012 -0.075
## Nuclear_electricity_net_generation_BKWH                           0.023  0.025
## Renewable_electricity_installed_capacity_MK                       0.563 -0.155
## Renewable_electricity_net_generation_BKWH                        -0.292 -0.370
## Solar_electricity_installed_capacity_MK                          -0.031  0.023
## Solar_electricity_net_generation_BKWH                             0.061 -0.011
## Tide_and_wave_electricity_installed_capacity_MK                  -0.005 -0.002
## Tide_and_wave_electricity_net_generation_BKWH                     0.008  0.001
## Wind_electricity_installed_capacity_MK                           -0.140  0.045
## Wind_electricity_net_generation_BKWH                              0.351 -0.036
##                                                                    PC51   PC52
## Bunker_fuel_consumption_TBPD                                      0.002  0.001
## Bunker_residual_fuel_oil_consumption_TBPD                        -0.002  0.000
## Crude_oil_including_lease_condensate_exports_TBPD                -0.002  0.001
## Crude_oil_including_lease_condensate_imports_TBPD                 0.006 -0.002
## Crude_oil_including_lease_condensate_production_TBPD             -0.008  0.014
## Crude_oil_including_lease_condensate_reserves_BB                  0.000  0.000
## Distillate_fuel_oil_consumption_TBPD                              0.014  0.006
## Distillate_fuel_oil_production_TBPD                              -0.011 -0.021
## Jet_fuel_consumption_TBPD                                        -0.001  0.001
## Jet_fuel_production_TBPD                                          0.000 -0.006
## Kerosene_consumption_TBPD                                         0.000  0.001
## Kerosene_production_TBPD                                          0.000 -0.001
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                  0.004  0.003
## Liquefied_petroleum_gases_.LPG._production_TBPD                   0.004 -0.005
## Motor_gasoline_consumption_TBPD                                   0.029  0.011
## Motor_gasoline_production_TBPD                                    0.013 -0.030
## Natural_gas_plant_liquids_production_TBPD                        -0.002  0.003
## Other_liquids_production_TBPD                                    -0.002  0.001
## Other_refined_products_consumption_TBPD                           0.009  0.006
## Other_refined_products_production_TBPD                           -0.005 -0.013
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                  -0.016  0.000
## Petroleum_and_other_liquids_consumption_TBPD                     -0.041 -0.028
## Petroleum_and_other_liquids_production_TBPD                       0.015 -0.019
```

```
## Refined_petroleum_products_production_TBPD                        0.034  0.082
## Refinery_processing_gain_TBPD                                    -0.038 -0.006
## Residual_fuel_oil_consumption_TBPD                                0.004  0.001
## Residual_fuel_oil_production_TBPD                                -0.009 -0.007
## Biomass_and_waste_electricity_installed_capacity_MK              -0.009 -0.002
## Biomass_and_waste_electricity_net_generation_BKWH                 0.006  0.009
## Electricity_distribution_losses_BKWH                              0.005  0.043
## Electricity_exports_BKWH                                         -0.006 -0.014
## Electricity_imports_BKWH                                          0.006  0.013
## Electricity_installed_capacity_MK                                 0.412 -0.021
## Electricity_net_consumption_BKWH                                  0.050  0.584
## Electricity_net_generation_BKWH                                  -0.083  0.345
## Electricity_net_imports_BKWH                                     -0.008 -0.027
## Fossil_fuels_electricity_installed_capacity_MK                   -0.281  0.019
## Fossil_fuels_electricity_net_generation_BKWH                      0.017 -0.697
## Geothermal_electricity_installed_capacity_MK                     -0.003  0.001
## Geothermal_electricity_net_generation_BKWH                        0.002  0.001
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.011  0.000
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.001 -0.002
## Hydroelectricity_installed_capacity_MK                           -0.212 -0.028
## Hydroelectricity_net_generation_BKWH                             -0.480 -0.097
## Non.hydro_renewable_electricity_installed_capacity_MK            -0.060 -0.016
## Non.hydro_renewable_electricity_net_generation_BKWH              -0.177 -0.058
## Nuclear_electricity_installed_capacity_MK                        -0.031 -0.005
## Nuclear_electricity_net_generation_BKWH                           0.006 -0.123
## Renewable_electricity_installed_capacity_MK                       0.215  0.051
## Renewable_electricity_net_generation_BKWH                         0.611 -0.090
## Solar_electricity_installed_capacity_MK                          -0.014  0.000
## Solar_electricity_net_generation_BKWH                             0.003  0.003
## Tide_and_wave_electricity_installed_capacity_MK                  -0.002 -0.001
## Tide_and_wave_electricity_net_generation_BKWH                     0.002  0.000
## Wind_electricity_installed_capacity_MK                           -0.058 -0.006
## Wind_electricity_net_generation_BKWH                              0.022  0.020
##                                                                    PC53   PC54
## Bunker_fuel_consumption_TBPD                                      0.000  0.000
## Bunker_residual_fuel_oil_consumption_TBPD                         0.000 -0.001
## Crude_oil_including_lease_condensate_exports_TBPD                -0.002  0.000
## Crude_oil_including_lease_condensate_imports_TBPD                 0.001  0.000
## Crude_oil_including_lease_condensate_production_TBPD              0.033 -0.075
## Crude_oil_including_lease_condensate_reserves_BB                  0.000  0.000
## Distillate_fuel_oil_consumption_TBPD                             -0.021  0.100
## Distillate_fuel_oil_production_TBPD                               0.015 -0.210
## Jet_fuel_consumption_TBPD                                        -0.005  0.029
## Jet_fuel_production_TBPD                                          0.002 -0.055
## Kerosene_consumption_TBPD                                        -0.001  0.007
## Kerosene_production_TBPD                                          0.001 -0.013
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                 -0.005  0.035
## Liquefied_petroleum_gases_.LPG._production_TBPD                   0.001 -0.038
## Motor_gasoline_consumption_TBPD                                  -0.034  0.166
## Motor_gasoline_production_TBPD                                    0.018 -0.317
## Natural_gas_plant_liquids_production_TBPD                         0.005 -0.014
## Other_liquids_production_TBPD                                     0.003 -0.006
## Other_refined_products_consumption_TBPD                          -0.017  0.087
## Other_refined_products_production_TBPD                            0.009 -0.137
```

```
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                  0.007 -0.003
## Petroleum_and_other_liquids_consumption_TBPD                     0.071 -0.408
## Petroleum_and_other_liquids_production_TBPD                     -0.037  0.092
## Refined_petroleum_products_production_TBPD                      -0.049  0.758
## Refinery_processing_gain_TBPD                                    0.006 -0.004
## Residual_fuel_oil_consumption_TBPD                              -0.004  0.023
## Residual_fuel_oil_production_TBPD                                0.004 -0.059
## Biomass_and_waste_electricity_installed_capacity_MK             -0.004  0.001
## Biomass_and_waste_electricity_net_generation_BKWH                0.005 -0.004
## Electricity_distribution_losses_BKWH                            -0.043 -0.009
## Electricity_exports_BKWH                                         0.002  0.005
## Electricity_imports_BKWH                                        -0.002 -0.005
## Electricity_installed_capacity_MK                                0.065 -0.056
## Electricity_net_consumption_BKWH                                -0.588 -0.117
## Electricity_net_generation_BKWH                                  0.782  0.047
## Electricity_net_imports_BKWH                                     0.014  0.008
## Fossil_fuels_electricity_installed_capacity_MK                  -0.043  0.039
## Fossil_fuels_electricity_net_generation_BKWH                    -0.112  0.053
## Geothermal_electricity_installed_capacity_MK                     0.000  0.000
## Geothermal_electricity_net_generation_BKWH                       0.001 -0.001
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.002  0.002
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH   -0.001  0.000
## Hydroelectricity_installed_capacity_MK                          -0.012  0.007
## Hydroelectricity_net_generation_BKWH                            -0.060  0.011
## Non.hydro_renewable_electricity_installed_capacity_MK            0.025 -0.005
## Non.hydro_renewable_electricity_net_generation_BKWH             -0.038  0.018
## Nuclear_electricity_installed_capacity_MK                       -0.007  0.006
## Nuclear_electricity_net_generation_BKWH                         -0.018  0.009
## Renewable_electricity_installed_capacity_MK                      0.001  0.009
## Renewable_electricity_net_generation_BKWH                        0.043  0.000
## Solar_electricity_installed_capacity_MK                         -0.008  0.002
## Solar_electricity_net_generation_BKWH                            0.002 -0.003
## Tide_and_wave_electricity_installed_capacity_MK                  0.000  0.001
## Tide_and_wave_electricity_net_generation_BKWH                    0.000 -0.001
## Wind_electricity_installed_capacity_MK                          -0.026  0.008
## Wind_electricity_net_generation_BKWH                             0.014 -0.011
##                                                                   PC55   PC56
## Bunker_fuel_consumption_TBPD                                     0.000  0.001
## Bunker_residual_fuel_oil_consumption_TBPD                        0.000 -0.001
## Crude_oil_including_lease_condensate_exports_TBPD                0.000  0.000
## Crude_oil_including_lease_condensate_imports_TBPD                0.001  0.000
## Crude_oil_including_lease_condensate_production_TBPD             0.005 -0.617
## Crude_oil_including_lease_condensate_reserves_BB                 0.000  0.000
## Distillate_fuel_oil_consumption_TBPD                            -0.191 -0.015
## Distillate_fuel_oil_production_TBPD                             -0.116  0.024
## Jet_fuel_consumption_TBPD                                       -0.053 -0.005
## Jet_fuel_production_TBPD                                        -0.032  0.006
## Kerosene_consumption_TBPD                                       -0.012 -0.001
## Kerosene_production_TBPD                                        -0.008  0.002
## Liquefied_petroleum_gases_.LPG._consumption_TBPD                -0.066 -0.005
## Liquefied_petroleum_gases_.LPG._production_TBPD                 -0.021  0.005
## Motor_gasoline_consumption_TBPD                                 -0.312 -0.024
## Motor_gasoline_production_TBPD                                  -0.179  0.039
## Natural_gas_plant_liquids_production_TBPD                        0.001 -0.117
```

```
## Other_liquids_production_TBPD                                    0.000 -0.050
## Other_refined_products_consumption_TBPD                         -0.164 -0.013
## Other_refined_products_production_TBPD                          -0.075  0.017
## Petroleum_and_other_liquids_CO2_emissions_MMTCD                  0.000  0.001
## Petroleum_and_other_liquids_consumption_TBPD                     0.773  0.059
## Petroleum_and_other_liquids_production_TBPD                     -0.006  0.764
## Refined_petroleum_products_production_TBPD                       0.418 -0.089
## Refinery_processing_gain_TBPD                                    0.005 -0.039
## Residual_fuel_oil_consumption_TBPD                              -0.042 -0.003
## Residual_fuel_oil_production_TBPD                               -0.032  0.006
## Biomass_and_waste_electricity_installed_capacity_MK              0.001 -0.001
## Biomass_and_waste_electricity_net_generation_BKWH                0.002  0.005
## Electricity_distribution_losses_BKWH                             0.001  0.000
## Electricity_exports_BKWH                                         0.003  0.001
## Electricity_imports_BKWH                                        -0.003 -0.001
## Electricity_installed_capacity_MK                                0.032 -0.003
## Electricity_net_consumption_BKWH                                 0.021 -0.004
## Electricity_net_generation_BKWH                                 -0.046  0.042
## Electricity_net_imports_BKWH                                     0.003  0.001
## Fossil_fuels_electricity_installed_capacity_MK                  -0.021  0.002
## Fossil_fuels_electricity_net_generation_BKWH                     0.017 -0.028
## Geothermal_electricity_installed_capacity_MK                     0.000  0.000
## Geothermal_electricity_net_generation_BKWH                       0.000  0.001
## Hydroelectric_pumped_storage_electricity_installed_capacity_MK -0.001  0.000
## Hydroelectric_pumped_storage_electricity_net_generation_BKWH    0.000  0.000
## Hydroelectricity_installed_capacity_MK                          -0.014 -0.002
## Hydroelectricity_net_generation_BKWH                             0.003  0.004
## Non.hydro_renewable_electricity_installed_capacity_MK           -0.009  0.005
## Non.hydro_renewable_electricity_net_generation_BKWH             -0.008 -0.017
## Nuclear_electricity_installed_capacity_MK                       -0.002 -0.001
## Nuclear_electricity_net_generation_BKWH                          0.002 -0.004
## Renewable_electricity_installed_capacity_MK                      0.010  0.005
## Renewable_electricity_net_generation_BKWH                        0.003 -0.013
## Solar_electricity_installed_capacity_MK                          0.000 -0.001
## Solar_electricity_net_generation_BKWH                            0.001  0.002
## Tide_and_wave_electricity_installed_capacity_MK                  0.000  0.000
## Tide_and_wave_electricity_net_generation_BKWH                    0.000  0.000
## Wind_electricity_installed_capacity_MK                           0.001 -0.006
## Wind_electricity_net_generation_BKWH                             0.005  0.013
```

```
round(acp_escalado$x, 2)
```

```
##                      PC1    PC2   PC3   PC4    PC5   PC6   PC7   PC8   PC9
## Albania            -2.15  -0.25  0.48 -0.60  -0.02  0.00  0.01 -0.09  0.36
## Algeria            -1.24   0.85 -0.45 -0.47   0.26 -0.33  0.00 -0.33  0.64
## Angola             -1.84   0.81 -0.71 -0.39   0.24 -0.32  0.14 -0.29  0.16
## Argentina          -0.58   0.22  0.07 -0.60  -0.28  0.57  0.51  0.10  0.44
## Australia           0.10  -0.68  0.35 -0.45  -0.33 -0.15 -0.21 -0.18  0.22
## Austria            -0.52  -0.78  1.25 -0.01   0.52 -0.09  0.94  1.86 -0.59
## Azerbaijan         -1.94   0.22 -0.09 -0.50   0.18 -0.19 -0.10 -0.27  0.34
## Bahrain            -1.94  -0.09  0.27 -0.48  -0.01 -0.01 -0.24 -0.27  0.55
## Bangladesh         -2.02  -0.29  0.37 -0.50  -0.22  0.00 -0.14 -0.11  0.39
## Belarus            -1.72  -0.12  0.38 -0.39   0.06  0.00  0.14  0.15  0.43
## Belgium            -0.14   0.03  0.96  0.07  -0.74  1.23  1.10  1.14 -0.02
```

```
## Bolivia             -2.14  -0.24  0.38 -0.56   0.03 -0.04 -0.24 -0.23  0.43
## BosniaHerzegovina   -2.12  -0.39  0.38 -0.35   0.33 -0.07 -0.35  0.07  0.14
## Brazil               6.92  -0.19 -0.54 -3.02   0.53  1.77  5.51  2.77 -0.44
## Brunei              -2.21  -0.19  0.34 -0.53   0.03 -0.04 -0.23 -0.23  0.39
## Bulgaria            -1.72  -0.60  0.34  0.13   0.80 -0.20 -0.77  0.38 -0.12
## Burma               -2.14  -0.33  0.36 -0.59   0.05 -0.02 -0.24 -0.25  0.39
## Cameroon            -2.15  -0.23  0.37 -0.56   0.01 -0.02 -0.18 -0.22  0.40
## Canada               4.38   0.23 -4.14  1.04   5.60 -1.78 -1.40  2.19 -3.33
## Chile               -1.48  -0.46  0.29 -0.62  -0.05  0.05 -0.08 -0.10  0.35
## China               34.38 -12.83 -7.51 -4.22   1.74  1.75  0.10 -2.49 -0.53
## Colombia            -1.39   0.18 -0.50 -0.53   0.20 -0.12  0.00 -0.25  0.30
## Congo-Brazzaville   -2.19  -0.12  0.25 -0.53   0.06 -0.10 -0.19 -0.24  0.38
## CostaRica           -2.03  -0.13  0.68 -0.59  -0.08 -0.15 -0.28 -0.63 -0.19
## Croatia             -1.94  -0.27  0.63 -0.43   0.22  0.01  0.21  0.44  0.05
## Cuba                -1.94  -0.17  0.18 -0.40  -0.19  0.29 -0.22 -0.19  0.35
## CzechRepublic       -1.26  -0.83  0.57  0.65   1.53 -0.35 -0.86  1.31 -0.73
## CoteDiViore         -2.15  -0.28  0.35 -0.49   0.04 -0.04 -0.31 -0.21  0.40
## Denmark             -1.29  -0.40  0.71 -0.41   0.37  0.09  0.44  0.89 -0.25
## DominicanRepublic   -2.10  -0.26  0.33 -0.52  -0.05  0.07 -0.28 -0.26  0.40
## Ecuador             -1.80   0.15 -0.14 -0.45  -0.01  0.04 -0.04 -0.24  0.31
## Egypt               -0.86   0.30 -0.76 -0.19  -0.40  0.71 -0.32 -0.45  0.33
## ElSalvador          -2.04  -0.09  0.72 -0.58  -0.13 -0.14 -0.25 -0.62 -0.19
## Estonia             -2.09  -0.34  0.45 -0.34   0.34  0.01 -0.30  0.18  0.10
## Ethiopia            -2.14  -0.35  0.34 -0.50   0.01 -0.05 -0.32 -0.18  0.36
## Finland             -1.09  -0.09  1.30 -0.76  -0.11  0.19  1.55  1.23  0.11
## France               4.00  -2.54  1.81  9.31   7.54 -0.55 -2.94  0.51 -0.11
## Gabon               -2.18  -0.13  0.26 -0.52   0.02 -0.06 -0.17 -0.22  0.37
## Germany              1.68  -2.71  1.66  3.97   1.62 -0.17  2.09  0.11 -0.77
## Ghana               -2.14  -0.27  0.30 -0.54   0.07 -0.06 -0.25 -0.23  0.36
## Greece              -1.07  -0.30  0.46 -0.31  -0.56  0.43  0.48  0.16  0.20
## Guatemala           -2.04  -0.30  0.46 -0.55   0.06  0.00 -0.15 -0.10  0.25
## Honduras            -2.15  -0.30  0.39 -0.53   0.00  0.03 -0.21 -0.17  0.35
## Hungary             -1.63  -0.06  1.07 -0.55   0.03  0.09  0.92  0.86  0.12
## India                8.75  -2.84 -2.28  0.41  -3.69 -2.14 -0.56  1.40  2.24
## Indonesia            0.73   1.58  1.64 -0.64  -1.33 -1.38 -0.64 -3.16 -2.79
## Iran                 1.12   2.30 -3.91  1.41  -1.08 -0.68 -0.27  0.56 -0.41
## Iraq                -0.53   2.51 -2.70  0.16  -0.42 -1.22  1.60  0.42 -0.41
## Ireland             -1.86  -0.39  0.45 -0.47  -0.19 -0.15 -0.17  0.01  0.39
## Israel              -1.79  -0.39  0.24 -0.25   0.13 -0.15 -0.56 -0.13  0.37
## Italy                4.34  -2.24  4.11  4.40  -1.89 -0.41  6.94 -1.32 -2.67
## Jamaica             -2.17  -0.24  0.34 -0.50  -0.05  0.10 -0.27 -0.24  0.38
## Japan                8.51  -3.65 -2.19  5.13 -10.10 -4.88 -2.70  1.73 -0.35
## Jordan              -2.07  -0.23  0.32 -0.49  -0.10  0.07 -0.26 -0.23  0.41
## Kazakhstan          -1.46   0.81 -0.89 -0.34   0.39 -0.50  0.12 -0.19  0.16
## Kenya               -1.81   0.25  1.18 -0.58  -0.45 -0.45 -0.48 -1.46 -1.13
## Kuwait              -0.81   2.06 -2.66  0.71  -0.97 -1.30  0.18  0.36 -0.17
## Latvia              -2.11  -0.27  0.53 -0.50   0.09  0.05  0.01  0.11  0.22
## Libya               -1.86   0.29 -0.29 -0.40   0.02 -0.27 -0.09 -0.20  0.18
## Lithuania           -1.87  -0.17  0.70 -0.55  -0.14  0.03  0.31  0.17  0.40
## Malaysia            -0.94   0.14  0.04 -0.61  -0.02  0.12 -0.13 -0.21  0.61
## Mexico               1.39   2.13 -0.44 -0.58  -0.21 -0.87 -0.13 -2.21 -1.44
## Morocco             -1.72  -0.17  0.47 -0.56  -0.23  0.15  0.16 -0.03  0.42
## Netherlands          0.56   0.55  0.44  0.65  -0.82  2.70  0.95  2.01 -0.65
## NetherlandsAntilles -2.02  -0.12  0.20 -0.36  -0.17  0.39 -0.30 -0.24  0.41
```

```
## NewZealand          -1.32   0.50  1.65 -0.77  -0.52 -0.65 -0.54 -2.30 -2.15
## Nicaragua           -2.10  -0.16  0.60 -0.57  -0.09 -0.11 -0.28 -0.53  0.01
## Nigeria             -1.50   1.17 -1.35  0.00  -0.36 -1.11  0.06  0.03  0.13
## Norway              -0.72  -0.14 -1.17 -0.03   1.96 -0.59 -0.55  0.58 -0.94
## Oman                -1.83   0.41 -0.27 -0.40   0.05  0.08 -0.09 -0.29  0.35
## Pakistan            -1.38  -0.33 -0.21 -0.39  -0.31  0.39 -0.24 -0.28  0.25
## PapuaNewGuinea      -2.20  -0.27  0.36 -0.52  -0.02 -0.02 -0.26 -0.23  0.40
## Peru                -1.74  -0.19  0.24 -0.58   0.01 -0.01 -0.21 -0.23  0.43
## Philippines         -0.75   1.20  2.48 -0.72  -1.18 -1.11 -0.87 -3.79 -3.84
## Poland              -0.70  -0.50  0.66 -0.34   0.34 -0.05  0.27  0.80  0.08
## Portugal            -1.18  -0.59  0.52 -0.44   0.22  0.06  0.02  0.35 -0.02
## Qatar               -1.34   1.03 -0.17 -0.61   0.31 -0.25 -0.03 -0.17  0.43
## Romania             -1.43  -0.73  0.27 -0.20   0.63 -0.14 -0.59  0.15 -0.06
## Russia               8.67   6.78 -8.19  1.24   2.55  0.45  1.05 -1.41  2.64
## SaudiArabia          3.66   8.54 -8.55  1.69  -0.47 -2.56  1.57 -0.04 -0.96
## Senegal             -2.19  -0.26  0.40 -0.53  -0.02  0.03 -0.24 -0.22  0.40
## Serbia              -1.91  -0.36  0.51 -0.41   0.23 -0.02 -0.05  0.21  0.18
## Singapore            0.87   2.30 -2.30  2.87  -4.19 11.12 -2.56  0.54 -1.95
## Slovakia            -1.75  -0.40  0.77  0.11   0.58 -0.06  0.19  0.64 -0.07
## Slovenia            -2.05  -0.39  0.50 -0.21   0.51 -0.07 -0.27  0.39 -0.05
## SouthAfrica         -0.48  -0.35  0.57  0.17   0.31  0.19 -0.55  0.77  0.08
## SouthKorea           3.47  -1.12  1.29  8.84   0.02  1.64  2.31 -3.88  3.55
## Spain                3.14  -2.16 -0.27  1.82  -1.80  0.02 -1.00  2.24 -1.38
## SriLanka            -2.09  -0.27  0.31 -0.49  -0.08  0.11 -0.29 -0.22  0.38
## Sudan               -2.05  -0.22  0.27 -0.56   0.04 -0.02 -0.19 -0.23  0.39
## Suriname            -2.23  -0.27  0.41 -0.55   0.00 -0.01 -0.22 -0.22  0.40
## Sweden              -0.14  -0.95  0.53  0.34   2.02  0.26 -0.33  1.98 -0.98
## Switzerland         -0.83  -0.93  1.03  0.93   1.34 -0.49 -0.24  2.37 -1.20
## Syria               -2.03  -0.23  0.22 -0.50  -0.03  0.05 -0.24 -0.28  0.42
## Taiwan              -0.21  -0.20  0.31 -0.14  -0.25  0.41 -0.63 -0.27  0.95
## Thailand             0.50   0.31  0.47 -0.43  -0.64  0.43  0.79  0.60  0.72
## TrinidadandTobago   -2.10  -0.13  0.29 -0.48  -0.03  0.05 -0.25 -0.24  0.45
## Tunisia             -2.10  -0.25  0.34 -0.51   0.00 -0.03 -0.25 -0.21  0.40
## Turkey              -0.18  -0.45  0.67 -0.13   0.08  0.09  0.42 -1.15 -0.37
## Turkmenistan        -1.97  -0.15  0.16 -0.35   0.13 -0.09 -0.44 -0.15  0.37
## Ukraine             -1.28  -0.46  0.42  0.08   0.97 -0.22 -1.14  0.00  0.43
## UnitedArabEmirates   0.05   3.49 -2.85  0.93  -0.83  3.15 -0.41  0.05 -1.08
## UnitedKingdom        3.09  -0.56  1.14  0.04  -1.52 -0.73  1.56  2.37  0.38
## UnitedStates        44.17   9.42  9.81 -1.52   0.29 -0.35 -1.67  0.74  0.85
## Uruguay             -2.05  -0.36  0.36 -0.53   0.11  0.03 -0.24 -0.13  0.31
## Uzbekistan          -1.98  -0.35  0.42 -0.38   0.33 -0.06 -0.17  0.22  0.11
## Venezuela           -0.24   2.65 -3.59 -0.09   0.89 -1.25  1.00 -0.17 -1.02
## Vietnam             -1.44  -0.41 -0.05 -0.77   0.20  0.06 -0.02 -0.25  0.21
## Yemen               -2.09  -0.17  0.27 -0.49  -0.05 -0.05 -0.24 -0.24  0.41
## Zambia              -2.17  -0.37  0.33 -0.53   0.12 -0.03 -0.30 -0.20  0.33
## Zimbabwe            -2.18  -0.32  0.42 -0.53   0.07 -0.04 -0.21 -0.14  0.34
##                      PC10  PC11  PC12  PC13  PC14  PC15  PC16  PC17  PC18  PC19
## Albania             -0.20 -0.06  0.05  0.01  0.01 -0.03 -0.04  0.09  0.11  0.10
## Algeria             -0.09 -0.22 -0.10 -0.20 -0.09  0.34 -0.26 -0.09 -0.07  0.08
## Angola              -0.54 -0.21 -0.13 -0.74 -0.03  0.45  0.56  0.45  0.05  0.02
## Argentina            0.16 -0.22  0.45  0.15  0.49 -0.25 -0.01 -0.18  0.40  0.19
## Australia           -0.81 -0.44 -0.05 -0.34 -0.78  0.32 -0.02 -0.40 -0.14  0.19
## Austria              0.34 -1.12  0.76  0.43  1.20 -0.29  0.57  0.55 -0.71 -0.35
## Azerbaijan          -0.37 -0.08 -0.04 -0.30 -0.11  0.21  0.15  0.25 -0.03  0.01
```

```
## Bahrain             -0.15 -0.04  0.02 -0.05 -0.10  0.05 -0.18  0.12 -0.09 -0.05
## Bangladesh          -0.18  0.00  0.02  0.02 -0.02 -0.04  0.04 -0.01  0.17  0.03
## Belarus              0.28 -0.33  0.00  0.39  0.20  0.01 -0.26 -0.12  0.10 -0.09
## Belgium              0.23 -0.86 -0.02  0.59  0.30 -0.11  0.25 -0.04  0.17  0.37
## Bolivia             -0.22  0.02  0.02 -0.12 -0.15 -0.07 -0.10  0.02  0.00  0.02
## BosniaHerzegovina   -0.05 -0.08  0.09  0.00  0.03  0.08 -0.15 -0.03 -0.01 -0.01
## Brazil               2.62  3.49  1.22 -2.03 -0.16 -1.39  0.01  0.01 -0.06  0.22
## Brunei              -0.27 -0.01  0.01 -0.14 -0.14 -0.02 -0.04  0.07 -0.01  0.04
## Bulgaria             0.11 -0.39  0.05  0.04 -0.02  0.10 -0.18 -0.15 -0.20 -0.09
## Burma               -0.23  0.05  0.05 -0.11 -0.13 -0.07 -0.09  0.06  0.06  0.06
## Cameroon            -0.23 -0.01  0.02 -0.09 -0.10 -0.03 -0.05  0.07  0.03  0.05
## Canada              -0.20  1.71  0.84  0.25 -0.40  1.09 -1.50  0.64  0.66 -0.17
## Chile               -0.09  0.33 -0.02 -0.26 -0.27 -0.10 -0.04 -0.14 -0.09  0.02
## China               -1.22 -0.84  0.20  0.14  0.44 -0.08  0.39 -0.05 -0.09 -0.08
## Colombia             0.02 -0.01  0.06 -0.52 -0.19  0.10  0.01  0.40  0.17  0.00
## Congo-Brazzaville   -0.28 -0.02  0.00 -0.17 -0.14  0.00  0.00  0.11 -0.01  0.03
## CostaRica            0.04  0.07 -0.03 -0.06 -0.05 -0.08 -0.04  0.11  0.01  0.02
## Croatia              0.05 -0.30  0.12  0.30  0.34  0.15 -0.11  0.04  0.21  0.03
## Cuba                -0.08  0.10  0.07 -0.16 -0.11 -0.20  0.00 -0.30  0.03 -0.11
## CzechRepublic        0.47 -0.43  0.04  0.09  0.08  0.45 -0.20 -0.44 -0.41 -0.21
## CoteDiViore         -0.19  0.03  0.02 -0.09 -0.12 -0.03 -0.11  0.02 -0.02  0.01
## Denmark              0.05 -0.21 -0.44  0.12  0.06  0.18  0.08 -0.10 -0.02 -0.40
## DominicanRepublic   -0.22  0.03  0.04 -0.09 -0.09 -0.08 -0.04 -0.11  0.02  0.06
## Ecuador             -0.16 -0.05  0.06 -0.18 -0.11 -0.07  0.10 -0.09  0.10  0.01
## Egypt                0.12  0.00  0.10 -0.05  0.16 -0.22  0.19 -1.19  0.42 -0.04
## ElSalvador           0.07  0.11 -0.04 -0.10 -0.07 -0.10  0.00  0.05 -0.07  0.01
## Estonia             -0.05 -0.01  0.01 -0.04 -0.01  0.07 -0.11 -0.08 -0.07 -0.09
## Ethiopia            -0.18  0.08  0.08 -0.10 -0.19  0.00 -0.05  0.09  0.08 -0.02
## Finland              0.32 -0.07  0.02  0.39  0.41  0.03  0.34  0.05  0.21  0.15
## France               1.00  0.54 -0.23 -0.37 -0.19 -0.90  1.34 -0.36  0.23  0.73
## Gabon               -0.26 -0.03  0.00 -0.16 -0.12 -0.01  0.01  0.11  0.01  0.03
## Germany             -0.96 -0.46 -0.99 -0.77 -1.52  0.55 -0.19 -0.97 -0.67 -0.75
## Ghana               -0.23  0.04  0.04 -0.14 -0.13 -0.01 -0.04  0.07  0.03  0.06
## Greece              -0.10 -0.79 -0.07  0.31 -0.21 -0.05 -0.28 -0.20  0.31  0.16
## Guatemala           -0.10  0.18  0.00 -0.21 -0.20 -0.07 -0.01 -0.05 -0.14 -0.01
## Honduras            -0.21  0.04  0.03 -0.09 -0.10 -0.07 -0.02 -0.05  0.05  0.02
## Hungary              0.13 -0.35  0.08  0.54  0.60  0.15  0.11  0.08  0.35  0.17
## India                2.24  3.25 -2.55  1.47  0.00  1.67 -0.01  0.03 -0.09  0.54
## Indonesia            2.06  0.99 -0.34 -0.01  0.40 -0.16  0.22 -0.25 -0.31 -0.19
## Iran                 0.13  0.65  0.40  1.42 -0.28 -0.90 -0.01 -0.96  0.78 -1.05
## Iraq                -0.80 -0.03  0.11  1.04  0.02 -0.30  1.04  0.29  0.42 -0.09
## Ireland             -0.18 -0.06  0.00  0.06 -0.06 -0.03  0.16  0.17  0.13 -0.24
## Israel              -0.03 -0.07  0.01 -0.01 -0.22  0.11 -0.28 -0.16 -0.08 -0.13
## Italy               -0.54 -1.96 -0.30  0.26 -0.96  0.69 -0.46 -0.01  0.46  0.54
## Jamaica             -0.22  0.01  0.04 -0.11 -0.12 -0.12 -0.05 -0.07  0.04  0.01
## Japan                1.06 -0.77  2.27 -0.88 -0.52 -0.24 -0.14  0.01 -0.04  0.11
## Jordan              -0.20  0.02  0.04 -0.06 -0.07 -0.11 -0.03 -0.15  0.08  0.01
## Kazakhstan          -0.38 -0.22 -0.04 -0.22 -0.10  0.35  0.30  0.25  0.01 -0.08
## Kenya                0.56  0.23 -0.11  0.00 -0.02 -0.09  0.07  0.26 -0.05 -0.05
## Kuwait              -1.02  0.31 -1.06 -0.31  0.63 -1.17 -0.67  0.65 -0.09 -0.07
## Latvia              -0.11 -0.10  0.05  0.06  0.06  0.03 -0.05  0.04  0.08  0.04
## Libya               -0.59  0.22  0.13  0.35 -0.32 -0.22  0.08 -0.04 -0.06  0.01
## Lithuania           -0.11 -0.40  0.16  0.27  0.29 -0.11  0.04  0.21  0.03  0.02
## Malaysia             0.00  0.25  0.00 -0.23 -0.26  0.10 -0.09 -0.14 -0.26  0.04
```

```
## Mexico               1.21  0.12 -0.21 -0.30  0.55  0.39  0.22 -1.00  0.30 -0.03
## Morocco             -0.11 -0.20  0.08  0.20  0.26 -0.06  0.06 -0.14  0.09  0.17
## Netherlands          0.98 -0.61 -0.07  1.17  1.10  0.59 -0.32 -0.38  0.15 -0.21
## NetherlandsAntilles -0.05 -0.05  0.01 -0.03 -0.16 -0.16 -0.16 -0.08  0.00 -0.05
## NewZealand           0.99  0.36 -0.25  0.02  0.10 -0.15  0.02  0.45 -0.10 -0.07
## Nicaragua           -0.04  0.07 -0.04 -0.09 -0.10 -0.11 -0.02  0.04 -0.04  0.00
## Nigeria             -0.58  0.23  0.26 -0.57 -0.41  0.83  1.00  0.91  0.42 -0.44
## Norway               0.17 -0.15  0.47 -0.84  0.42  0.91 -0.49  0.64  0.16 -0.03
## Oman                -0.24 -0.19 -0.09 -0.34 -0.17  0.17  0.18  0.22  0.00 -0.03
## Pakistan            -0.03  0.24  0.20 -0.09  0.08 -0.33  0.11 -0.66  0.55 -0.05
## PapuaNewGuinea      -0.23  0.02  0.02 -0.12 -0.11 -0.10 -0.07  0.01  0.03  0.02
## Peru                -0.15  0.11  0.05 -0.17 -0.15  0.02 -0.21 -0.07  0.01  0.15
## Philippines          1.97  0.50 -0.35  0.24  0.37 -0.25  0.29  0.44 -0.30 -0.04
## Poland               0.28 -0.26 -0.06  0.44  0.54  0.18  0.13 -0.28 -0.34 -0.19
## Portugal            -0.06 -0.21 -0.18  0.10  0.10 -0.02  0.01 -0.01 -0.10 -0.21
## Qatar               -0.76  0.15  0.29 -0.58 -0.08  0.18 -0.25  0.11 -0.18  0.10
## Romania             -0.12 -0.21 -0.16 -0.02 -0.17  0.15 -0.23 -0.10  0.03 -0.06
## Russia               5.17 -3.26 -0.42 -0.05 -0.84 -0.25 -0.42  0.47 -0.11 -0.05
## SaudiArabia         -3.04  0.26 -0.50 -1.30  1.53  0.47  0.47 -0.98 -0.34  0.33
## Senegal             -0.23  0.00  0.02 -0.09 -0.11 -0.08 -0.06  0.00  0.02  0.04
## Serbia              -0.03 -0.27  0.17  0.17  0.28  0.00 -0.01  0.12  0.00 -0.01
## Singapore            0.10  0.53  0.05 -0.04 -0.46  0.19  0.09  0.03  0.03  0.08
## Slovakia             0.16 -0.29  0.15  0.28  0.35  0.19 -0.05 -0.07  0.14  0.09
## Slovenia             0.05 -0.20  0.07  0.13  0.15  0.18 -0.15 -0.07  0.07 -0.02
## SouthAfrica          0.10 -0.45  0.36  0.28  0.54 -0.16  0.20  0.01 -0.25 -0.34
## SouthKorea          -0.91  2.12  0.82  0.30  1.43  0.03 -0.64  0.74 -0.16 -0.59
## Spain               -1.13 -1.16 -3.19 -0.77  0.74 -1.91 -0.91  0.62  0.17  0.02
## SriLanka            -0.19  0.04  0.04 -0.10 -0.14 -0.08 -0.06  0.00  0.06  0.02
## Sudan               -0.20  0.08  0.03 -0.09 -0.17 -0.09 -0.07  0.01 -0.02  0.07
## Suriname            -0.23 -0.01  0.03 -0.10 -0.12 -0.08 -0.07  0.02  0.02  0.03
## Sweden               1.21  0.80 -0.40 -0.35 -0.27  0.13 -0.02 -0.34 -0.55 -0.45
## Switzerland          0.76 -1.47  1.18  0.67  1.21  0.39  0.13  0.19 -0.53  0.32
## Syria               -0.13  0.01  0.04 -0.02 -0.11 -0.15 -0.08 -0.16  0.05  0.02
## Taiwan              -0.05 -0.09  0.37  0.10  0.17 -0.19  0.00 -0.11 -0.58  0.11
## Thailand             0.48  0.18 -0.01  0.06  0.29  0.34 -0.49 -0.67 -0.77  0.42
## TrinidadandTobago   -0.13 -0.06  0.01 -0.09 -0.17 -0.09 -0.15  0.01 -0.01 -0.02
## Tunisia             -0.20  0.00 -0.01 -0.07 -0.09 -0.03 -0.04 -0.02  0.00  0.04
## Turkey               0.49  0.47 -0.20  0.50  0.54  0.07  0.24  0.00  0.43  0.28
## Turkmenistan        -0.05 -0.01 -0.01 -0.12 -0.12  0.03 -0.15 -0.14 -0.02 -0.06
## Ukraine              0.04 -0.14 -0.01  0.01 -0.19 -0.30  0.40  0.04  0.17  0.52
## UnitedArabEmirates  -1.54  0.05  0.05 -0.26 -0.56  0.95  0.56  0.70 -0.35  0.28
## UnitedKingdom        0.23  0.10 -1.10  0.05 -0.71  0.14  1.64  0.64  0.02 -0.71
## UnitedStates        -1.14 -0.05  0.19  0.08 -0.20  0.00 -0.22  0.02  0.10 -0.06
## Uruguay             -0.15  0.13 -0.01 -0.18 -0.20 -0.07 -0.08 -0.01 -0.08 -0.03
## Uzbekistan           0.01 -0.12  0.05  0.19  0.16  0.17 -0.11 -0.03  0.11 -0.02
## Venezuela           -1.93  1.14  0.71  2.99 -1.55 -1.39 -0.12  0.24 -0.81  0.43
## Vietnam             -0.13  0.14  0.24 -0.08  0.03  0.05 -0.08  0.19  0.31  0.25
## Yemen               -0.23  0.04  0.02 -0.10 -0.10 -0.06 -0.02 -0.07  0.02  0.05
## Zambia              -0.18  0.05  0.05 -0.12 -0.13 -0.05 -0.13  0.07  0.04  0.04
## Zimbabwe            -0.18  0.02  0.04 -0.08 -0.10 -0.03 -0.07  0.04  0.02  0.03
##                      PC20  PC21  PC22  PC23  PC24  PC25  PC26  PC27  PC28  PC29
## Albania              0.01  0.01  0.01  0.05 -0.01 -0.01  0.05  0.00  0.05  0.01
## Algeria             -0.24 -0.18 -0.04  0.29 -0.10  0.01  0.09 -0.12 -0.15  0.08
## Angola               0.01  0.16  0.01 -0.42  0.10  0.16 -0.11  0.01 -0.05 -0.05
```

```
## Argentina           -0.22 -0.06  0.00  0.23 -0.01 -0.06 -0.05 -0.13  0.01 -0.01
## Australia            0.20 -0.18 -0.41 -0.37  0.07 -0.35 -0.27 -0.46 -0.01 -0.02
## Austria             -0.27 -0.73  0.29  0.12  0.28 -0.17  0.03  0.32  0.09 -0.04
## Azerbaijan          -0.03  0.12  0.02 -0.19  0.00  0.05 -0.04  0.00  0.03  0.00
## Bahrain              0.09  0.09 -0.07 -0.05  0.00  0.05 -0.11  0.03 -0.14  0.04
## Bangladesh          -0.06  0.03  0.04  0.05 -0.04 -0.11  0.01  0.01  0.02  0.00
## Belarus              0.09  0.16 -0.06 -0.04  0.11  0.12 -0.13  0.02  0.00 -0.07
## Belgium              0.44  0.28  0.01  0.27  0.35  0.10 -0.01  0.02  0.01  0.05
## Bolivia              0.00 -0.01  0.02  0.04 -0.02 -0.01  0.04 -0.04  0.05 -0.01
## BosniaHerzegovina   -0.07  0.05  0.01 -0.02 -0.06  0.01  0.04 -0.03  0.03  0.01
## Brazil              -0.32  0.09 -0.10 -0.05  0.01  0.14 -0.06 -0.06  0.01  0.02
## Brunei               0.00  0.02  0.04  0.03 -0.02  0.02  0.03 -0.01  0.04  0.01
## Bulgaria            -0.17  0.07  0.05 -0.01  0.13  0.04 -0.03  0.07  0.01  0.05
## Burma               -0.01 -0.04  0.02  0.01  0.00 -0.02  0.03 -0.03  0.07  0.01
## Cameroon             0.02  0.00  0.02  0.01  0.00  0.02  0.00 -0.01  0.03  0.01
## Canada               0.50 -0.18  0.01  0.10  0.27 -0.07 -0.19 -0.10  0.06  0.03
## Chile                0.19 -0.12  0.05 -0.04 -0.03 -0.04  0.02  0.03  0.07  0.06
## China                0.06  0.16  0.01  0.04 -0.06 -0.03  0.05 -0.01 -0.03  0.01
## Colombia            -0.02 -0.03 -0.12 -0.26  0.14  0.02 -0.01  0.00  0.07 -0.02
## Congo-Brazzaville    0.01  0.03  0.03 -0.01 -0.01  0.04  0.01 -0.01  0.03  0.00
## CostaRica            0.02  0.02  0.04  0.07 -0.03  0.03  0.03 -0.03  0.03  0.05
## Croatia             -0.02  0.08 -0.06  0.02 -0.07 -0.02  0.08  0.00  0.02 -0.01
## Cuba                 0.09 -0.04  0.15  0.02 -0.03  0.03 -0.11  0.04 -0.02 -0.08
## CzechRepublic       -0.22  0.44  0.09 -0.05  0.05 -0.11 -0.07  0.07  0.03  0.17
## CoteDiViore          0.01  0.01  0.01  0.01  0.00  0.03  0.00  0.00  0.00  0.01
## Denmark              0.17  0.00 -0.07 -0.02 -0.19  0.15  0.17 -0.09 -0.04 -0.10
## DominicanRepublic    0.03 -0.05  0.04  0.01 -0.01  0.00  0.01  0.02 -0.01  0.00
## Ecuador              0.05 -0.08  0.01 -0.13  0.07  0.07  0.00  0.00 -0.02  0.00
## Egypt               -0.04 -0.42  0.20 -0.18 -0.04  0.02 -0.26  0.16 -0.31  0.02
## ElSalvador           0.05  0.02  0.09  0.06 -0.03  0.00  0.05  0.00  0.01  0.04
## Estonia             -0.09  0.11  0.00  0.00 -0.05  0.00  0.06 -0.01  0.01  0.00
## Ethiopia            -0.01 -0.04 -0.01  0.07  0.02  0.00  0.05 -0.05  0.04 -0.02
## Finland              0.52  0.19  0.12  0.11 -0.29 -0.08  0.06  0.17  0.13  0.17
## France              -0.05  0.15 -0.12  0.08  0.03  0.01  0.05 -0.03 -0.06  0.00
## Gabon                0.01  0.04  0.02 -0.01  0.00  0.04  0.03 -0.01  0.04  0.00
## Germany             -0.30  0.01 -0.36  0.04 -0.13  0.17 -0.12  0.21  0.20  0.14
## Ghana               -0.01 -0.01  0.02 -0.01  0.03  0.01  0.03 -0.02  0.05  0.01
## Greece               0.15  0.12 -0.16  0.00  0.16  0.18 -0.17  0.11 -0.18 -0.09
## Guatemala            0.05 -0.02  0.10  0.00 -0.04 -0.03  0.08  0.01  0.04  0.04
## Honduras             0.00 -0.03  0.07  0.03 -0.04  0.02  0.00  0.01  0.02  0.01
## Hungary              0.16  0.25 -0.01  0.09 -0.08 -0.11  0.15  0.04 -0.01  0.00
## India               -0.53 -0.06  0.29 -0.03  0.01  0.08 -0.09  0.04  0.06 -0.02
## Indonesia           -0.08  0.13 -0.16 -0.24  0.39 -0.28  0.32 -0.21 -0.11  0.08
## Iran                -0.47  0.06  0.19  0.17 -0.25 -0.24  0.16 -0.23  0.16 -0.17
## Iraq                -0.03  0.22 -0.06 -0.38  0.16  0.08 -0.12 -0.05  0.02 -0.11
## Ireland             -0.01 -0.12 -0.04  0.19  0.05  0.10  0.14 -0.07  0.00 -0.03
## Israel              -0.01  0.16 -0.06 -0.04  0.08 -0.02 -0.05  0.00 -0.01  0.01
## Italy               -0.22  0.02  0.46 -0.04  0.05 -0.15  0.06 -0.03  0.01 -0.08
## Jamaica              0.01 -0.02  0.06  0.03 -0.03  0.03 -0.03  0.00  0.00  0.00
## Japan                0.22  0.05 -0.18  0.04 -0.04  0.07  0.05  0.11 -0.03  0.03
## Jordan               0.01 -0.04  0.05 -0.01 -0.01 -0.02 -0.05  0.00 -0.02 -0.02
## Kazakhstan          -0.10  0.20  0.05 -0.24 -0.06  0.05 -0.04  0.09 -0.02 -0.06
## Kenya                0.02  0.09  0.05  0.17 -0.03  0.04  0.01 -0.04  0.02  0.06
## Kuwait               0.02  0.44 -0.01 -0.36 -0.49 -0.40 -0.17  0.17 -0.09  0.33
```

```
## Latvia                0.00  0.05 -0.01  0.04 -0.04 -0.03  0.08  0.00  0.05  0.01
## Libya                -0.05  0.01  0.05  0.03 -0.03 -0.03  0.03 -0.04 -0.01 -0.02
## Lithuania             0.01 -0.07  0.00  0.09  0.05  0.05 -0.01  0.07  0.01  0.01
## Malaysia             -0.05 -0.09 -0.13 -0.32  0.11 -0.38  0.10 -0.04 -0.04 -0.22
## Mexico                0.08 -0.59  0.11 -0.22  0.15  0.21  0.17  0.07 -0.07  0.33
## Morocco               0.08 -0.15 -0.06  0.02  0.07  0.03  0.09 -0.01 -0.05 -0.01
## Netherlands           0.37  0.55 -0.47 -0.01  0.07  0.11 -0.03  0.01 -0.25  0.09
## NetherlandsAntilles   0.08  0.00  0.02  0.01  0.04  0.14 -0.07  0.02 -0.01 -0.01
## NewZealand            0.14  0.06  0.06  0.13 -0.01  0.06  0.01 -0.09 -0.06  0.29
## Nicaragua             0.04  0.02  0.05  0.08 -0.08  0.07 -0.05  0.03  0.06 -0.08
## Nigeria              -0.13  0.32 -0.05 -0.23  0.20  0.21  0.28  0.08 -0.08 -0.10
## Norway               -0.24 -0.46  0.01  0.09  0.01 -0.03 -0.03  0.20 -0.08  0.00
## Oman                 -0.03  0.09  0.00 -0.24  0.03  0.09 -0.02 -0.01  0.04  0.00
## Pakistan              0.03 -0.27  0.31 -0.12 -0.07 -0.09 -0.39  0.21 -0.26 -0.08
## PapuaNewGuinea        0.00  0.01  0.05  0.04 -0.03  0.01  0.02 -0.02  0.03  0.01
## Peru                  0.03 -0.20  0.00  0.16  0.02 -0.04  0.04  0.01  0.02  0.02
## Philippines           0.08  0.22  0.03  0.29 -0.27  0.11 -0.44  0.07  0.07 -0.35
## Poland               -0.05  0.02 -0.04 -0.07 -0.06  0.24  0.02 -0.13 -0.02  0.08
## Portugal              0.11 -0.23 -0.05  0.01 -0.06  0.32  0.06 -0.14 -0.10 -0.02
## Qatar                -0.66  0.29 -0.23  0.61 -0.09  0.12 -0.03 -0.18 -0.30 -0.07
## Romania              -0.10  0.06  0.00  0.02  0.05  0.13  0.03 -0.01  0.01 -0.04
## Russia                0.04 -0.06 -0.02  0.04 -0.03  0.00  0.00  0.01  0.03  0.00
## SaudiArabia           0.19  0.09  0.08  0.27  0.07  0.09  0.01 -0.03  0.19 -0.05
## Senegal               0.01 -0.01  0.04  0.04 -0.01  0.00  0.01 -0.02  0.03 -0.01
## Serbia               -0.13 -0.07  0.05  0.06  0.07 -0.05  0.02  0.08  0.06  0.05
## Singapore            -0.09 -0.06  0.11 -0.08  0.07  0.07  0.02 -0.03  0.19  0.06
## Slovakia             -0.05  0.20 -0.25 -0.02 -0.21  0.03  0.08  0.10  0.10 -0.01
## Slovenia             -0.08  0.14 -0.02 -0.01 -0.04 -0.04  0.06 -0.01  0.02  0.04
## SouthAfrica          -0.97  0.35 -0.12  0.08  0.58 -0.30 -0.22  0.15  0.00  0.02
## SouthKorea            0.25 -0.12  0.06 -0.05  0.02  0.02 -0.01 -0.09 -0.08 -0.01
## Spain                -0.10 -0.31 -0.06 -0.06  0.23  0.15  0.10 -0.10 -0.05 -0.15
## SriLanka              0.01 -0.04  0.01  0.03  0.00 -0.01  0.02 -0.02  0.02 -0.02
## Sudan                 0.00 -0.04  0.05  0.00  0.00  0.00  0.04 -0.01  0.05 -0.03
## Suriname              0.01  0.01  0.04  0.04 -0.03  0.02  0.03 -0.01  0.04  0.01
## Sweden                0.70  0.14  0.42 -0.19 -0.24 -0.10  0.33  0.23 -0.16 -0.31
## Switzerland          -0.45 -0.33  0.17 -0.45 -0.53  0.17 -0.09 -0.47  0.04 -0.01
## Syria                 0.01 -0.05  0.06  0.02  0.00  0.03 -0.05  0.02 -0.02 -0.01
## Taiwan                0.58  0.15  0.26 -0.04  0.21  0.02 -0.35 -0.08  0.31 -0.05
## Thailand              0.47 -0.04 -0.30  0.12  0.19 -0.30  0.13 -0.07 -0.14 -0.20
## TrinidadandTobago     0.03 -0.02 -0.01  0.07  0.00  0.04  0.00 -0.02  0.00  0.01
## Tunisia               0.00  0.00  0.01  0.02  0.01  0.01  0.04 -0.03  0.04  0.01
## Turkey               -0.01 -0.47 -1.17 -0.05 -0.30 -0.09  0.02  0.30  0.24 -0.10
## Turkmenistan         -0.07 -0.02  0.01 -0.01 -0.02  0.02  0.05 -0.01  0.02  0.01
## Ukraine               0.06  0.23  0.44  0.14 -0.03 -0.22  0.06  0.01  0.00  0.11
## UnitedArabEmirates   -0.22 -0.15 -0.05  0.17 -0.34 -0.27  0.03  0.11 -0.31 -0.04
## UnitedKingdom         0.61 -0.44 -0.14  0.40  0.02 -0.25 -0.25 -0.26 -0.07  0.18
## UnitedStates         -0.06 -0.01  0.03 -0.07 -0.04  0.03  0.01  0.05  0.02 -0.01
## Uruguay               0.05 -0.03  0.07  0.02 -0.05  0.04  0.02  0.01  0.05  0.01
## Uzbekistan           -0.14  0.11  0.01 -0.03 -0.07 -0.10  0.02  0.04  0.04  0.05
## Venezuela             0.01 -0.22 -0.12 -0.09  0.11  0.22  0.10  0.14 -0.13  0.09
## Vietnam               0.00 -0.15 -0.02 -0.09  0.07 -0.22  0.12  0.04  0.16 -0.02
## Yemen                 0.00 -0.03  0.02 -0.01  0.02  0.03  0.06 -0.02  0.01  0.02
## Zambia               -0.01 -0.02  0.04  0.03  0.00  0.01  0.01  0.00  0.04  0.02
## Zimbabwe             -0.01  0.01  0.03  0.03 -0.01 -0.01  0.05 -0.01  0.04  0.00
```

```
##                       PC30  PC31  PC32  PC33  PC34  PC35  PC36  PC37  PC38  PC39
## Albania              -0.02  0.04  0.01 -0.01 -0.01 -0.02  0.00  0.00  0.00  0.00
## Algeria               0.35  0.11 -0.09 -0.22  0.08  0.08 -0.02 -0.05  0.05 -0.02
## Angola                0.12  0.04 -0.04  0.04 -0.05  0.04 -0.05  0.01  0.03 -0.05
## Argentina            -0.03  0.17  0.03  0.03  0.02  0.01  0.04  0.09  0.05  0.03
## Australia             0.12  0.03 -0.01 -0.12 -0.13 -0.02  0.06  0.00 -0.02  0.00
## Austria               0.23  0.15  0.03  0.02 -0.10 -0.03 -0.02 -0.06  0.00  0.02
## Azerbaijan            0.03  0.07 -0.02  0.04 -0.05  0.04 -0.05  0.03 -0.04 -0.04
## Bahrain               0.10  0.01  0.03 -0.07 -0.02  0.03 -0.10 -0.04 -0.04  0.07
## Bangladesh           -0.04  0.00  0.02 -0.01  0.04  0.00  0.02 -0.01  0.03 -0.02
## Belarus               0.15 -0.17  0.05 -0.05  0.07 -0.04  0.12  0.03  0.00 -0.04
## Belgium               0.29  0.16 -0.03 -0.07  0.04 -0.06  0.02  0.04  0.02 -0.03
## Bolivia               0.02  0.02  0.00 -0.01 -0.02 -0.01  0.03 -0.01  0.00  0.01
## BosniaHerzegovina    -0.04  0.01 -0.01  0.00 -0.02  0.02  0.01  0.02  0.00 -0.01
## Brazil               -0.01  0.05 -0.02  0.02  0.00  0.03  0.04  0.01 -0.01  0.01
## Brunei                0.00  0.01  0.00 -0.01 -0.01 -0.01  0.02  0.00 -0.01 -0.01
## Bulgaria             -0.07  0.04 -0.03 -0.02 -0.03  0.08 -0.02  0.03 -0.01 -0.01
## Burma                 0.00  0.02 -0.01 -0.01 -0.01  0.00  0.02 -0.01 -0.01  0.02
## Cameroon              0.00  0.01  0.00  0.00 -0.01 -0.01  0.00 -0.01  0.00  0.00
## Canada               -0.06  0.06  0.01  0.00  0.05 -0.02  0.01 -0.06  0.02  0.00
## Chile                 0.07 -0.15 -0.13  0.03 -0.02  0.10  0.02 -0.06  0.02  0.00
## China                 0.01  0.00 -0.01  0.00  0.00 -0.01 -0.01  0.00  0.00 -0.01
## Colombia              0.05 -0.16 -0.09  0.01  0.05 -0.03 -0.12  0.06  0.01  0.07
## Congo-Brazzaville     0.01  0.01  0.00 -0.01 -0.02  0.00  0.02  0.00 -0.01 -0.01
## CostaRica            -0.02  0.00  0.04  0.05 -0.01  0.00  0.00  0.01  0.00  0.00
## Croatia              -0.07  0.02 -0.01  0.04  0.00 -0.04  0.01  0.01  0.03  0.00
## Cuba                  0.00  0.01  0.12 -0.02 -0.01  0.00  0.04 -0.01  0.02  0.01
## CzechRepublic        -0.10  0.12 -0.11  0.04 -0.06  0.20 -0.01  0.01  0.01 -0.02
## CoteDiViore           0.01 -0.01  0.00  0.00 -0.04  0.00 -0.03 -0.02 -0.01  0.01
## Denmark              -0.10 -0.01  0.05 -0.05  0.04 -0.02 -0.01  0.03  0.01  0.02
## DominicanRepublic    -0.08  0.00 -0.02  0.05  0.00 -0.01 -0.02  0.01 -0.01 -0.02
## Ecuador               0.00 -0.04 -0.06 -0.01  0.01 -0.02 -0.01 -0.02  0.02  0.01
## Egypt                -0.09 -0.09 -0.08  0.08 -0.01  0.00 -0.03 -0.03  0.04 -0.02
## ElSalvador            0.00  0.02  0.09  0.06 -0.02 -0.01  0.01  0.00  0.01 -0.02
## Estonia              -0.08  0.00  0.00 -0.01 -0.03  0.01  0.03  0.00  0.01 -0.01
## Ethiopia             -0.04  0.01 -0.01  0.02  0.01 -0.01 -0.02  0.02  0.01  0.00
## Finland               0.04 -0.24 -0.10  0.06  0.04  0.02 -0.05 -0.10 -0.06 -0.02
## France                0.11 -0.06 -0.03  0.05  0.00 -0.02 -0.02  0.01 -0.01 -0.01
## Gabon                 0.01  0.01  0.00  0.00 -0.01 -0.01  0.00  0.01  0.00  0.00
## Germany               0.04  0.14  0.02  0.05  0.06 -0.11 -0.01 -0.03  0.04  0.04
## Ghana                -0.01  0.01 -0.02  0.00 -0.01 -0.01  0.01 -0.01  0.01  0.00
## Greece                0.16  0.04  0.10  0.00  0.00  0.01  0.02 -0.01 -0.07  0.03
## Guatemala             0.00 -0.02  0.06  0.04  0.00  0.01  0.00 -0.02  0.01  0.01
## Honduras             -0.03  0.03  0.01  0.02 -0.01 -0.01  0.01 -0.01  0.00  0.00
## Hungary              -0.02  0.11  0.00  0.01  0.05 -0.05  0.01 -0.01  0.01 -0.03
## India                 0.06  0.01  0.04  0.02 -0.02 -0.03 -0.01  0.00  0.00  0.01
## Indonesia             0.12 -0.07  0.04  0.03  0.00 -0.04  0.05 -0.03  0.02  0.02
## Iran                  0.25 -0.01 -0.03  0.06  0.00  0.05 -0.05  0.04 -0.06  0.01
## Iraq                 -0.08  0.06  0.14 -0.10  0.04  0.06 -0.09 -0.07  0.12 -0.01
## Ireland              -0.05 -0.03 -0.03 -0.01  0.03 -0.03  0.01  0.02  0.01  0.02
## Israel                0.02 -0.03 -0.03  0.01 -0.03  0.05  0.07  0.01 -0.02 -0.02
## Italy                -0.03 -0.11  0.02 -0.03 -0.01  0.04 -0.02  0.02 -0.01 -0.01
## Jamaica              -0.03  0.00  0.02  0.02  0.00 -0.01  0.01  0.01 -0.01 -0.01
## Japan                -0.04 -0.01  0.00 -0.01  0.00  0.00  0.01 -0.01  0.01  0.00
```

```
## Jordan                0.00 -0.02 -0.02  0.06  0.00 -0.01  0.00 -0.01  0.02  0.01
## Kazakhstan            0.08 -0.06  0.01  0.05 -0.05  0.02  0.12  0.01  0.00 -0.02
## Kenya                -0.01  0.01  0.07  0.10  0.01  0.00 -0.02  0.02  0.02  0.00
## Kuwait                0.12  0.05 -0.04 -0.10  0.05 -0.09 -0.07  0.06  0.01 -0.01
## Latvia               -0.04  0.01 -0.02  0.01  0.00  0.00  0.00  0.00  0.00  0.00
## Libya                 0.03  0.03 -0.02  0.03  0.01  0.02  0.05 -0.06  0.02  0.02
## Lithuania             0.06 -0.02  0.01  0.02 -0.04 -0.05 -0.01 -0.02  0.00  0.02
## Malaysia             -0.04  0.03 -0.02  0.05  0.02 -0.06 -0.05 -0.07 -0.07 -0.01
## Mexico               -0.14  0.20 -0.04 -0.17  0.11  0.00  0.00  0.01 -0.09 -0.03
## Morocco              -0.11 -0.01 -0.06  0.03  0.00 -0.03 -0.07  0.02  0.03  0.02
## Netherlands          -0.06  0.00 -0.08  0.07 -0.13 -0.04  0.02 -0.02 -0.04  0.06
## NetherlandsAntilles   0.07 -0.07  0.04 -0.07  0.03 -0.01  0.03  0.01  0.02  0.00
## NewZealand            0.10 -0.09  0.34  0.20 -0.05  0.03 -0.03  0.01  0.02  0.00
## Nicaragua            -0.03  0.04 -0.10 -0.05 -0.02 -0.01  0.02 -0.01 -0.01  0.00
## Nigeria               0.02  0.12 -0.08  0.06  0.08 -0.01  0.01  0.00 -0.05  0.02
## Norway                0.17 -0.19 -0.04 -0.05 -0.06  0.02 -0.02  0.11 -0.02 -0.01
## Oman                  0.06 -0.01 -0.02 -0.05 -0.01  0.00  0.02  0.01  0.00 -0.01
## Pakistan              0.02  0.05  0.11  0.13  0.01 -0.03  0.02  0.00  0.00  0.00
## PapuaNewGuinea        0.00  0.02 -0.01  0.02 -0.02 -0.01  0.00  0.00 -0.01 -0.02
## Peru                  0.00 -0.14 -0.05 -0.01  0.04 -0.04 -0.02 -0.01  0.01  0.02
## Philippines          -0.06  0.07 -0.30 -0.07  0.00 -0.03 -0.01  0.00  0.00  0.00
## Poland                0.06 -0.25 -0.16 -0.04  0.09  0.06  0.04 -0.03  0.03  0.02
## Portugal             -0.05 -0.07  0.05 -0.02  0.02  0.01 -0.03  0.05 -0.01  0.03
## Qatar                -0.07 -0.04  0.17 -0.11 -0.04 -0.02 -0.12 -0.08 -0.04 -0.06
## Romania              -0.04  0.07 -0.02  0.07 -0.09  0.03 -0.05  0.05  0.00  0.04
## Russia               -0.08  0.00  0.01  0.03 -0.01 -0.01  0.01 -0.02  0.01  0.00
## SaudiArabia          -0.04 -0.09  0.03  0.01 -0.08 -0.03  0.03  0.01  0.00  0.01
## Senegal              -0.01  0.01  0.00  0.03  0.01 -0.01  0.00  0.01 -0.01 -0.01
## Serbia               -0.03  0.05 -0.03  0.03 -0.03  0.00  0.02  0.00  0.01 -0.02
## Singapore             0.02 -0.03  0.01 -0.08 -0.04 -0.03 -0.05  0.00  0.02 -0.03
## Slovakia              0.04  0.06  0.09  0.07  0.05 -0.03  0.04 -0.01 -0.01 -0.05
## Slovenia             -0.08  0.03 -0.05  0.02 -0.02  0.02  0.00 -0.01  0.02  0.00
## SouthAfrica          -0.21 -0.21  0.05 -0.11  0.10 -0.04  0.03  0.02 -0.01 -0.01
## SouthKorea           -0.07  0.04 -0.02 -0.03 -0.02  0.03  0.03 -0.01  0.00  0.00
## Spain                -0.08  0.04 -0.02  0.06  0.03  0.07  0.04 -0.04 -0.02  0.00
## SriLanka             -0.02 -0.02 -0.02  0.02  0.02 -0.01 -0.01  0.01  0.00  0.00
## Sudan                -0.01  0.03 -0.01  0.06 -0.05 -0.03  0.01  0.00  0.01 -0.01
## Suriname             -0.01  0.01  0.00  0.00 -0.01 -0.01  0.01  0.00 -0.01 -0.01
## Sweden               -0.05 -0.06  0.19 -0.22 -0.04  0.00  0.00  0.03  0.01 -0.02
## Switzerland          -0.05  0.01  0.05  0.02  0.04 -0.06  0.01  0.00 -0.01  0.00
## Syria                 0.01 -0.04  0.02 -0.03  0.03  0.01  0.03  0.01  0.00  0.01
## Taiwan                0.00  0.04  0.19  0.04  0.17  0.08 -0.05  0.02 -0.08  0.01
## Thailand              0.03  0.06 -0.08  0.13  0.10  0.03 -0.09  0.06  0.05 -0.03
## TrinidadandTobago     0.03 -0.07  0.03 -0.09  0.04 -0.01  0.03  0.00  0.00  0.01
## Tunisia              -0.04  0.00 -0.02 -0.01  0.02 -0.01  0.01  0.01  0.01 -0.01
## Turkey               -0.03 -0.01  0.17 -0.10 -0.02  0.11  0.01  0.01  0.00  0.01
## Turkmenistan         -0.05 -0.01 -0.06  0.05 -0.05 -0.01 -0.03  0.02  0.00 -0.03
## Ukraine              -0.14  0.08  0.03 -0.22 -0.05  0.03  0.03  0.00  0.00  0.12
## UnitedArabEmirates   -0.02  0.06  0.00  0.15  0.15  0.07  0.10  0.00 -0.02  0.05
## UnitedKingdom        -0.14 -0.04 -0.04 -0.03 -0.03  0.00  0.00  0.04 -0.02 -0.04
## UnitedStates          0.01  0.00  0.00  0.00 -0.01  0.00  0.00  0.00  0.01  0.00
## Uruguay               0.00 -0.03 -0.01  0.01 -0.01 -0.01  0.01 -0.02  0.00  0.01
## Uzbekistan           -0.11  0.04  0.02 -0.05 -0.01  0.02  0.02  0.00  0.00 -0.03
## Venezuela            -0.07 -0.03 -0.06 -0.01 -0.04 -0.01  0.02  0.04 -0.03 -0.02
```

```
## Vietnam             -0.09 -0.06 -0.05  0.02 -0.01 -0.03 -0.02 -0.01 -0.01 -0.01
## Yemen               -0.02  0.03 -0.07  0.03 -0.02 -0.01 -0.01 -0.01  0.01  0.01
## Zambia              -0.02  0.00 -0.01  0.00 -0.02 -0.03  0.01 -0.01  0.00 -0.01
## Zimbabwe            -0.02  0.02  0.00 -0.01 -0.02 -0.02  0.02 -0.01  0.01  0.00
##                      PC40  PC41  PC42  PC43  PC44  PC45  PC46  PC47 PC48 PC49
## Albania              0.00 -0.01 -0.01 -0.01  0.00  0.00  0.00  0.00    0    0
## Algeria             -0.02  0.01  0.01 -0.02  0.00  0.00  0.00  0.00    0    0
## Angola              -0.01  0.00 -0.01  0.00  0.01  0.00  0.00  0.00    0    0
## Argentina           -0.01 -0.05  0.00  0.04  0.01 -0.01  0.00  0.00    0    0
## Australia            0.00  0.02 -0.03 -0.01  0.00 -0.02  0.00  0.00    0    0
## Austria             -0.01  0.00  0.01  0.01  0.00 -0.01  0.00  0.00    0    0
## Azerbaijan          -0.02 -0.01  0.03  0.01 -0.01  0.00  0.00  0.00    0    0
## Bahrain             -0.04  0.00  0.02  0.01 -0.01  0.01  0.00  0.00    0    0
## Bangladesh          -0.04 -0.01 -0.01 -0.01 -0.01  0.00 -0.01  0.00    0    0
## Belarus              0.07  0.00  0.02  0.00  0.00  0.01  0.00  0.00    0    0
## Belgium             -0.01  0.04 -0.04  0.02 -0.01  0.01  0.00  0.00    0    0
## Bolivia             -0.01 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## BosniaHerzegovina    0.01 -0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Brazil              -0.01  0.01  0.00 -0.01  0.00  0.00  0.00  0.00    0    0
## Brunei              -0.01 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Bulgaria             0.00 -0.01 -0.02 -0.01  0.00  0.01  0.00 -0.01    0    0
## Burma                0.02  0.00 -0.02  0.00  0.00  0.00  0.00  0.00    0    0
## Cameroon            -0.01 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Canada              -0.01  0.00  0.00  0.01  0.01  0.00  0.00  0.00    0    0
## Chile                0.03 -0.02 -0.01  0.02  0.01  0.00 -0.01  0.00    0    0
## China                0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Colombia            -0.03 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Congo-Brazzaville   -0.01 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## CostaRica            0.01  0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Croatia              0.01  0.00 -0.02  0.00 -0.01  0.00  0.00  0.00    0    0
## Cuba                 0.00 -0.01  0.02  0.01  0.00  0.00 -0.01  0.00    0    0
## CzechRepublic        0.01  0.03  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## CoteDiViore          0.00  0.00 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Denmark             -0.08  0.06  0.01  0.01  0.01 -0.01  0.00  0.00    0    0
## DominicanRepublic    0.02  0.01  0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Ecuador              0.00 -0.01 -0.01  0.00  0.00  0.00 -0.01  0.00    0    0
## Egypt               -0.03  0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## ElSalvador           0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Estonia             -0.01 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Ethiopia            -0.01 -0.01  0.00 -0.01  0.00  0.00  0.00  0.00    0    0
## Finland              0.00 -0.01 -0.01 -0.01  0.01 -0.01 -0.01  0.00    0    0
## France              -0.01 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Gabon               -0.01 -0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Germany              0.03 -0.01  0.00 -0.01  0.00  0.00  0.00  0.00    0    0
## Ghana                0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Greece              -0.05 -0.04 -0.02 -0.01  0.00  0.00  0.01  0.00    0    0
## Guatemala            0.00  0.01  0.00  0.01  0.00  0.00  0.00  0.00    0    0
## Honduras             0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Hungary             -0.01  0.00  0.01 -0.03  0.00 -0.02  0.00  0.00    0    0
## India               -0.01  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Indonesia           -0.01 -0.01  0.03 -0.01  0.01  0.00  0.00  0.00    0    0
## Iran                 0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Iraq                 0.02 -0.02  0.01 -0.01 -0.01  0.00  0.00  0.00    0    0
## Ireland             -0.01 -0.01  0.00 -0.01 -0.01 -0.01  0.00  0.00    0    0
```

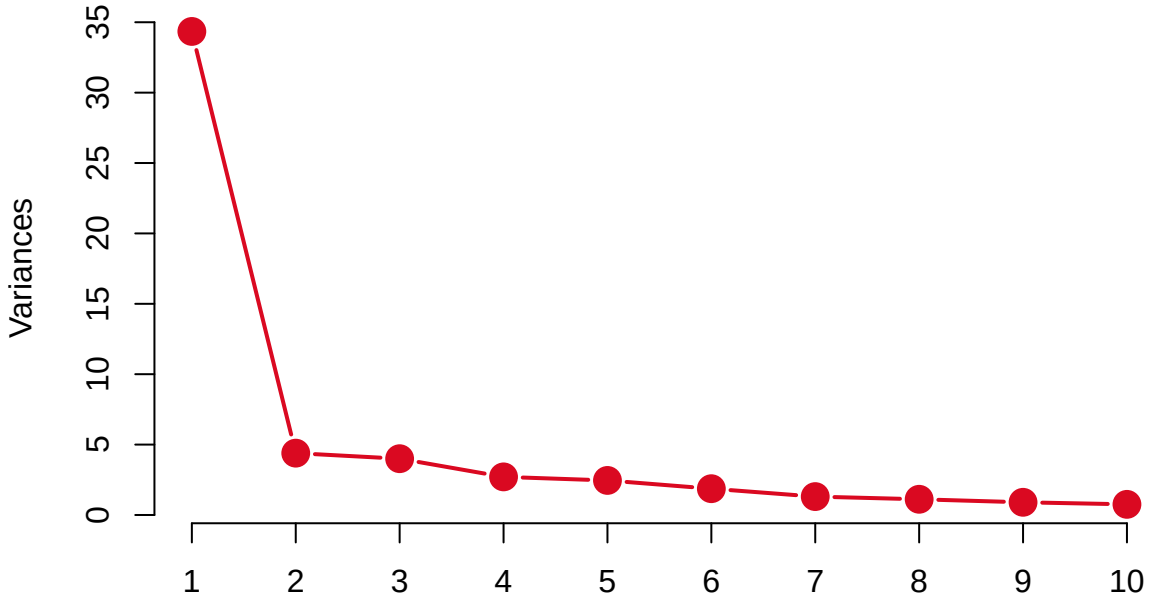
```
## Israel              -0.01 -0.02  0.03  0.01 -0.01  0.00  0.00  0.00    0    0
## Italy                0.00  0.00  0.01  0.01  0.00  0.00  0.00  0.00    0    0
## Jamaica              0.01  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Japan               -0.01  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Jordan               0.01  0.00 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Kazakhstan           0.00  0.01  0.01  0.01  0.00  0.00 -0.01 -0.01    0    0
## Kenya                0.00  0.00 -0.01  0.00  0.00  0.00  0.01  0.00    0    0
## Kuwait               0.01  0.00  0.00  0.01  0.00  0.00  0.00  0.00    0    0
## Latvia               0.01  0.00 -0.02  0.00  0.00  0.00  0.00  0.00    0    0
## Libya                0.00  0.02 -0.02  0.00  0.00  0.00  0.00  0.00    0    0
## Lithuania            0.02  0.00 -0.01  0.00  0.01  0.01  0.01 -0.01    0    0
## Malaysia             0.01  0.01 -0.02  0.02 -0.01  0.01  0.00  0.00    0    0
## Mexico               0.01 -0.02 -0.01  0.01  0.00  0.00  0.00  0.00    0    0
## Morocco              0.02  0.01  0.01  0.00  0.01  0.00  0.00  0.00    0    0
## Netherlands          0.01 -0.02  0.01  0.00  0.00  0.00 -0.01  0.00    0    0
## NetherlandsAntilles  0.02 -0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## NewZealand           0.00  0.02 -0.03  0.00 -0.01  0.00  0.00  0.00    0    0
## Nicaragua           -0.02  0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Nigeria              0.06  0.02  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Norway               0.03  0.00  0.00 -0.02 -0.01  0.00  0.00  0.00    0    0
## Oman                -0.03 -0.02  0.02  0.01  0.02  0.00  0.01  0.00    0    0
## Pakistan             0.04  0.03  0.01 -0.01  0.00  0.00  0.01  0.01    0    0
## PapuaNewGuinea       0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Peru                -0.02  0.00  0.03  0.00 -0.01  0.00  0.00  0.00    0    0
## Philippines          0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Poland               0.01  0.02 -0.01  0.02 -0.01 -0.02  0.01  0.00    0    0
## Portugal            -0.02  0.03  0.00  0.00  0.00  0.01  0.00  0.00    0    0
## Qatar                0.04  0.00  0.00  0.02  0.00 -0.01  0.00  0.00    0    0
## Romania              0.03  0.01 -0.01 -0.01  0.00  0.00  0.00  0.00    0    0
## Russia               0.00  0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## SaudiArabia         -0.01  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Senegal              0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Serbia              -0.01  0.00  0.00  0.00 -0.01  0.00  0.00  0.00    0    0
## Singapore            0.01  0.01  0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Slovakia            -0.02  0.03  0.04  0.00  0.01  0.01  0.00  0.00    0    0
## Slovenia            -0.01 -0.01 -0.01 -0.01  0.00  0.00  0.00  0.00    0    0
## SouthAfrica         -0.03  0.01 -0.02  0.00  0.01  0.00  0.00  0.00    0    0
## SouthKorea           0.00  0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Spain                0.01  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## SriLanka             0.02  0.00 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Sudan               -0.01  0.00  0.01  0.00  0.01  0.00  0.00  0.00    0    0
## Suriname             0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Sweden               0.00 -0.02 -0.03  0.00  0.00  0.00  0.00  0.00    0    0
## Switzerland          0.02 -0.01  0.00 -0.01  0.00  0.01  0.00  0.00    0    0
## Syria                0.04  0.00  0.02  0.00  0.00  0.00  0.00  0.00    0    0
## Taiwan               0.00  0.00  0.01 -0.01  0.00 -0.01  0.00  0.00    0    0
## Thailand             0.04  0.00  0.01  0.00  0.00  0.00  0.00  0.00    0    0
## TrinidadandTobago    0.01 -0.02  0.00  0.00 -0.01  0.00  0.00  0.00    0    0
## Tunisia              0.00  0.00  0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Turkey               0.02  0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Turkmenistan         0.00 -0.01  0.00  0.00  0.00  0.00  0.01  0.00    0    0
## Ukraine              0.05  0.03  0.03  0.00  0.00  0.00  0.00  0.00    0    0
## UnitedArabEmirates  -0.03 -0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## UnitedKingdom        0.01 -0.01  0.01  0.00  0.00  0.01  0.00  0.00    0    0
```

```
## UnitedStates         0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Uruguay              0.00  0.00  0.01  0.01  0.00  0.00  0.00  0.00    0    0
## Uzbekistan          -0.02 -0.01  0.02  0.00 -0.01  0.01  0.01  0.00    0    0
## Venezuela            0.00  0.01  0.00  0.00  0.00  0.00  0.00  0.00    0    0
## Vietnam              0.01  0.02  0.00  0.00 -0.01  0.00  0.00  0.00    0    0
## Yemen                0.00  0.01 -0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Zambia              -0.01  0.00  0.01  0.00  0.00  0.00  0.00  0.00    0    0
## Zimbabwe            -0.02  0.00  0.00  0.00  0.00  0.01  0.00  0.00    0    0
##                     PC50 PC51 PC52 PC53 PC54 PC55 PC56
## Albania                0    0    0    0    0    0    0
## Algeria                0    0    0    0    0    0    0
## Angola                 0    0    0    0    0    0    0
## Argentina              0    0    0    0    0    0    0
## Australia              0    0    0    0    0    0    0
## Austria                0    0    0    0    0    0    0
## Azerbaijan             0    0    0    0    0    0    0
## Bahrain                0    0    0    0    0    0    0
## Bangladesh             0    0    0    0    0    0    0
## Belarus                0    0    0    0    0    0    0
## Belgium                0    0    0    0    0    0    0
## Bolivia                0    0    0    0    0    0    0
## BosniaHerzegovina      0    0    0    0    0    0    0
## Brazil                 0    0    0    0    0    0    0
## Brunei                 0    0    0    0    0    0    0
## Bulgaria               0    0    0    0    0    0    0
## Burma                  0    0    0    0    0    0    0
## Cameroon               0    0    0    0    0    0    0
## Canada                 0    0    0    0    0    0    0
## Chile                  0    0    0    0    0    0    0
## China                  0    0    0    0    0    0    0
## Colombia               0    0    0    0    0    0    0
## Congo-Brazzaville      0    0    0    0    0    0    0
## CostaRica              0    0    0    0    0    0    0
## Croatia                0    0    0    0    0    0    0
## Cuba                   0    0    0    0    0    0    0
## CzechRepublic          0    0    0    0    0    0    0
## CoteDiViore            0    0    0    0    0    0    0
## Denmark                0    0    0    0    0    0    0
## DominicanRepublic      0    0    0    0    0    0    0
## Ecuador                0    0    0    0    0    0    0
## Egypt                  0    0    0    0    0    0    0
## ElSalvador             0    0    0    0    0    0    0
## Estonia                0    0    0    0    0    0    0
## Ethiopia               0    0    0    0    0    0    0
## Finland                0    0    0    0    0    0    0
## France                 0    0    0    0    0    0    0
## Gabon                  0    0    0    0    0    0    0
## Germany                0    0    0    0    0    0    0
## Ghana                  0    0    0    0    0    0    0
## Greece                 0    0    0    0    0    0    0
## Guatemala              0    0    0    0    0    0    0
## Honduras               0    0    0    0    0    0    0
## Hungary                0    0    0    0    0    0    0
## India                  0    0    0    0    0    0    0
```

```
## Indonesia              0    0    0    0    0    0    0
## Iran                   0    0    0    0    0    0    0
## Iraq                   0    0    0    0    0    0    0
## Ireland                0    0    0    0    0    0    0
## Israel                 0    0    0    0    0    0    0
## Italy                  0    0    0    0    0    0    0
## Jamaica                0    0    0    0    0    0    0
## Japan                  0    0    0    0    0    0    0
## Jordan                 0    0    0    0    0    0    0
## Kazakhstan             0    0    0    0    0    0    0
## Kenya                  0    0    0    0    0    0    0
## Kuwait                 0    0    0    0    0    0    0
## Latvia                 0    0    0    0    0    0    0
## Libya                  0    0    0    0    0    0    0
## Lithuania              0    0    0    0    0    0    0
## Malaysia               0    0    0    0    0    0    0
## Mexico                 0    0    0    0    0    0    0
## Morocco                0    0    0    0    0    0    0
## Netherlands            0    0    0    0    0    0    0
## NetherlandsAntilles    0    0    0    0    0    0    0
## NewZealand             0    0    0    0    0    0    0
## Nicaragua              0    0    0    0    0    0    0
## Nigeria                0    0    0    0    0    0    0
## Norway                 0    0    0    0    0    0    0
## Oman                   0    0    0    0    0    0    0
## Pakistan               0    0    0    0    0    0    0
## PapuaNewGuinea         0    0    0    0    0    0    0
## Peru                   0    0    0    0    0    0    0
## Philippines            0    0    0    0    0    0    0
## Poland                 0    0    0    0    0    0    0
## Portugal               0    0    0    0    0    0    0
## Qatar                  0    0    0    0    0    0    0
## Romania                0    0    0    0    0    0    0
## Russia                 0    0    0    0    0    0    0
## SaudiArabia            0    0    0    0    0    0    0
## Senegal                0    0    0    0    0    0    0
## Serbia                 0    0    0    0    0    0    0
## Singapore              0    0    0    0    0    0    0
## Slovakia               0    0    0    0    0    0    0
## Slovenia               0    0    0    0    0    0    0
## SouthAfrica            0    0    0    0    0    0    0
## SouthKorea             0    0    0    0    0    0    0
## Spain                  0    0    0    0    0    0    0
## SriLanka               0    0    0    0    0    0    0
## Sudan                  0    0    0    0    0    0    0
## Suriname               0    0    0    0    0    0    0
## Sweden                 0    0    0    0    0    0    0
## Switzerland            0    0    0    0    0    0    0
## Syria                  0    0    0    0    0    0    0
## Taiwan                 0    0    0    0    0    0    0
## Thailand               0    0    0    0    0    0    0
## TrinidadandTobago      0    0    0    0    0    0    0
## Tunisia                0    0    0    0    0    0    0
## Turkey                 0    0    0    0    0    0    0
```

```
## Turkmenistan           0    0    0    0    0    0    0
## Ukraine                0    0    0    0    0    0    0
## UnitedArabEmirates     0    0    0    0    0    0    0
## UnitedKingdom          0    0    0    0    0    0    0
## UnitedStates           0    0    0    0    0    0    0
## Uruguay                0    0    0    0    0    0    0
## Uzbekistan             0    0    0    0    0    0    0
## Venezuela              0    0    0    0    0    0    0
## Vietnam                0    0    0    0    0    0    0
## Yemen                  0    0    0    0    0    0    0
## Zambia                 0    0    0    0    0    0    0
## Zimbabwe               0    0    0    0    0    0    0
```

```r
screeplot(acp_escalado, col = rojo, pch = 16,
          type = "lines", cex = 2, lwd = 2, main = "")
```



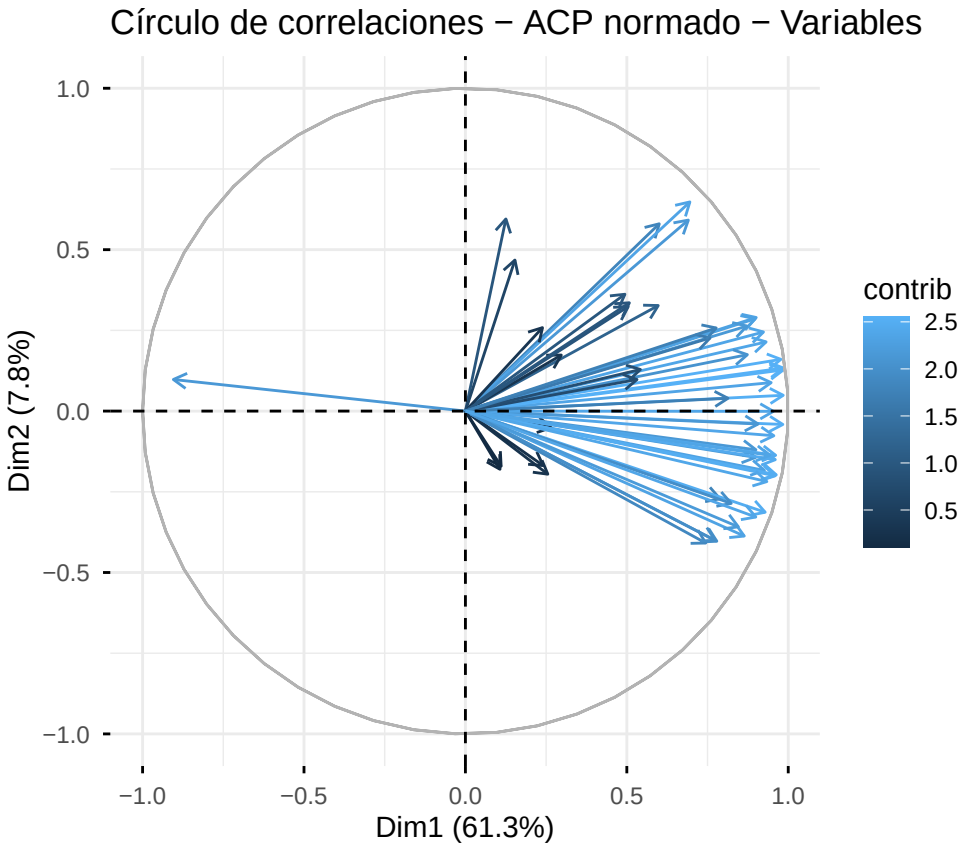
Pregunta 5. Analice la representación de las variables originales sobre las dos primeras componentes en el primer plano factorial. ¿Que conclusiones podemos sacar de este análisis?

Se desarrolla el círculo de correlaciones con factoextra

```r
install.packages("factoextra")
library(factoextra)

fviz_pca_var(
```

```
  acp_escalado,
  axes = c(1, 2),
  col.var = "contrib",
  label="none",
  repel = TRUE,
  title = "Círculo de correlaciones - ACP normado - Variables"
)
```



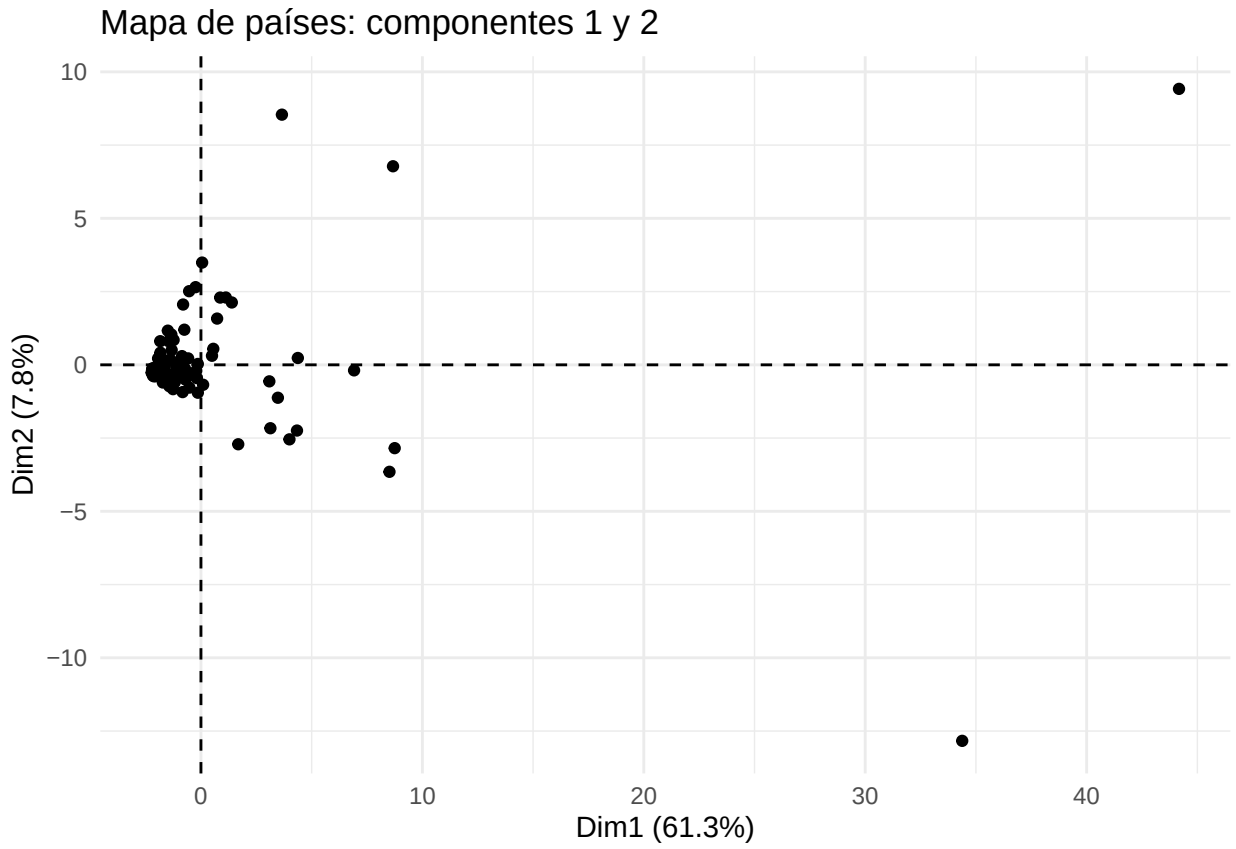
1. En cuanto a la longitud de las flechas podemos confirmar que la mayoría al ser largas y cercanas a la circunferencia tienen una mayor explicación y están mejor o más representadas por PC1 Y PC2
2. Como los angulos entre las variables es pequeño, menor a 90°, se concluye que se tiene una alta correlación positiva entre la gran mayoría de variables, solo una tiene una correlacion negativa con las demás porque su angulo es grande,mayor a 90°.
3. La gran mayoría de variables presentan un color azul claro, lo que muestra una mayor contribución en la construcción de las componentes principales.
4. Debido a que la mayoría de variables apuntan hacia la derecha hay una asociación positiva de estas con la primera componente principal.

Pregunta 6. Ahora examine con detenimiento el mapa de individuos sobre los planos factoriales que conforman las componentes (1,2) y (2,3). ¿Qué conclusiones puede sacar según la cercanía de algunas UE?

Plano factorial (1, 2)

```r
grafico_12 <- fviz_pca_ind(
  acp_escalado,
  axes = c(1, 2),
  geom = "point",label="all",
  title = "Mapa de países: componentes 1 y 2"
) +
  theme_minimal()

print(grafico_12)
```

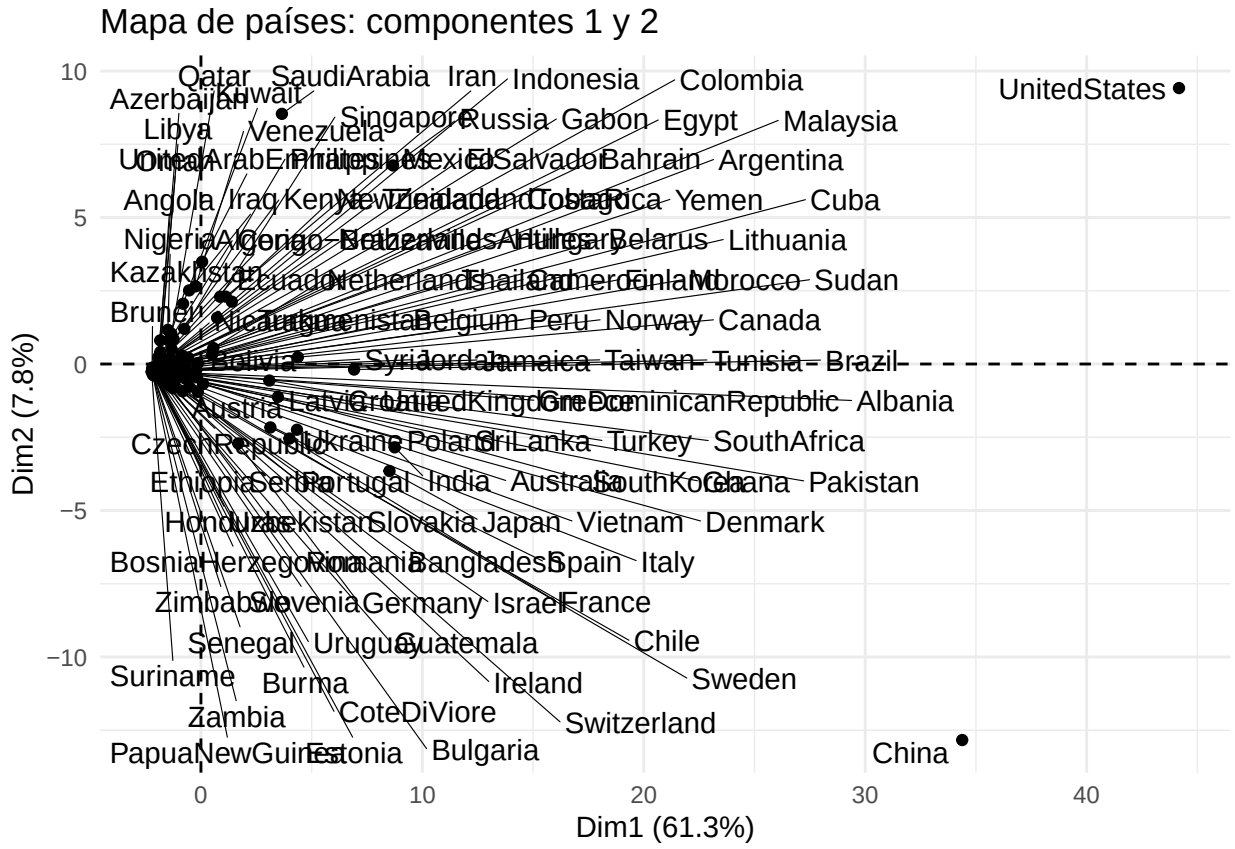


Para ver unicamente los nombres de los mas extremos, como hay muchos paises muy cerca se dificulta la lectura de los labels de los 111 países

:

```r
grafico_12 <- fviz_pca_ind(
  acp_escalado,
  axes = c(1, 2),
  geom = c("point", "text"),
  label = "all",
  repel = TRUE,
  title = "Mapa de países: componentes 1 y 2"
) +
  theme_minimal()

print(grafico_12)
```

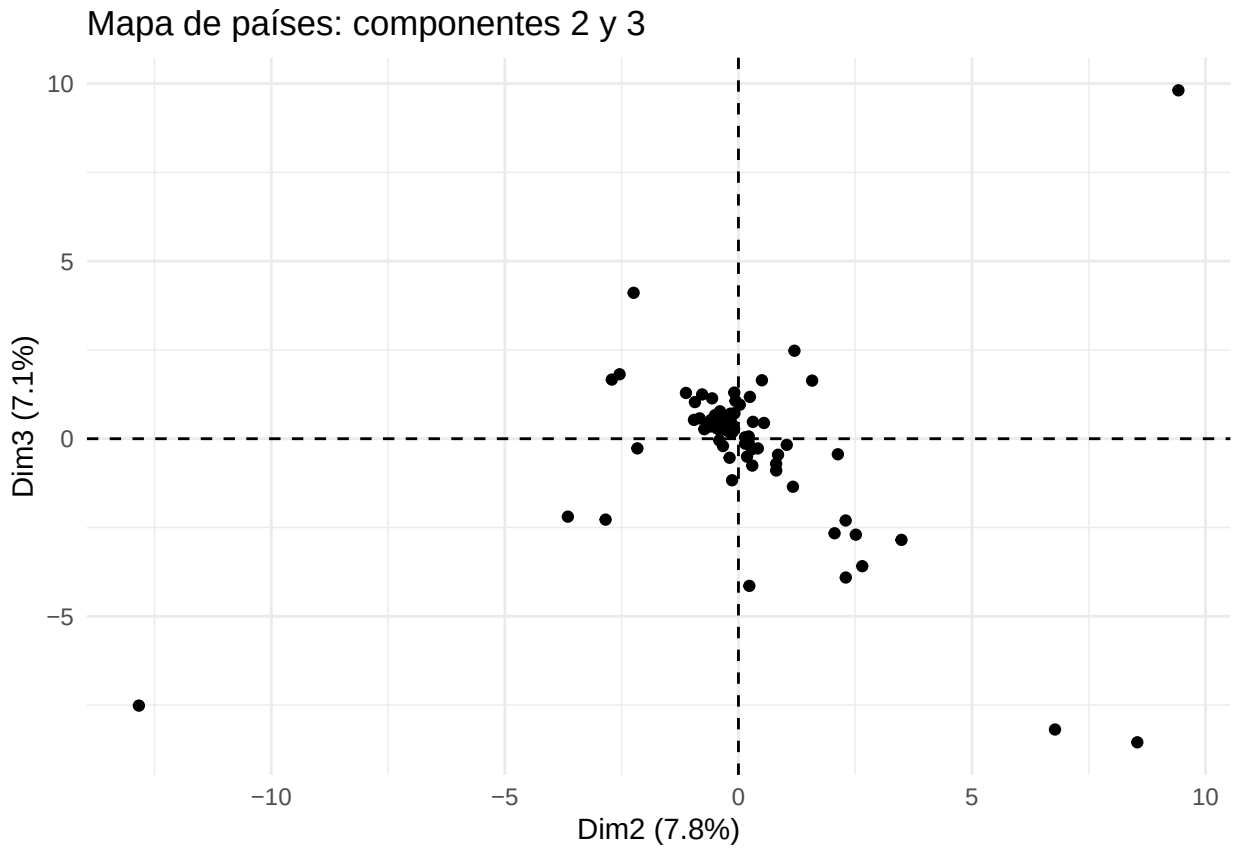


1. Los países cercanos presentan perfiles energéticos similares en las dimensiones CP1 Y CP2, mientras que los alejados como China y United States, tienen perfiles diferentes en esas dimensiones
2. Los países que se encuentran cerca al origen presentan valores próximos al perfil promedio en las dimensiones representadas

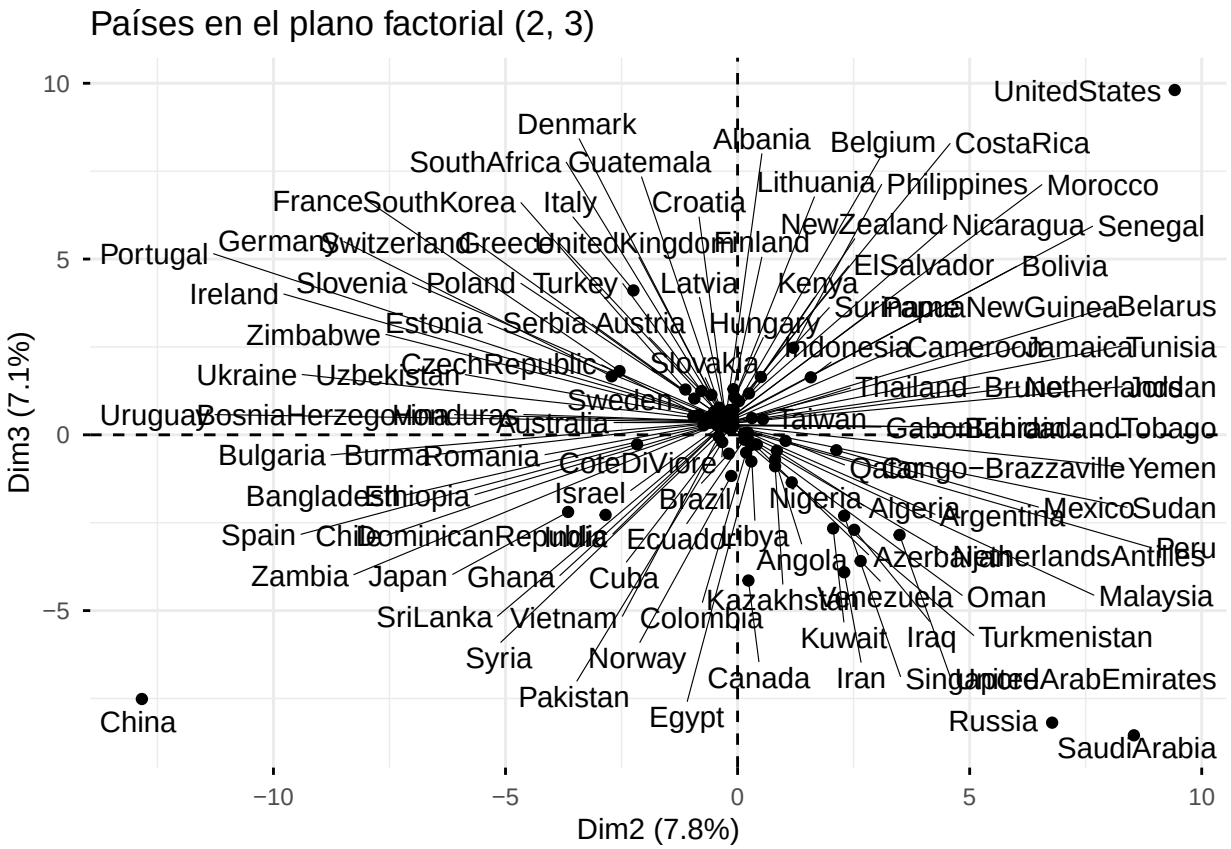
Plano factorial (2, 3)

```
grafico_23 <- fviz_pca_ind(
  acp_escalado,
  axes = c(2, 3),
  geom = "point",
  title = "Mapa de países: componentes 2 y 3"
) +
  theme_minimal()

print(grafico_23)
```




```
fviz_pca_ind(
  acp_escalado,
  axes = c(2, 3),
  label = "all",
  repel = TRUE,
  title = "Países en el plano factorial (2, 3)"
)
```



1. Este gráfico muestra PC2 Y PC3 dejando por fuera a PC1, y lo que nos permite afirmar es que hay países con caracteristicas relacionadas en la segunda y tercera componente que no muestra necesariamente la primera.
2. se observan las relaciones entre los paises y se identifican similitudes en las variables por su cercanía en el plano factorial (2,3)

Pregunta 7. Como hemos observado en clase, el ACP es una técnica bastante sensible a datos atípicos, ejecute nuevamente el ACP retirando del conjunto de datos a Estados Unidos, China, Arabia Saudi y Rusia. ¿En que cambia el ACP al excluir estos países?. ¿Se perciben clusters de países con mayor claridad?. Calcule únicamente el ACP normado.

Se filtran los 4 países mencionados de la siguiente forma:

```
# Crear una lista con los paises a eliminar
paises_eliminar <- c("UnitedStates", "China",
                     "SaudiArabia", "Russia")

# Crear una nueva base sin los 4 paises
data1_filtrada <- data1[
  !rownames(data1) %in% paises_eliminar,]

# Crear una nueva base sin esos países
data_sin_extremos <- data_sin_dup[
  !rownames(data_sin_dup) %in% paises_eliminar,
]
```

```r
# Comprobar las dimensiones
dim(data_sin_extremos)
```

```
## [1] 107  56
```

Ahora ejecutamos únicamente el ACP normado:

```r
acp_sin_extremos <- prcomp(
  data_sin_extremos,
  center = TRUE,
  scale. = TRUE
)

# Resumen de la varianza explicada
summary(acp_sin_extremos)
```

```
## Importance of components:
##                           PC1    PC2     PC3     PC4     PC5     PC6     PC7
## Standard deviation     5.1932 2.4137 2.14540 1.94202 1.67946 1.49228 1.43014
## Proportion of Variance 0.4816 0.1040 0.08219 0.06735 0.05037 0.03977 0.03652
## Cumulative Proportion  0.4816 0.5856 0.66783 0.73517 0.78554 0.82531 0.86183
##                           PC8    PC9   PC10    PC11    PC12    PC13    PC14
## Standard deviation     1.26498 1.1761 0.9287 0.79705 0.77169 0.65814 0.60643
## Proportion of Variance 0.02857 0.0247 0.0154 0.01134 0.01063 0.00773 0.00657
## Cumulative Proportion  0.89040 0.9151 0.9305 0.94185 0.95249 0.96022 0.96679
##                           PC15    PC16   PC17   PC18    PC19    PC20    PC21
## Standard deviation     0.56904 0.50399 0.4909 0.4362 0.37675 0.34466 0.33584
## Proportion of Variance 0.00578 0.00454 0.0043 0.0034 0.00253 0.00212 0.00201
## Cumulative Proportion  0.97257 0.97711 0.9814 0.9848 0.98734 0.98946 0.99148
##                           PC22   PC23    PC24    PC25    PC26    PC27    PC28
## Standard deviation     0.28375 0.2702 0.24583 0.21235 0.20252 0.17344 0.16514
## Proportion of Variance 0.00144 0.0013 0.00108 0.00081 0.00073 0.00054 0.00049
## Cumulative Proportion  0.99292 0.9942 0.99530 0.99610 0.99684 0.99737 0.99786
##                           PC29    PC30    PC31    PC32    PC33    PC34    PC35
## Standard deviation     0.16024 0.13442 0.13271 0.11474 0.10805 0.09966 0.08107
## Proportion of Variance 0.00046 0.00032 0.00031 0.00024 0.00021 0.00018 0.00012
## Cumulative Proportion  0.99832 0.99864 0.99896 0.99919 0.99940 0.99958 0.99969
##                           PC36    PC37    PC38    PC39    PC40    PC41    PC42
## Standard deviation     0.07068 0.06052 0.05086 0.04239 0.03507 0.03277 0.02904
## Proportion of Variance 0.00009 0.00007 0.00005 0.00003 0.00002 0.00002 0.00002
## Cumulative Proportion  0.99978 0.99985 0.99989 0.99993 0.99995 0.99997 0.99998
##                           PC43    PC44     PC45     PC46     PC47     PC48
## Standard deviation     0.02365 0.01532 0.008267 0.005231 0.004773 0.002896
## Proportion of Variance 0.00001 0.00000 0.000000 0.000000 0.000000 0.000000
## Cumulative Proportion  0.99999 1.00000 1.000000 1.000000 1.000000 1.000000
##                           PC49     PC50     PC51     PC52      PC53      PC54
## Standard deviation     0.002604 0.002244 0.001473 0.001092 0.0008538 0.0004512
## Proportion of Variance 0.000000 0.000000 0.000000 0.000000 0.0000000 0.0000000
## Cumulative Proportion  1.000000 1.000000 1.000000 1.000000 1.0000000 1.0000000
##                           PC55      PC56
## Standard deviation     0.0004243 0.0001739
## Proportion of Variance 0.0000000 0.0000000
## Cumulative Proportion  1.0000000 1.0000000
```

Y comparamos la varianza:

```
# Varianza explicada por cada componente
var_original <- summary(acp_escalado)$importance
var_sin_extremos <- summary(acp_sin_extremos)$importance
# Comparar los primeros cinco componentes
comparacion <- data.frame(
  Componente = paste0("PC", 1:5),
  Original = round(
    var_original[2, 1:5] * 100, 2
  ),
  Sin_cuatro_paises = round(
    var_sin_extremos[2, 1:5] * 100, 2
  ),
  Acumulada_original = round(
    var_original[3, 1:5] * 100, 2
  ),
  Acumulada_sin_extremos = round(
    var_sin_extremos[3, 1:5] * 100, 2
  )
)

print(comparacion)
```

```
##     Componente Original Sin_cuatro_paises Acumulada_original
## PC1        PC1    61.33             48.16              61.33
## PC2        PC2     7.84             10.40              69.18
## PC3        PC3     7.14              8.22              76.31
## PC4        PC4     4.83              6.73              81.14
## PC5        PC5     4.38              5.04              85.53
##     Acumulada_sin_extremos
## PC1                  48.16
## PC2                  58.56
## PC3                  66.78
## PC4                  73.52
## PC5                  78.55
```

```
# Porcentaje de varianza explicado por PC1
pc1_original <- summary(acp_escalado)$importance[2, 1] * 100

pc1_sin_extremos <- summary(acp_sin_extremos)$importance[2, 1] * 100

# Comparar ambos resultados
cat("PC1 original:", round(pc1_original, 2), "%\n")
```

```
## PC1 original: 61.33 %
```

```
cat("PC1 sin los cuatro países:", round(pc1_sin_extremos, 2), "%\n")
```

```
## PC1 sin los cuatro países: 48.16 %
```

```r
# Varianza explicada por las dos primeras componentes

summary(acp_escalado)$importance[, 1:2]
```

```
##                             PC1      PC2
## Standard deviation     5.860516 2.095872
## Proportion of Variance 0.613320 0.078440
## Cumulative Proportion  0.613320 0.691760
```

```r
summary(acp_sin_extremos)$importance[, 1:2]
```

```
##                             PC1      PC2
## Standard deviation     5.193203 2.413747
## Proportion of Variance 0.481600 0.104040
## Cumulative Proportion  0.481600 0.585630
```

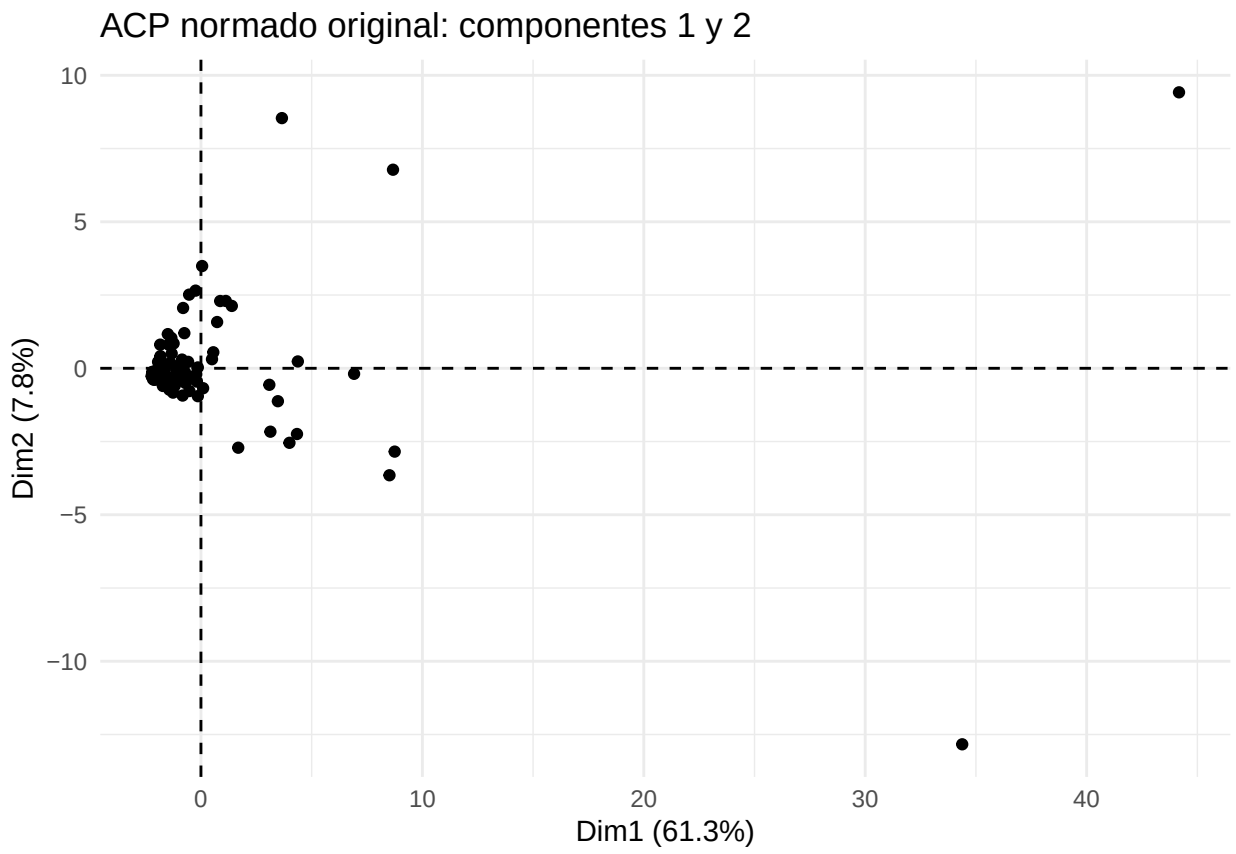
Observamos que al eliminar los 4 países el ACP muestra una disminución en la proporción de varianza que explica el modelo, de 69,17% a 58,56%, concentrado principalmente en la disminución de la primera componente.

Para poder revisar si hay clusteres de países excluyendo los 4 países desarrollamos:

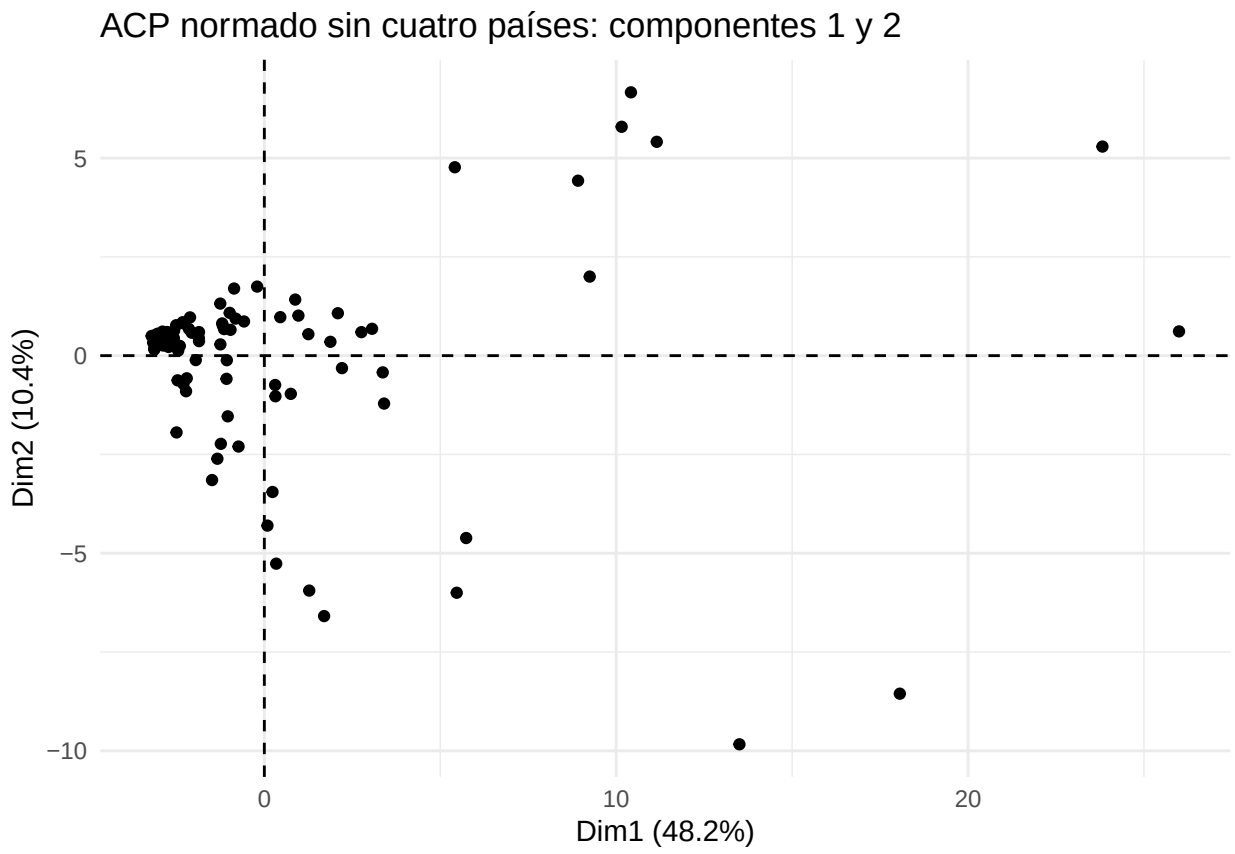
```r
# ACP original
grafico_original <- fviz_pca_ind(
  acp_escalado,
  axes = c(1, 2),
  geom = "point",
  label = "none",
  title = "ACP normado original: componentes 1 y 2"
) +
  theme_minimal()

# ACP sin los cuatro países
grafico_sin_extremos <- fviz_pca_ind(
  acp_sin_extremos,
  axes = c(1, 2),
  geom = "point",
  label = "none",
  title = "ACP normado sin cuatro países: componentes 1 y 2"
) +
  theme_minimal()

print(grafico_original)
```




```
print(grafico_sin_extremos)
```



Como mencionamos, PC1 pierde importancia relativa mientras que PC2 pasa a explicar del 7,85% al 10,40% de la variabilidad total. Por ello sería prudente incluir mas componentes para representar de manera mayor la información de las variables originales.

Clustering

Pregunta 8. De acuerdo a lo aprendido en la clase de agrupamiento, utilice el primer plano factorial para determinar de manera aproximada el número de grupos en el análisis. ¿Cuántos clusters espera que existan en el conjunto de datos?

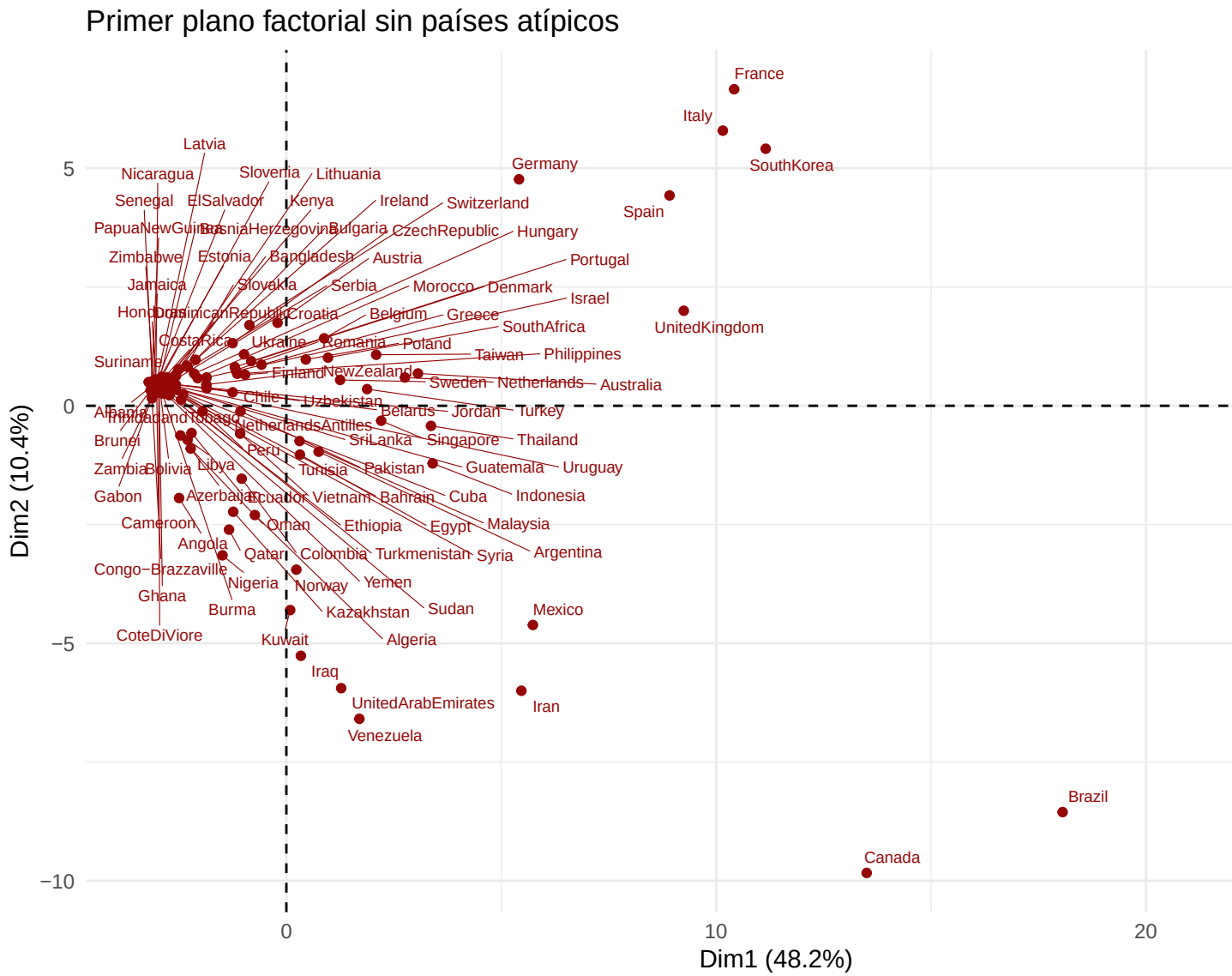
Para los puntos 8 a 11 trabajamos con `data_sin_extremos` (sin los 4 países atípicos del punto 7 y sin la columna duplicada) y con el ACP normado `acp_sin_extremos`. Para el clustering usamos la misma tabla estandarizada, ya que las variables están en unidades muy distintas (TBPD, BKWH, MK, etc.).

```
# Tabla estandarizada para el clustering
datos_esc <- scale(data_sin_extremos)

# Varianza explicada por el primer plano factorial
summary(acp_sin_extremos)$importance[, 1:2]
```

```
##                             PC1      PC2
## Standard deviation     5.193203 2.413747
## Proportion of Variance 0.481600 0.104040
## Cumulative Proportion  0.481600 0.585630
```

```r
# Mapa de países en el primer plano factorial
fviz_pca_ind(
  acp_sin_extremos,
  axes = c(1, 2),
  repel = TRUE,
  labelsize = 2.5,
  col.ind = rojo_oscuro,
  title = "Primer plano factorial sin países atípicos"
) +
  theme_minimal()
```



El primer plano factorial resume cerca del 59 % de la variabilidad total. El PC1 explica cerca del 48 % y el PC2 cerca del 10 %, así que es una buena aproximación para ver cómo se agrupan los países. El PC1 funciona como un índice de *tamaño energético*: a la derecha están los países que más consumen y producen petróleo y electricidad. El PC2 separa a los grandes *productores y exportadores de crudo* (Iraq, Venezuela,

Irán, Kuwait, Nigeria, Canadá) de los países *industriales importadores* con generación nuclear y renovable (Francia, Alemania, Italia, España, Corea del Sur).

Visualmente se distinguen aproximadamente **4 grupos**:

1. Una nube grande y compacta de países con bajo peso energético, en el extremo izquierdo del PC1. Aquí está la mayoría.
2. Los exportadores de petróleo, que se separan por el PC2.
3. Las potencias industriales importadoras de Europa, más Corea del Sur.
4. Los gigantes extremos (India y Japón, Brasil y Canadá), muy alejados del resto sobre el PC1.

Por lo tanto, esperamos alrededor de **4 clusters**. Como el grupo de exportadores no se separa con total claridad de la nube central, el número real podría estar entre 3 y 5.

Pregunta 9. Mediante el método de reducción de varianza explicada, determine un número fijo de clusters para su análisis. No tiene que ser el mismo número que el elegido por su compañero, recuerde que ninguno conoce las etiquetas reales de los grupos de los países.

Siguiendo el método visto en clase, calculamos cuánta varianza "explican" los clusters para k = 1, ..., 10. La reducción es la diferencia entre la distancia total y la suma de distancias dentro de los grupos.

```
set.seed(123)

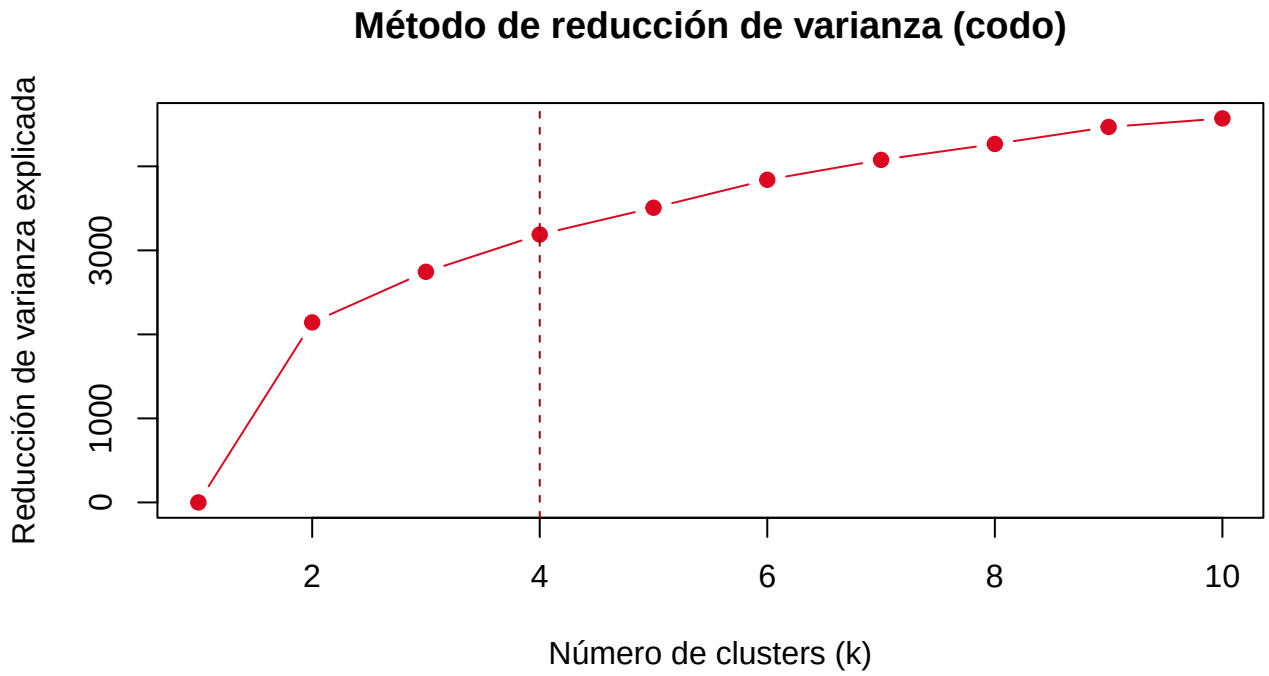
# Distancia cuadrada total (equivale a tener un solo grupo)
dist_total <- sum(dist(datos_esc)^2) / nrow(datos_esc)

reduccion <- numeric(10)
for (k in 1:10) {
  # nstart = 25: se prueban 25 arranques aleatorios y se toma el mejor
  km_k <- kmeans(datos_esc, centers = k, nstart = 25)
  reduccion[k] <- dist_total - sum(km_k$withinss)
}

tabla_codo <- data.frame(
  k = 1:10,
  reduccion = round(reduccion, 1),
  pct_explicado = round(100 * reduccion / dist_total, 1)
)
tabla_codo$ganancia_pct <- c(NA, diff(tabla_codo$pct_explicado))
tabla_codo
```

```
##     k reduccion pct_explicado ganancia_pct
## 1   1       0.0           0.0           NA
## 2   2    2142.6          36.1         36.1
## 3   3    2744.7          46.2         10.1
## 4   4    3188.7          53.7          7.5
## 5   5    3507.3          59.1          5.4
## 6   6    3840.0          64.7          5.6
## 7   7    4076.8          68.7          4.0
## 8   8    4266.5          71.9          3.2
## 9   9    4468.9          75.3          3.4
## 10 10    4570.4          77.0          1.7
```

```r
plot(1:10, reduccion, type = "b", pch = 19, col = rojo,
     xlab = "Número de clusters (k)",
     ylab = "Reducción de varianza explicada",
     main = "Método de reducción de varianza (codo)")
abline(v = 4, col = rojo_oscuro, lty = 2)
```



La varianza explicada por los clusters es:

k	Varianza explicada
2	36 %
3	46 %
4	54 %
5	61 %
6	66 %

El salto más grande ocurre al pasar de 1 a 2 grupos. Después las ganancias se reducen a cerca de 10, 8 y 7 puntos porcentuales, y a partir de k = 6 cada grupo adicional aporta menos de 5 puntos. El codo está entre k = 3 y k = 5.

Elegimos **k = 4** por tres razones:

- Coincide con lo que observamos en el primer plano factorial (pregunta 8).
- Es la solución más parsimoniosa antes de que la curva se aplane.
- Es **estable**: con distintas semillas, k-means siempre encuentra los mismos 4 grupos.

Con k = 5 aparece un grupo intermedio de exportadores de petróleo y economías medianas. Sin embargo, sus integrantes cambian según la semilla (entre 10 y 15 países), lo que indica que esa separación no es robusta.

Pregunta 10. Determine mediante k-means y aglomeración jerárquica (usando enlace promedio) la clasificación en grupos para los países de estudio. Considere todas las variables de estudio ¿son diferentes los resultados de los dos análisis cluster?

```
set.seed(123)
km4 <- kmeans(datos_esc, centers = 4, nstart = 25)

# Tamaño e integrantes de cada grupo
km4$size
```

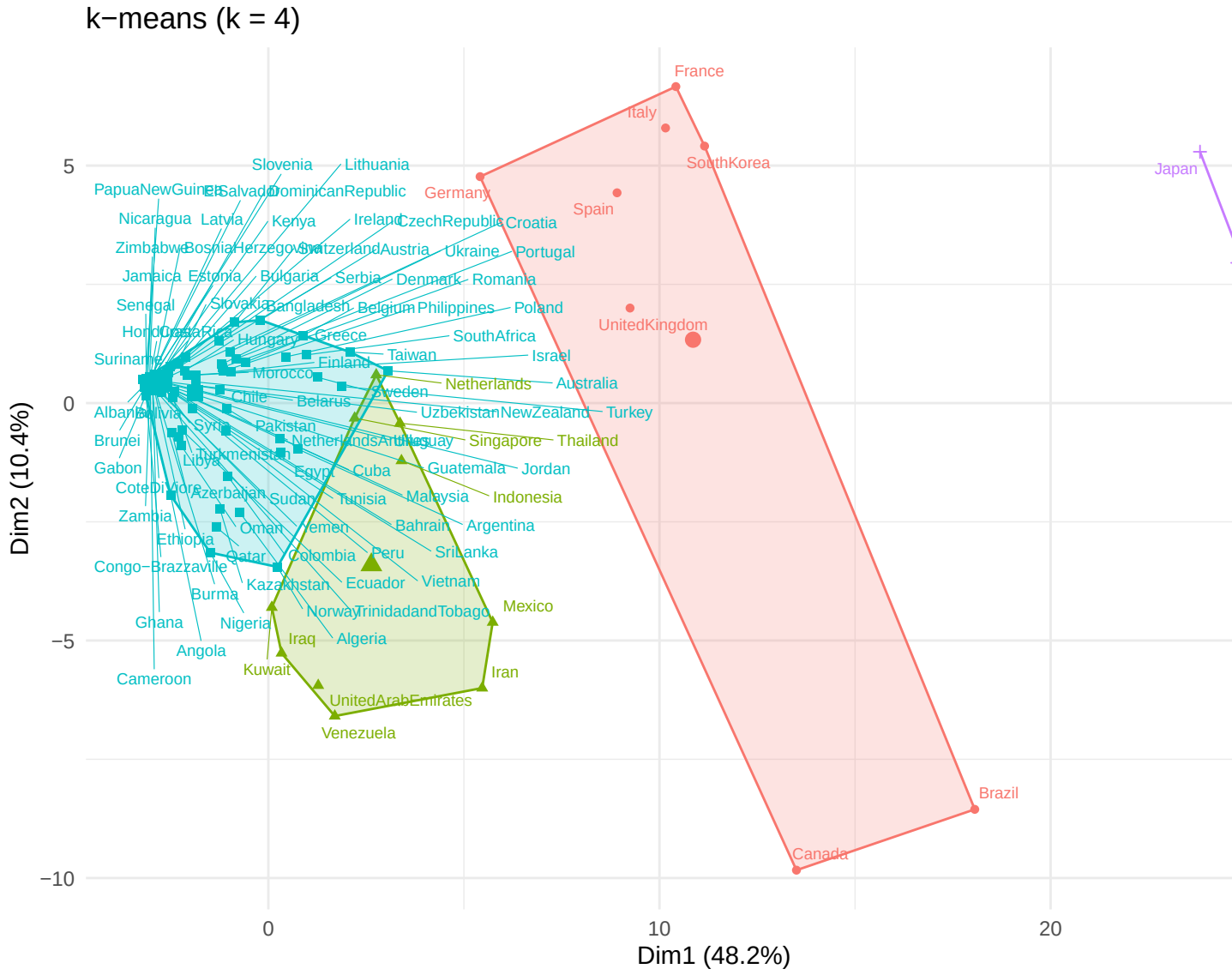
```
## [1]  8 10 87  2
```

```
split(rownames(datos_esc), km4$cluster)
```

```
## $`1`
## [1] "Brazil"        "Canada"        "France"        "Germany"
## [5] "Italy"         "SouthKorea"    "Spain"         "UnitedKingdom"
##
## $`2`
##  [1] "Indonesia"          "Iran"               "Iraq"
##  [4] "Kuwait"             "Mexico"             "Netherlands"
##  [7] "Singapore"          "Thailand"           "UnitedArabEmirates"
## [10] "Venezuela"
##
## $`3`
##  [1] "Albania"             "Algeria"             "Angola"
##  [4] "Argentina"           "Australia"           "Austria"
##  [7] "Azerbaijan"          "Bahrain"             "Bangladesh"
## [10] "Belarus"             "Belgium"             "Bolivia"
## [13] "BosniaHerzegovina"   "Brunei"              "Bulgaria"
## [16] "Burma"               "Cameroon"            "Chile"
## [19] "Colombia"            "Congo-Brazzaville"   "CostaRica"
## [22] "Croatia"             "Cuba"                "CzechRepublic"
## [25] "CoteDiViore"         "Denmark"             "DominicanRepublic"
## [28] "Ecuador"             "Egypt"               "ElSalvador"
## [31] "Estonia"             "Ethiopia"            "Finland"
## [34] "Gabon"               "Ghana"               "Greece"
## [37] "Guatemala"           "Honduras"            "Hungary"
## [40] "Ireland"             "Israel"              "Jamaica"
## [43] "Jordan"              "Kazakhstan"          "Kenya"
## [46] "Latvia"              "Libya"               "Lithuania"
## [49] "Malaysia"            "Morocco"             "NetherlandsAntilles"
## [52] "NewZealand"          "Nicaragua"           "Nigeria"
## [55] "Norway"              "Oman"                "Pakistan"
## [58] "PapuaNewGuinea"      "Peru"                "Philippines"
## [61] "Poland"              "Portugal"            "Qatar"
## [64] "Romania"             "Senegal"             "Serbia"
## [67] "Slovakia"            "Slovenia"            "SouthAfrica"
## [70] "SriLanka"            "Sudan"               "Suriname"
## [73] "Sweden"              "Switzerland"         "Syria"
```

```
## [76] "Taiwan"                "TrinidadandTobago"    "Tunisia"
## [79] "Turkey"                "Turkmenistan"         "Ukraine"
## [82] "Uruguay"               "Uzbekistan"           "Vietnam"
## [85] "Yemen"                 "Zambia"               "Zimbabwe"
##
## $`4`
## [1] "India" "Japan"
```

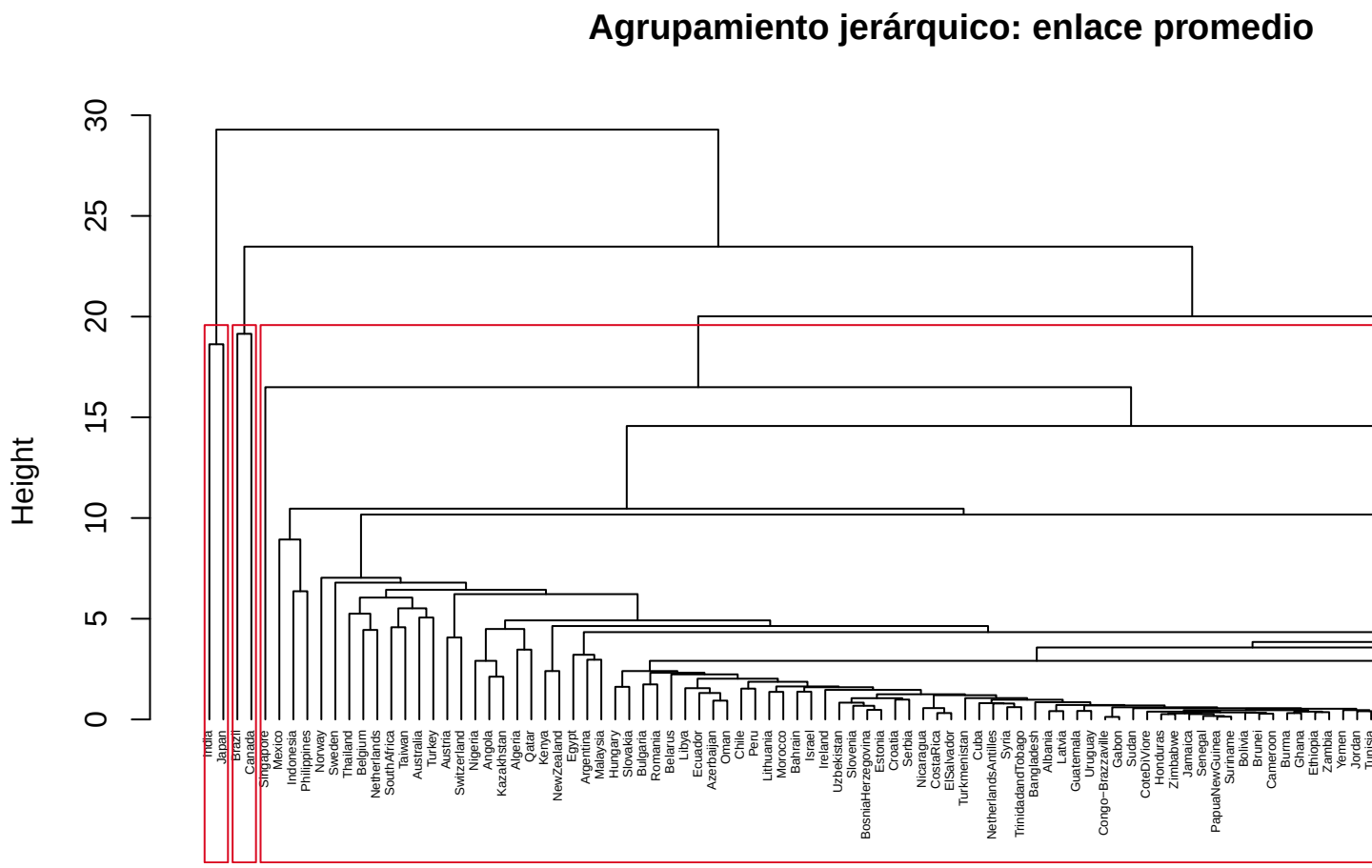
```
fviz_cluster(km4, data = datos_esc, repel = TRUE, labelsize = 7,
             ellipse.type = "convex", main = "k-means (k = 4)") +
  theme_minimal()
```




```
jer_prom <- hclust(dist(datos_esc), method = "average")

plot(jer_prom, cex = 0.45, hang = -1,
```

```
    main = "Agrupamiento jerárquico: enlace promedio", xlab = "", sub = "")
rect.hclust(jer_prom, k = 4, border = rojo)
```




```
grupos_jer <- cutree(jer_prom, k = 4)
table(grupos_jer)
```

```
## grupos_jer
##   1   2   3   4
## 101   2   2   2
```

```
split(rownames(datos_esc), grupos_jer)
```

```
## $`1`
```

```
##   [1] "Albania"             "Algeria"             "Angola"
##   [4] "Argentina"           "Australia"           "Austria"
##   [7] "Azerbaijan"          "Bahrain"             "Bangladesh"
##  [10] "Belarus"             "Belgium"             "Bolivia"
##  [13] "BosniaHerzegovina"   "Brunei"              "Bulgaria"
##  [16] "Burma"               "Cameroon"            "Chile"
##  [19] "Colombia"            "Congo-Brazzaville"   "CostaRica"
##  [22] "Croatia"             "Cuba"                "CzechRepublic"
##  [25] "CoteDiViore"         "Denmark"             "DominicanRepublic"
##  [28] "Ecuador"             "Egypt"               "ElSalvador"
##  [31] "Estonia"             "Ethiopia"            "Finland"
##  [34] "Gabon"               "Germany"             "Ghana"
##  [37] "Greece"              "Guatemala"           "Honduras"
##  [40] "Hungary"             "Indonesia"           "Iran"
##  [43] "Iraq"                "Ireland"             "Israel"
##  [46] "Italy"               "Jamaica"             "Jordan"
##  [49] "Kazakhstan"          "Kenya"               "Kuwait"
##  [52] "Latvia"              "Libya"               "Lithuania"
##  [55] "Malaysia"            "Mexico"              "Morocco"
##  [58] "Netherlands"         "NetherlandsAntilles" "NewZealand"
##  [61] "Nicaragua"           "Nigeria"             "Norway"
##  [64] "Oman"                "Pakistan"            "PapuaNewGuinea"
##  [67] "Peru"                "Philippines"         "Poland"
##  [70] "Portugal"            "Qatar"               "Romania"
##  [73] "Senegal"             "Serbia"              "Singapore"
##  [76] "Slovakia"            "Slovenia"            "SouthAfrica"
##  [79] "Spain"               "SriLanka"            "Sudan"
##  [82] "Suriname"            "Sweden"              "Switzerland"
##  [85] "Syria"               "Taiwan"              "Thailand"
##  [88] "TrinidadandTobago"   "Tunisia"             "Turkey"
##  [91] "Turkmenistan"        "Ukraine"             "UnitedArabEmirates"
##  [94] "UnitedKingdom"       "Uruguay"             "Uzbekistan"
##  [97] "Venezuela"           "Vietnam"             "Yemen"
## [100] "Zambia"              "Zimbabwe"
##
## $`2`
## [1] "Brazil" "Canada"
##
## $`3`
## [1] "France"     "SouthKorea"
##
## $`4`
## [1] "India" "Japan"
```

```r
# Tabla cruzada entre los dos métodos
table(KMeans = km4$cluster, Jerarquico = grupos_jer)
```

```
##       Jerarquico
## KMeans  1  2  3  4
##      1  4  2  2  0
##      2 10  0  0  0
##      3 87  0  0  0
##      4  0  0  0  2
```

Los números de los grupos son etiquetas arbitrarias, así que la comparación se hace por los países que forman cada grupo:

Grupo	k-means (k = 4)	Jerárquico, enlace promedio (k = 4)
Gigantes consumidores	India, Japón	India, Japón
Grandes productores con alta generación hidroeléctrica	Brasil, Canadá	Brasil, Canadá
Potencias industriales importadoras	Francia, Alemania, Italia, Corea del Sur, España, Reino Unido	Solo Francia y Corea del Sur
Resto de países	97 países	101 países (incluye Alemania, Italia, España y Reino Unido)

Los resultados son **parecidos pero no iguales**. Ambos métodos aíslan a India y Japón y a Brasil y Canadá, y dejan a la gran mayoría de países en un grupo común. La diferencia está en las potencias europeas. k-means las reúne en un grupo propio de 6 países. El enlace promedio solo separa a Francia y Corea del Sur, que son los dos países con mayor generación nuclear, y deja a Alemania, Italia, España y Reino Unido dentro del grupo grande.

Esto se explica por cómo funciona cada método:

- **k-means** minimiza la varianza dentro de los grupos, por lo que forma grupos compactos alrededor de centroides.
- **El enlace promedio** une los grupos según la distancia promedio entre todos sus puntos. Con datos tan asimétricos como estos (pocos países enormes y muchos pequeños), separa primero las observaciones más alejadas y deja un grupo grande que encadena al resto.

Para caracterizar a los países, k-means da una partición más interpretable. El dendrograma, en cambio, muestra con claridad cuáles son los países más atípicos.

> Para el punto 11, los grupos de k-means quedan en `km4$cluster`. Se pueden unir a la base con `data_sin_extremos$grupo <- factor(km4$cluster)`.

Pregunta 11. Grafique mediante boxplot la distribución de las 5 variables seleccionadas en el punto 2 diferenciando por los grupos encontrados mediante k-means (es decir, obtenga un boxplot por variable y por grupo: si por ejemplo k-means indica un total de 3 grupos, realice 3 boxplots, uno por grupo, para cada una de las 5 variables). ¿Observa diferencias importantes entre las distribuciones por grupos de la misma variable?